Virtuoso® Layout Pro: T3 Basic Commands

Course Version IC 25.1

Lab Manual **Revision 1.0**

cādence®

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Virtuoso Layout Pro: T3 Basic Commands

Module 1: About This Course

There are no labs for this module.

Module 2: Design Creation and Placement

Lab 2-1 Populating a Design with *Generate All From Source*

Objective: To use the Generate All From Source (GFS) command to populate a design.

An advantage of Virtuoso® EXL over the XL suite is the capability to automatically populate your design with all the components that exist in your schematic in one pass. This eliminates the initial LVS (Layout Versus Schematic) task when you are initially instantiating components into your layout. In the IC 25.1 version, instances and pins are placed within (rather than outside) the initial *prBoundary* for ease of use. Also, the devices are initially placed more intelligently based on connectivity and I/O pin access.

In this lab, you populate your layout canvas using the Generate All From Source (GFS) command. GFS takes every component from the schematic and places those components (instances and pins) either within the *prBoundary* or below the *prBoundary*.

When all devices populate the layout canvas, you run the Pin Placer to place your input, output, and rail pins.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. Untar the lab database file, using the following command.

   ```
   tar -xzvf <filename.tar.gz>
   ```

2. Update the following line in the /VLPT3_IC_25_1/. cshrc_VLPT3 file with the appropriate valid local tool paths for Virtuoso.

   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
   ```

3. Save and close the file.

4. Source the updated class setup file.

   ```
   source .cshrc_VLPT3
   ```

Invoking the virtuoso Executable

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially log in:

   ```
   cd ./VLPT3_IC_25_1/VLPT3_M02_DCP
   ```

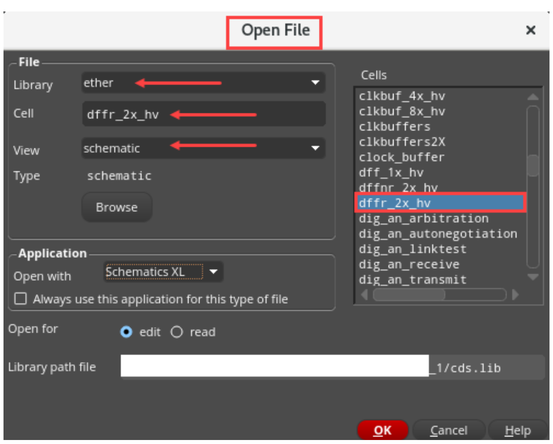
2. To start the Cadence® Virtuoso platform, enter this command in the same terminal window:

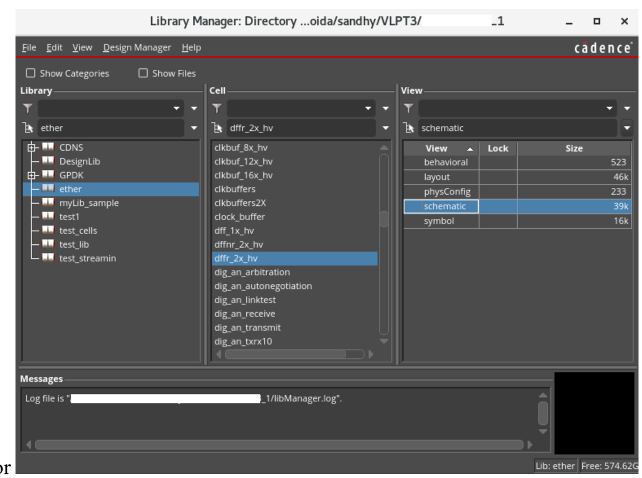
```
virtuoso &
```

The Command Interpreter Window (CIW) appears.

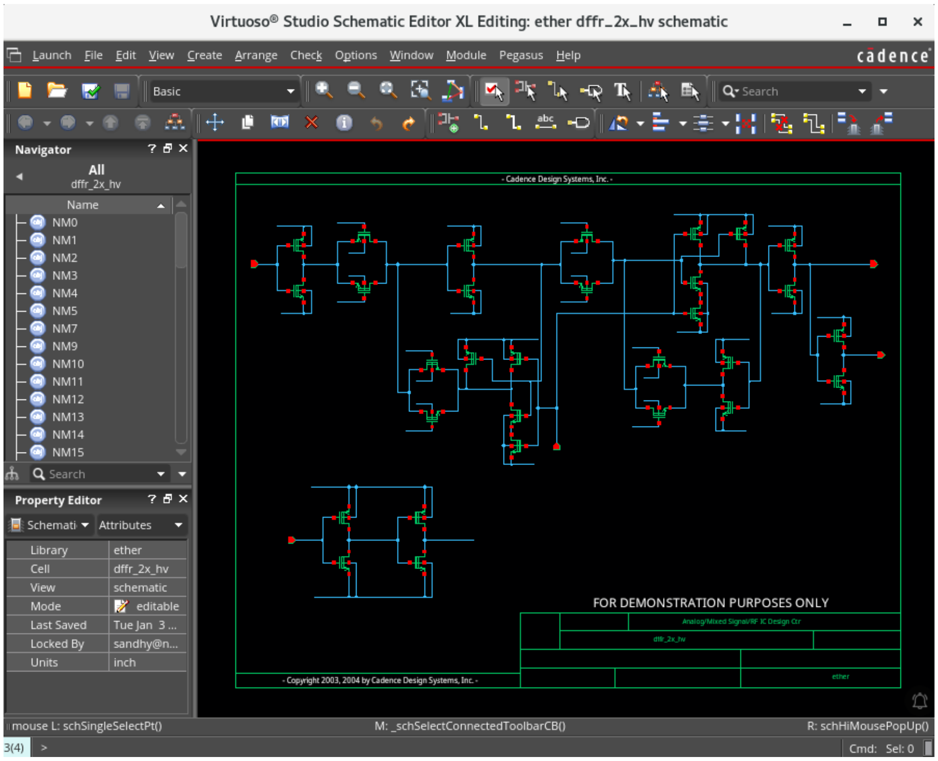
3. Open the following schematic in XL with the Library Manager or by choosing **File – Open** from the CIW or library manager.

Library	`ether`
Cell	`dffr_2x_hv`
View	`schematic`

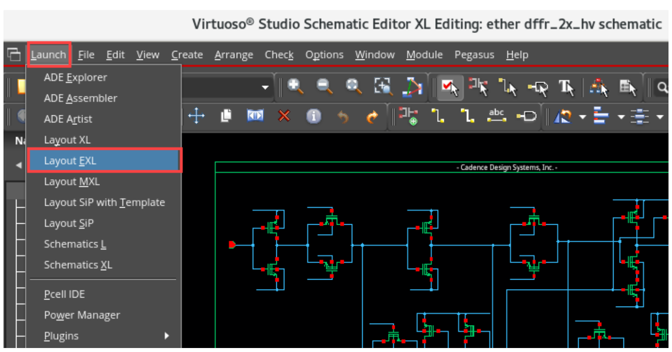


or


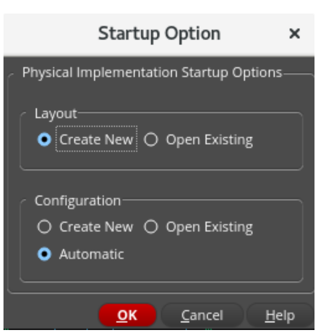
The *ether/dffr_2x_hv/schematic* window opens in XL mode.



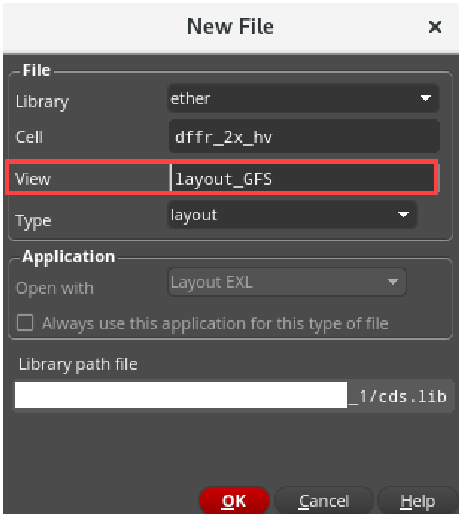
4. In the schematic, choose **Launch – Layout EXL** to bring up the **Startup Option** menu.



5. Select the options, as shown here, to create a new layout view and click **OK**.

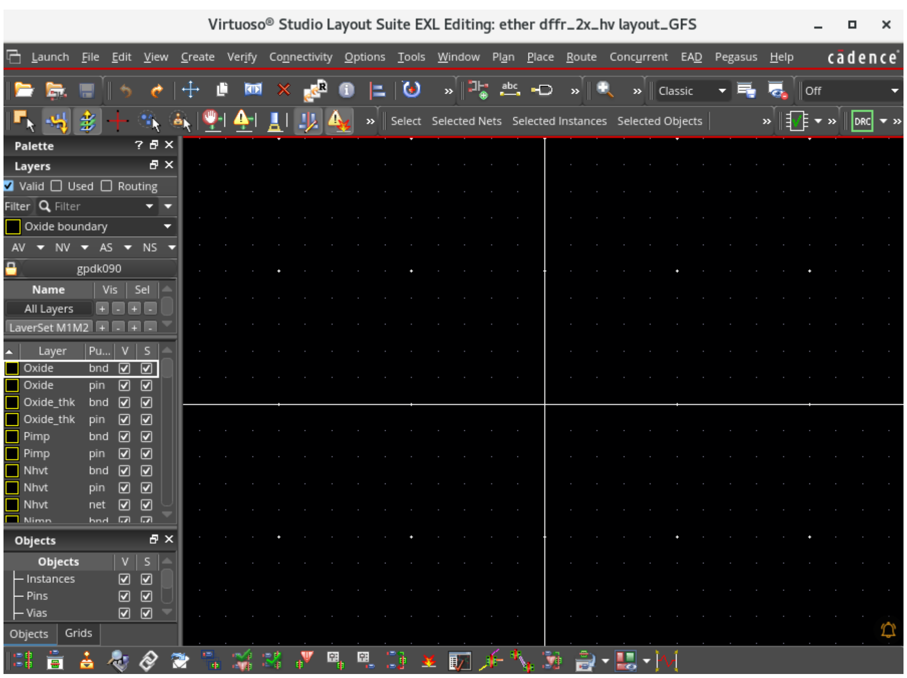


6. Fill out the New File form, as shown, by clicking in the **View** field and changing the view name to ***layout_GFS***.



7. Click **OK**.

   The *ether/dffr_2x_hv/layout_GFS* palette opens in EXL mode.

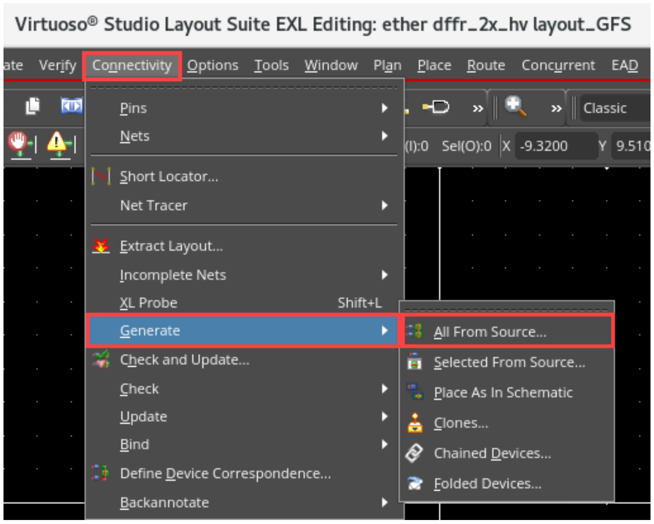


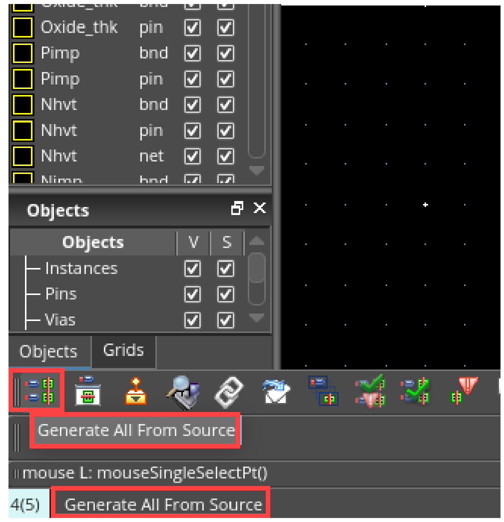
Note: In the layout canvas, from the **Window – Toolbars** menu, you can make the toolbars appear/disappear as per your requirement.

Note: In the layout canvas, from the **Window – Assistants** menu, you can enable/disable the assistants as per your requirement.

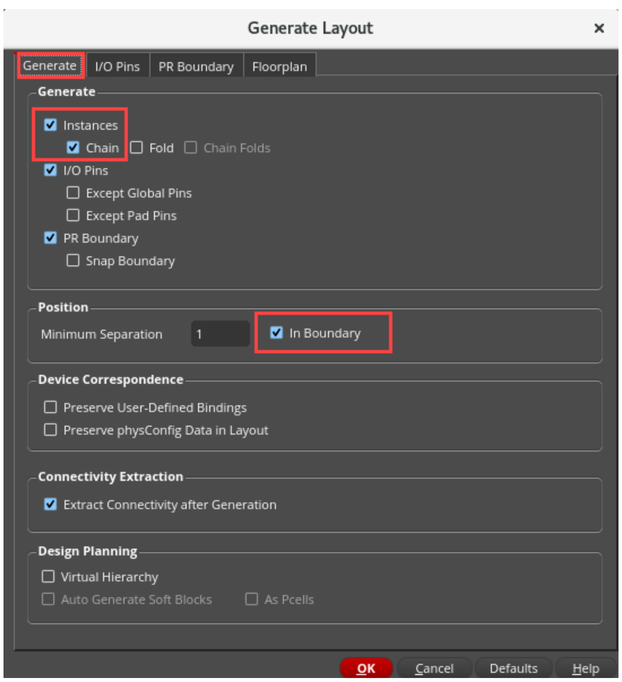
Note: In the layout canvas, from the Workspace Configuration cyclic field, you can choose the required workspace.

8. Generate all from the source by choosing either the **Connectivity – Generate – All From Source** menu command or clicking the **Generate All From Source (GFS)** icon.





The Generate Layout form appears, as shown here: select "**Chain**" and select "**In Boundary**".



Note: Notice the **In Boundary** option. If you choose it, instances and pins will be generated inside the *prboundary*. If you do not select this option, instances and pins will be generated below the *prboundary*.

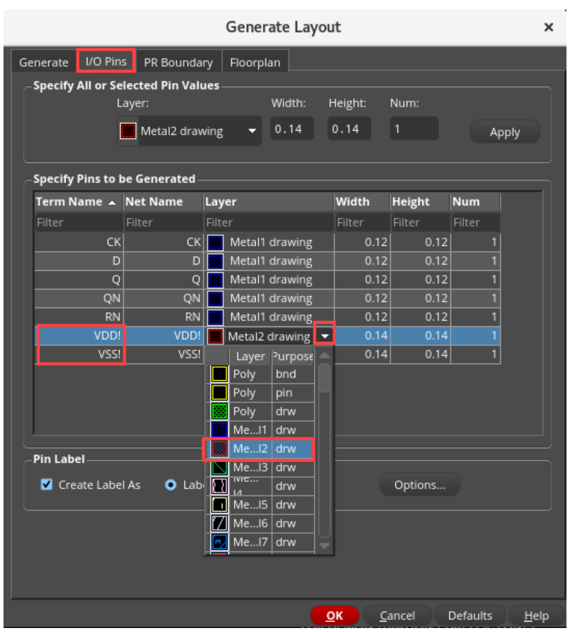
This is a user interface that has four tabs: **Generate**, **I/O Pins**, **PR Boundary** and **Floorplan**.

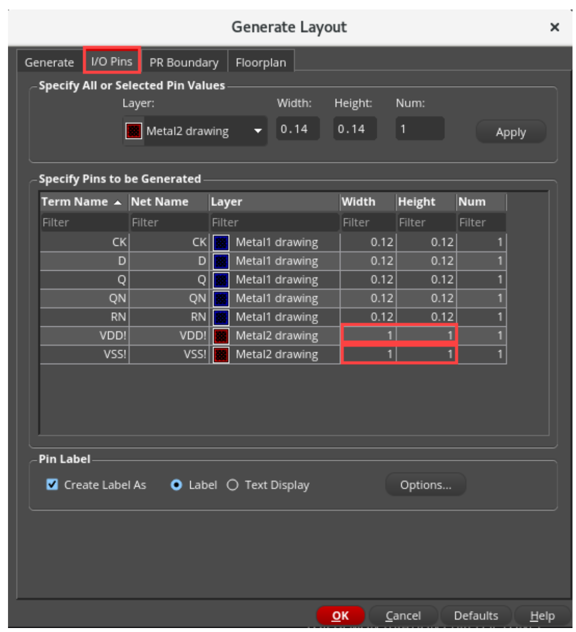
- The **first** tab (**Generate**) gives you the options to determine whether and how your instances, I/O pins, and PR Boundary will appear in your layout. You can also determine whether or not you want instances and pins to initially populate your layout inside or outside the PR Boundary, Preserve User-Defined Bindings, and whether or not to extract connectivity after generation.
- The **second** tab (**I/O Pins**) is for setting default values for pins, including layer and layer purpose, pin width and height and pin label.
- The **third** tab (**PR Boundary**) is used to define PR Boundary dimensions.
- The **fourth** tab (**Floorplan**) is used to preserve floorplanning objects, namely, rows and custom placement areas, blockages, area boundaries, track patterns and clusters.

9. Accept the setting by selecting the ***Chain*** and ***In Boundary*** options in the **Generate** tab, as shown in the above graphic, and click on the **I/O Pins** tab.

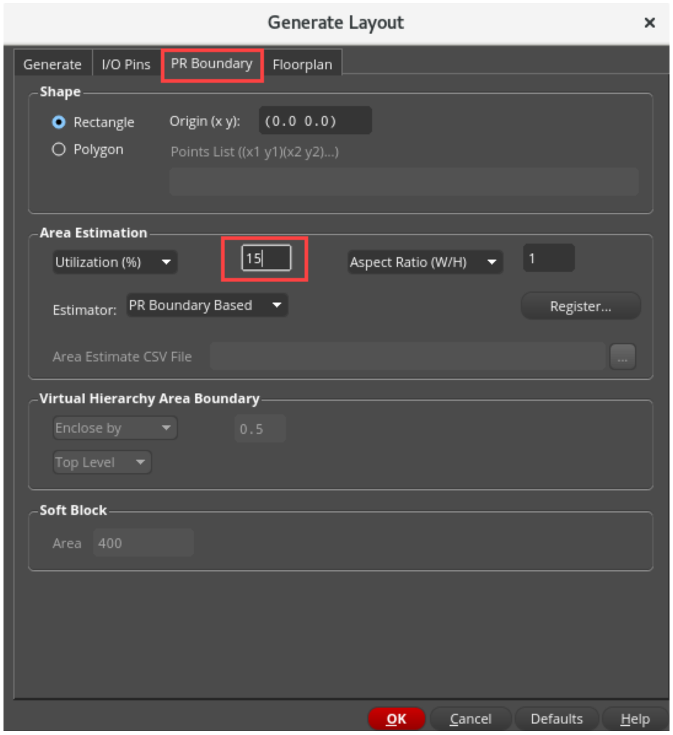
10. Under the I/O pins tab, change the *VDD!* and *VSS!* pin settings by drag-selecting VDD! and VSS!

11. Change the Layers of both pins to **Metal2 drawing** with a *Width* and *Height* of **1**.



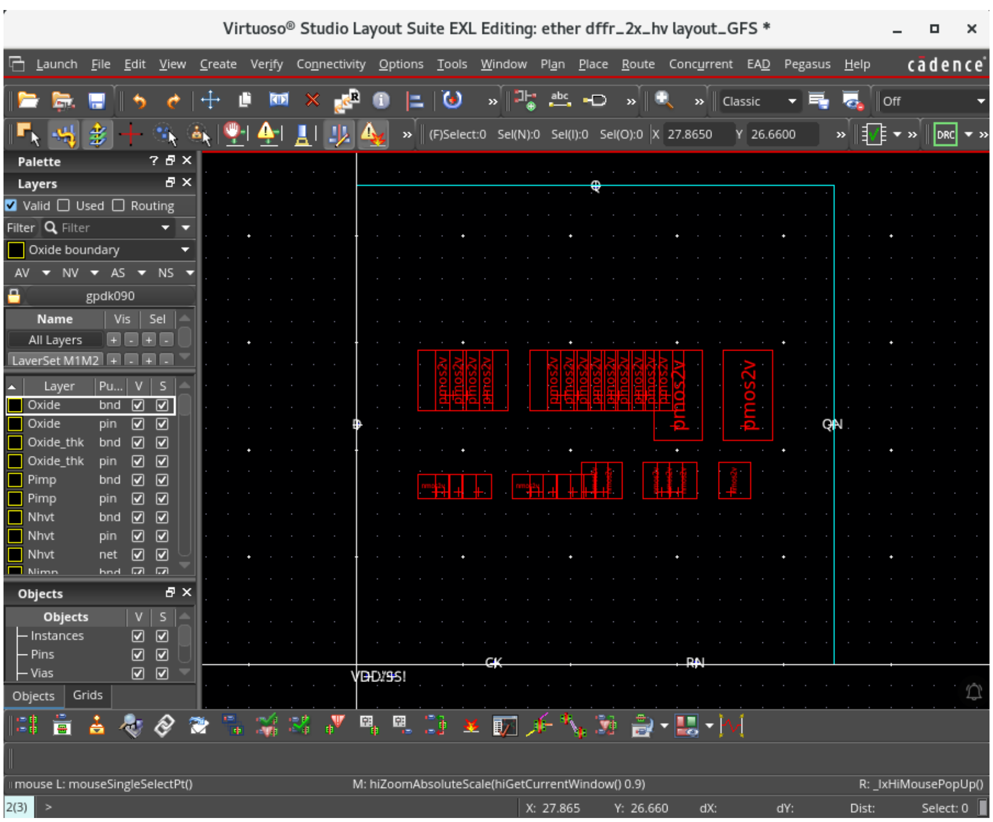


12. Under PR Boundary, keep the area estimation to 15 as shown below.

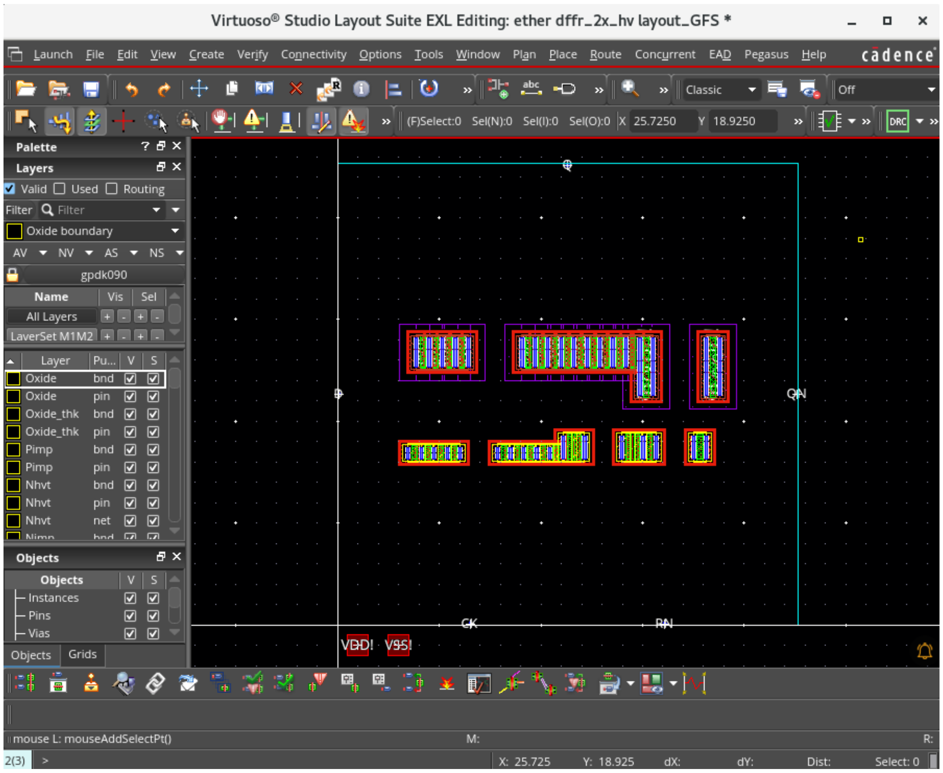


13. Click the **OK** button to effect the changes in the pin list.

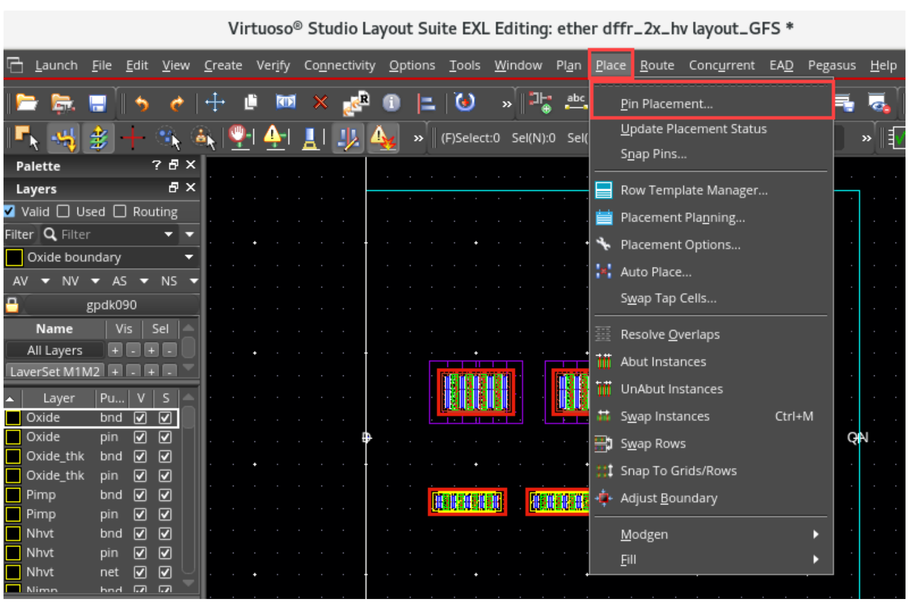
Your initial placement looks like this:



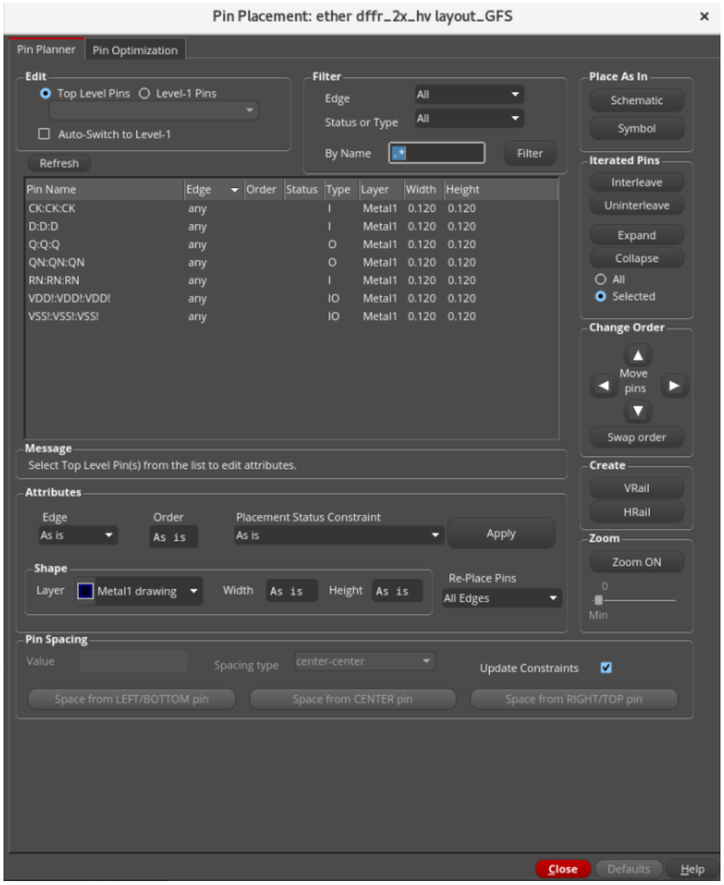
14. Shift+f to view the hierarchy.



15. Start the **Pin Placer** by choosing the **Place – Pin Placement** menu command in the layout canvas.



The Pin Placement form appears, as shown here:



You now adjust the settings in the Pin Placement form to place and stretch the rails across the *prBoundary* and place the input and output pins.

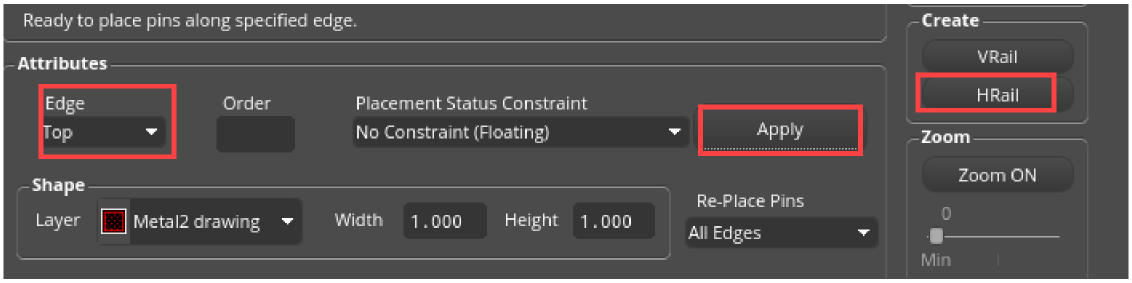
16. Click on the **VDD! VDD! VDD!** entry in the pin list.

Note: The reason for the pin name to appear three times in the pin list is due to its *Pin Terminal Name*, *Pin Net Name* and *Pin Fig Name*. The pin figure is the physical entity that you are placing.

17. Set the *Edge* to **Top** in the Attributes section.

18. Click the **Apply** button.

19. Click the **HRail** button to stretch the pin across the top of the *prBoundary*. Refer to the image below:



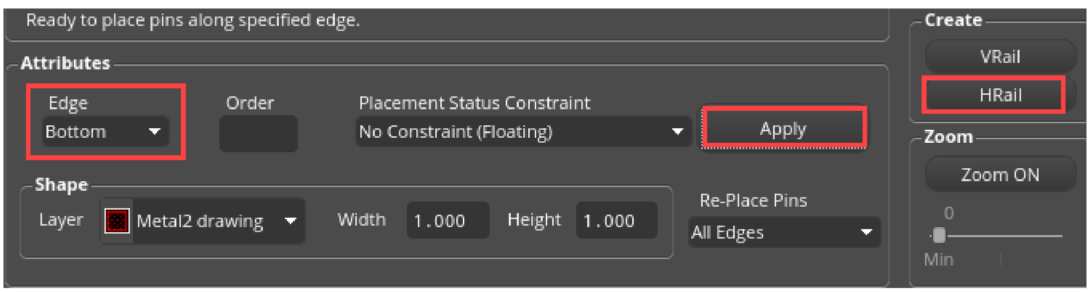
20. Click on the **VSS! VSS! VSS!** entry in the pin list.

21. Set the *Edge* to **Bottom** in the Attributes section.

22. Click the **Apply** button.

23. Click on the **HRail** button to stretch the pin across the bottom of the *prBoundary*.

    You have just generated two rail pins, VDD! and VSS!, across the top and bottom of your *prBoundary*.



24. Click on the **CK:CK:CK**, **D:D:D** and **RN:RN:RN** entries in the pin list.

25. Set the *Edge* to **Left** in the Attributes section, as these are input pins.

26. Click the **Apply** button.

    The input pins are now placed on the left side of the *prBoundary*.

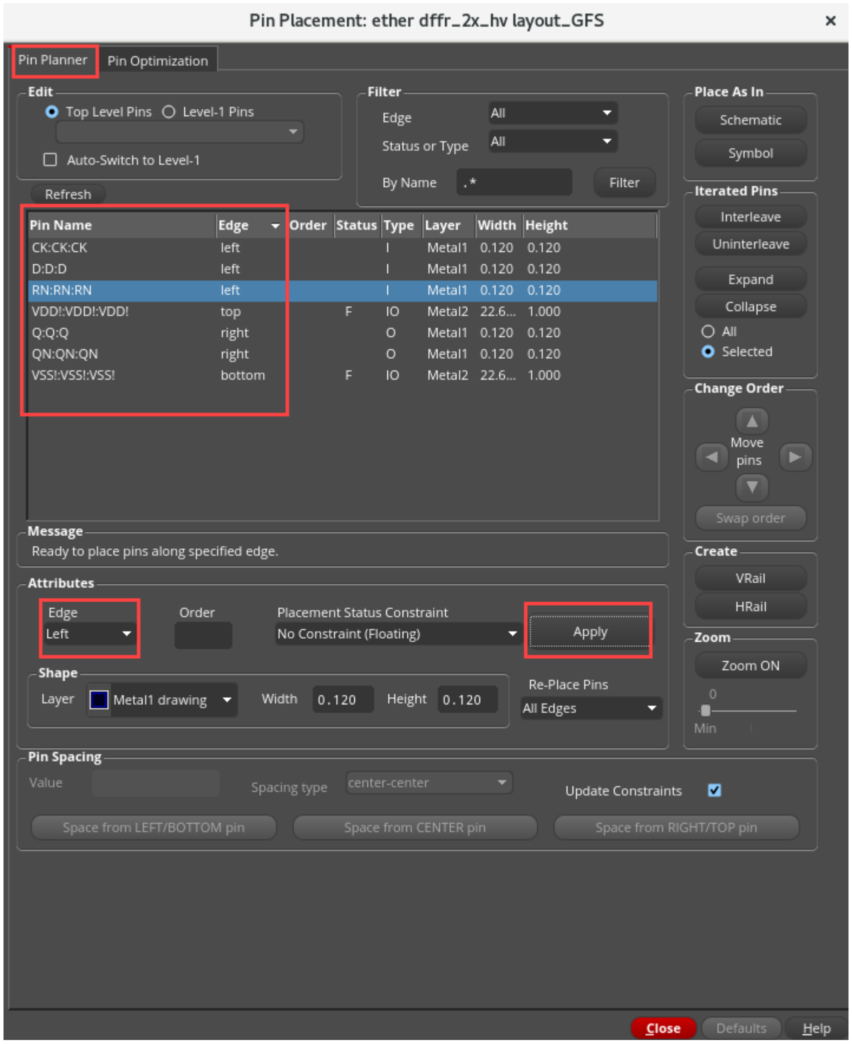
27. Click on the **Q:Q:Q** and **QN:QN:QN** entries in the pin list.

28. Set the *Edge* to **Right** in the Attributes section, as these are output pins.

29. Click the **Apply** button.

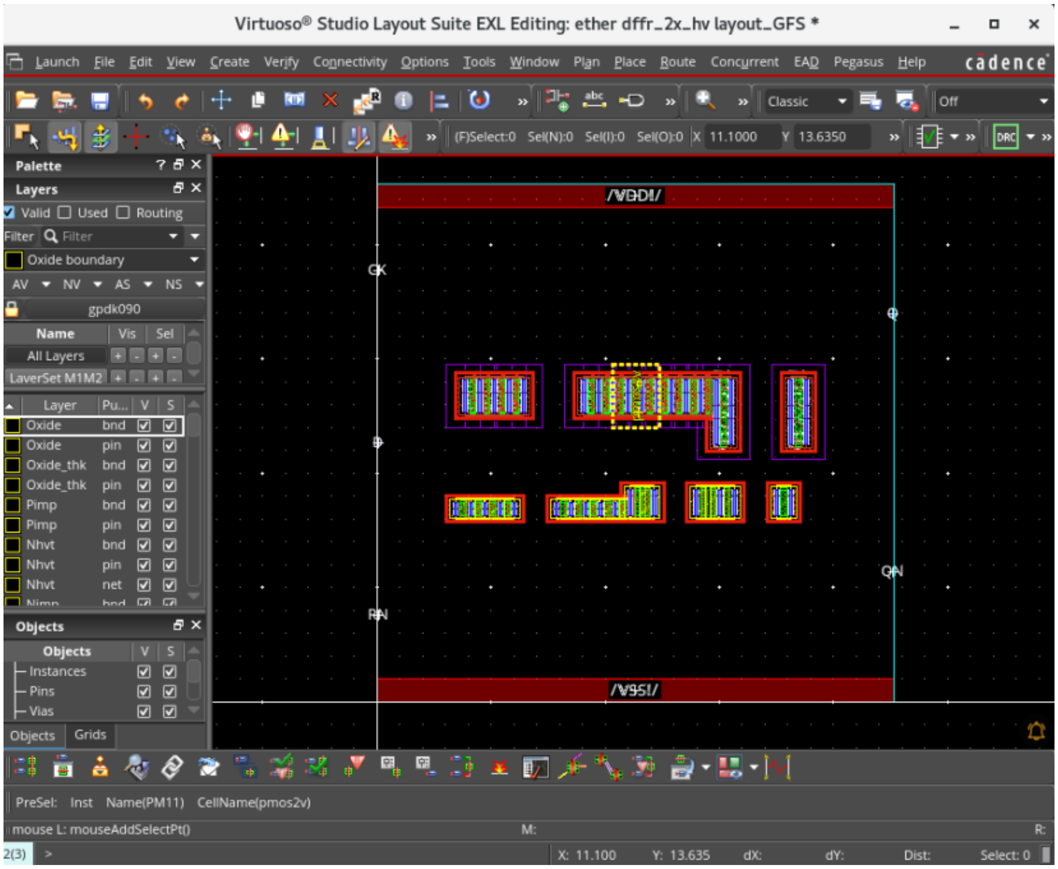
    The output pins are now placed on the right side of the *prBoundary*.

    The setting in the form looks like this:



30. Close the Pin Placement form.

    Your final placement looks like this:



    Notice how, in IC25.1, the devices (even without applying constraints) got placed so as to produce a placement with the shortest connections between the complementary devices!

    At this point, you could compact this layout by changing the folding on the two largest PMOS devices, and abutting them together. And, you could also stretch the *prBoundary* on all sides to substantially compact this block before routing it.

    However, for the sake of time, we'll stop here on the *Generate All From Source* and *Pin Placement* portions of the lab.

31. This is an optional exercise: Place the instances and pins with other Generate options, by selecting **Chain/Fold** and **Fold/Chain Folds** and observe the result.

32. Close the layout view. Do **NOT** save.

33. Close the schematic view.keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned:

- How to use the Generate All From Source (GFS) command to populate a design.

End of Lab

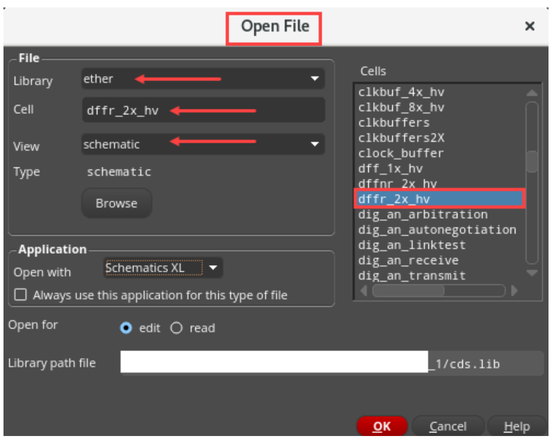
Lab 2-2 Populating a Design with *Generate Selected From Source*

Objective: To use the Generate Selected From Source (GSFS) command to populate a design.

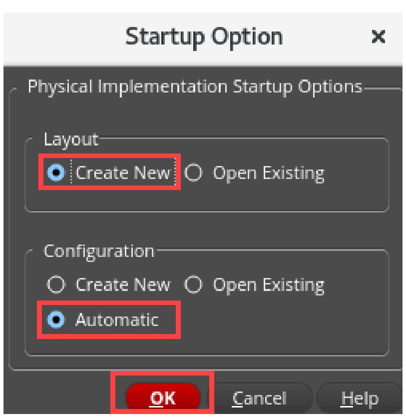
When you populate your design using the Generate Selected From Source (GSFS) command, you generate selected instances and pins together or individually depending on the options you choose in the Generate Selected Components form.

1. Start with the open schematic,open it in XL mode.

Library	`ether`
Cell	`dffr_2x_hv`
View	`schematic`

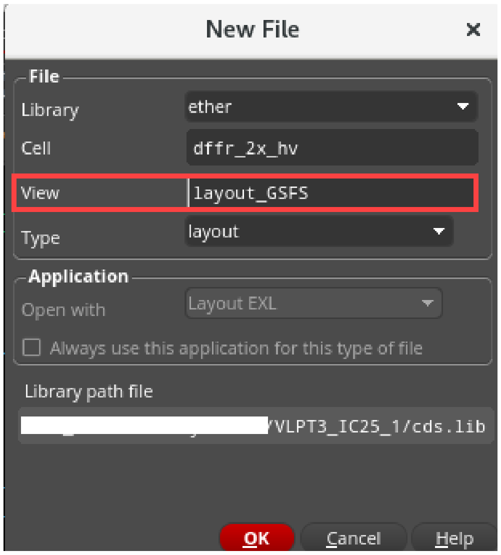


2. Choose **Launch – Layout EXL** and choose these options in the **Startup Option** form.



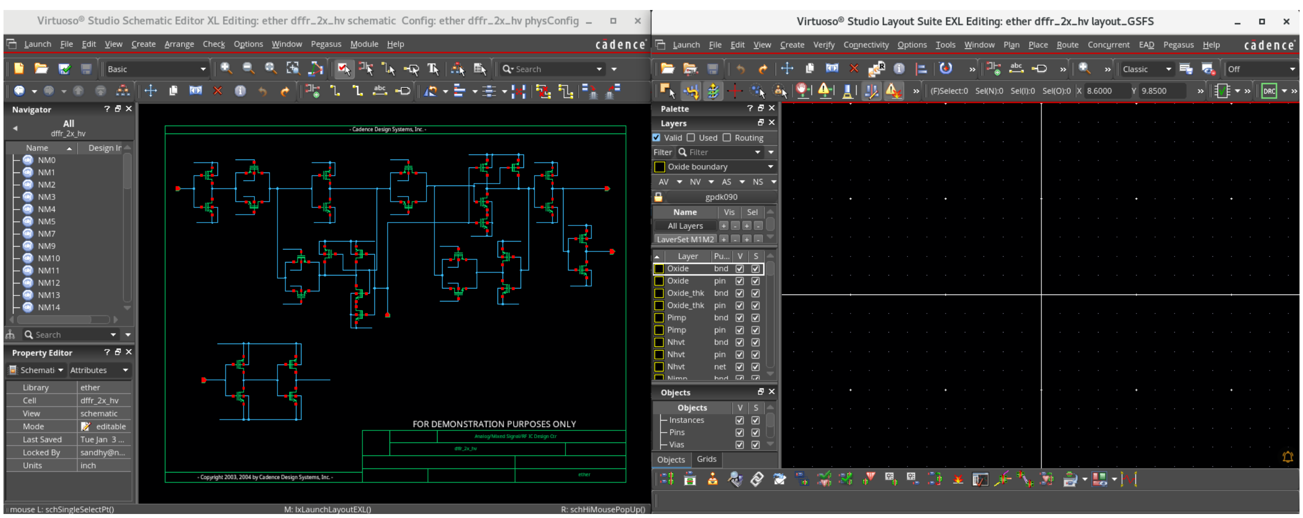
3. Click **OK**.

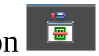
4. Name your new layout as ***layout_GSFS,*** as shown here:

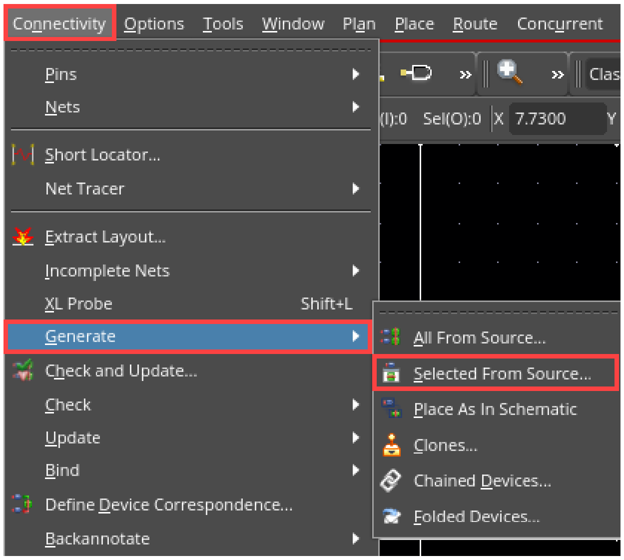


5. Click **OK**. You now, once again, have a new layout palette to work with.

6. Position your schematic and layout, as shown here, so that you can see both the windows.

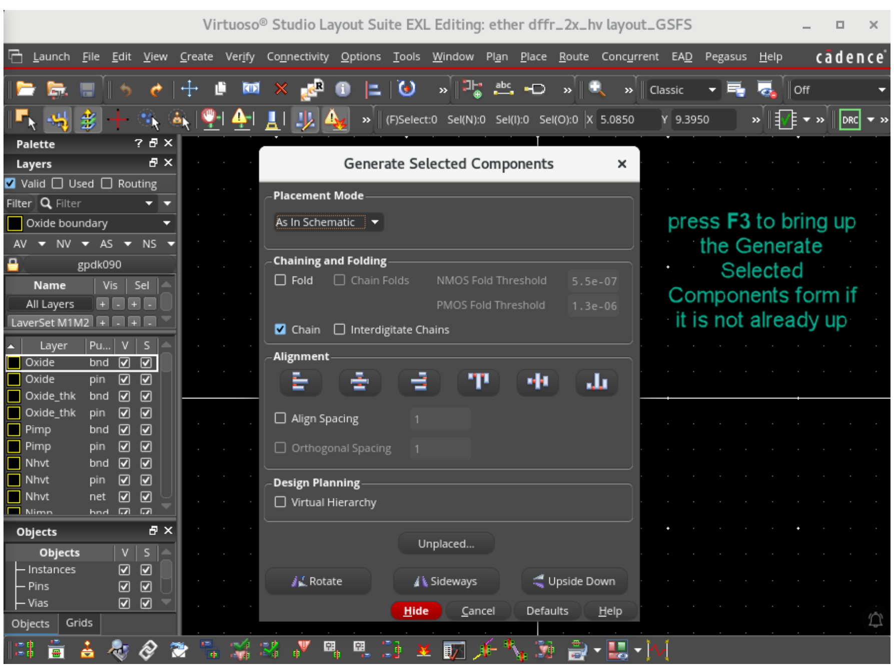


7. Click the **Generate Selected From Source (GSFS)** icon  or choose the **Connectivity – Generate – Selected From Source** menu command

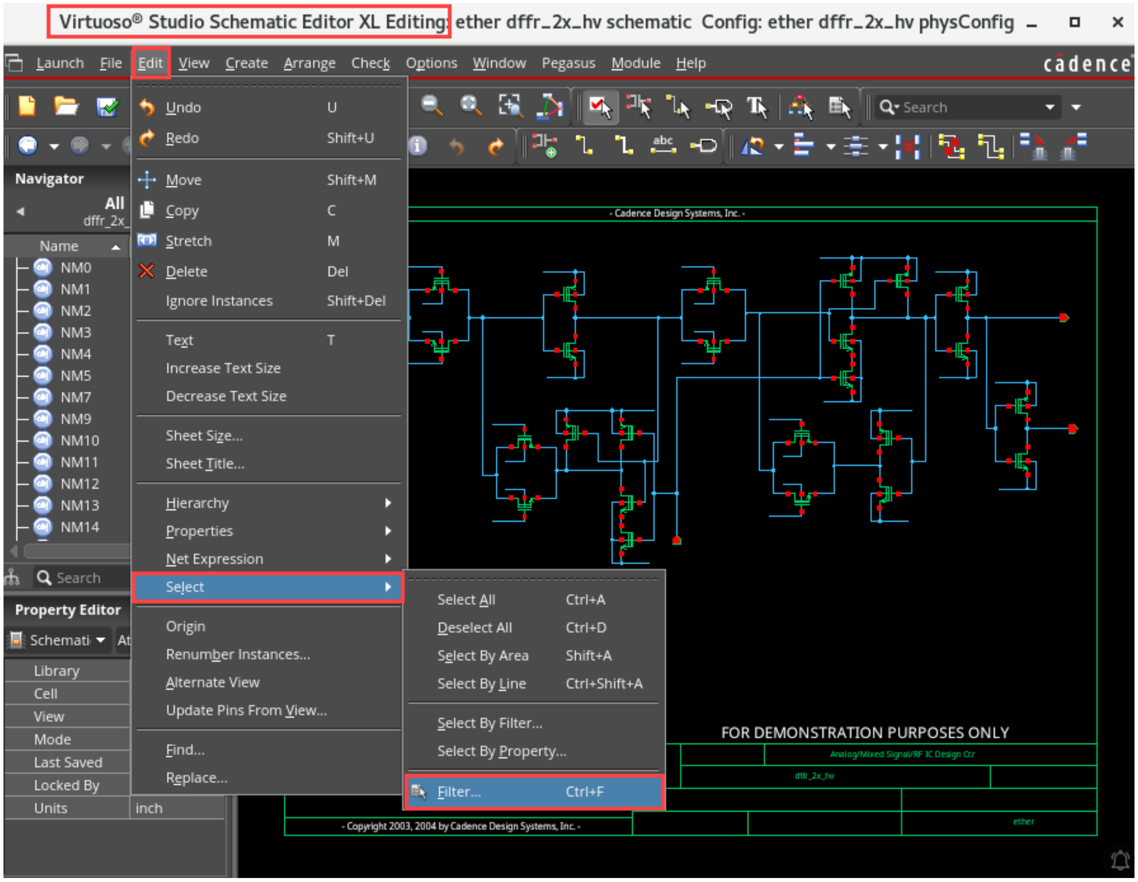


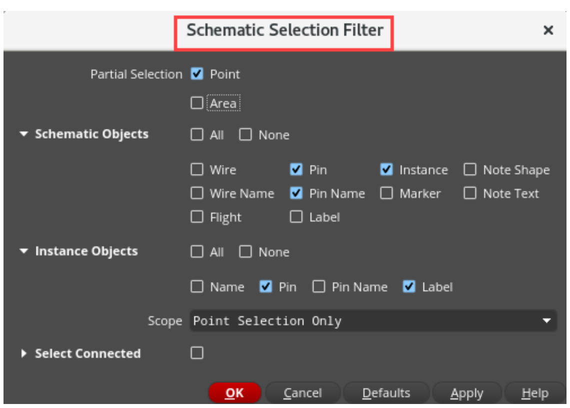
and press **F3** to bring up the Generate Selected Components form if it is not already up.

8. Select the options as shown here in the Generate Selected Components form.



9. In the schematic window, choose **Edit – Select – Filter** or press the bindkey **Ctrl+F** to set the select filter, as shown here:

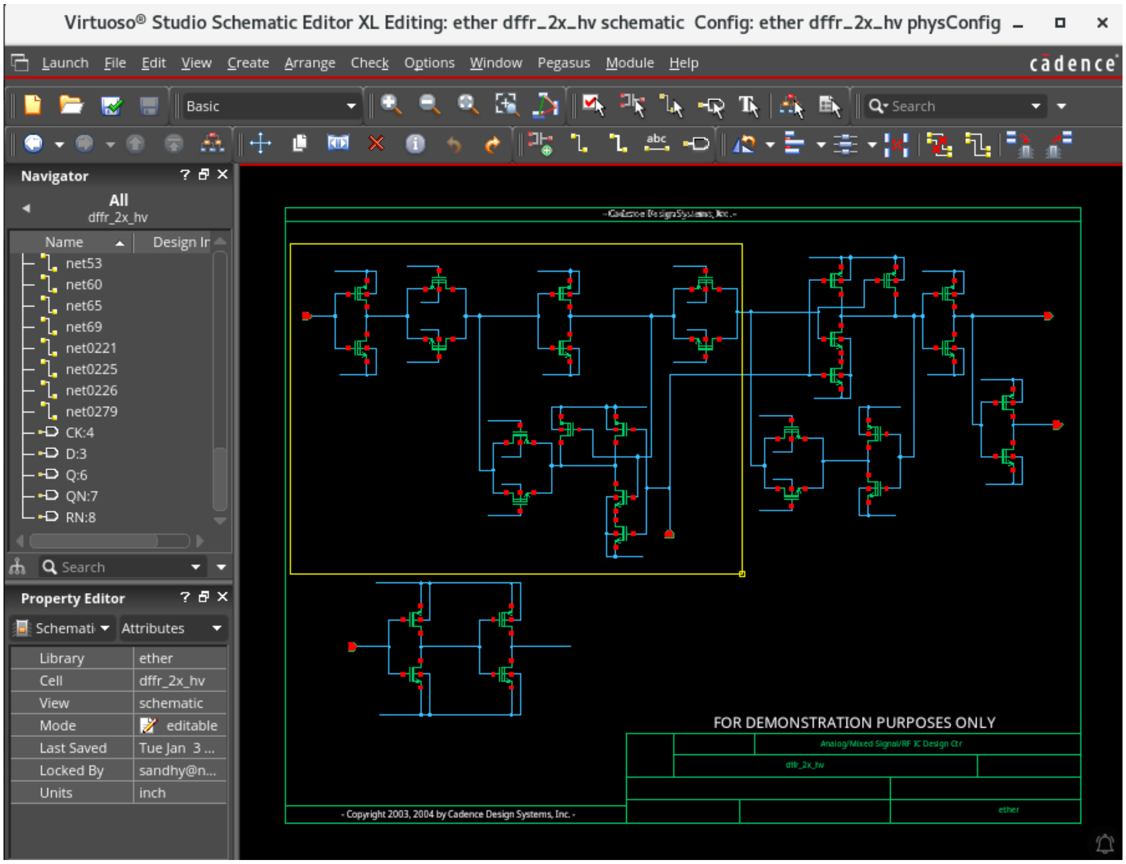




This makes it easier to cleanly select instances and pins in the schematic.

10. Click **OK**.

11. Area-select the components in your schematic, as shown here:



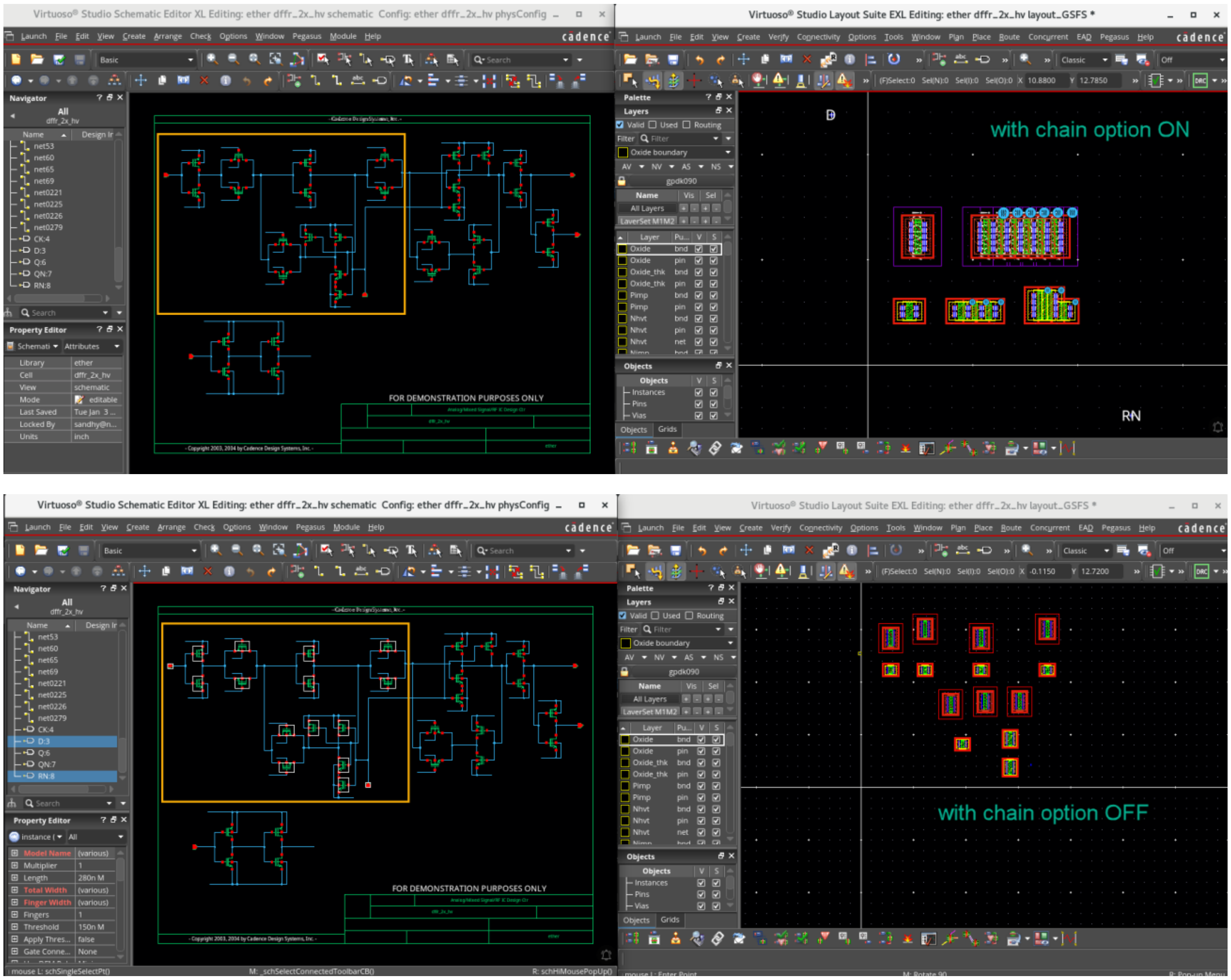
12. Move your cursor into the layout window, but **do NOT click to place**.

    The intent here is to show the difference in placement options when you click in the **Generate Selected Components** form **with** and **without** the *Chain* option.

13. Click on the ***Chain*** option, but do **NOT** click to place it in the layout.

    Do you see the difference?

    Answer: Without using the *Chain* option, components are simply attached to your cursor from your selections in the schematic until you click to place them in your layout. With the *Chain* option on, components are chained together in two compact rows, which helps you speed up the placement process.



14. This is an optional exercise: Select the instances and pins in the schematic and place them in layout canvas with other Chaining and Folding options, by selecting **Chain/Fold** and **Fold/Chain Folds** and observe the result.

15. Close the schematic and layout views. Do **NOT** save.

16. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned:

- How to use the Generate Selected From Source (GSFS) command to populate a design.

End of Lab

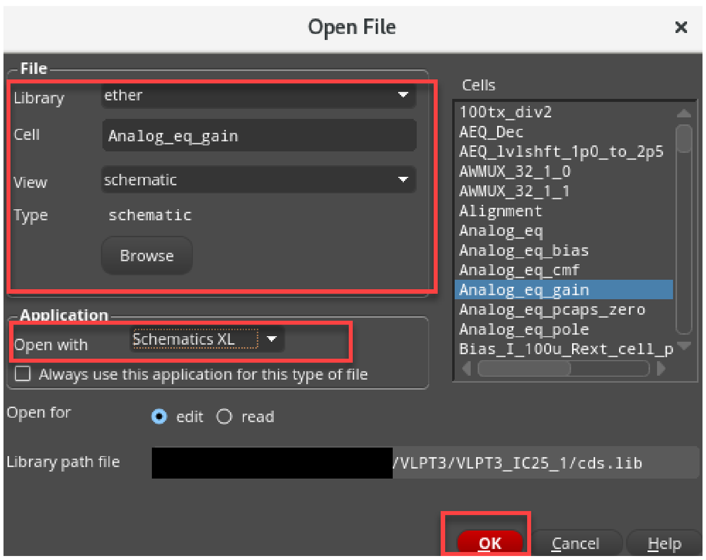
Lab 2-3 Placing Components with the Auto Place

Objective: To use the Virtuoso Automatic Placement to place components.

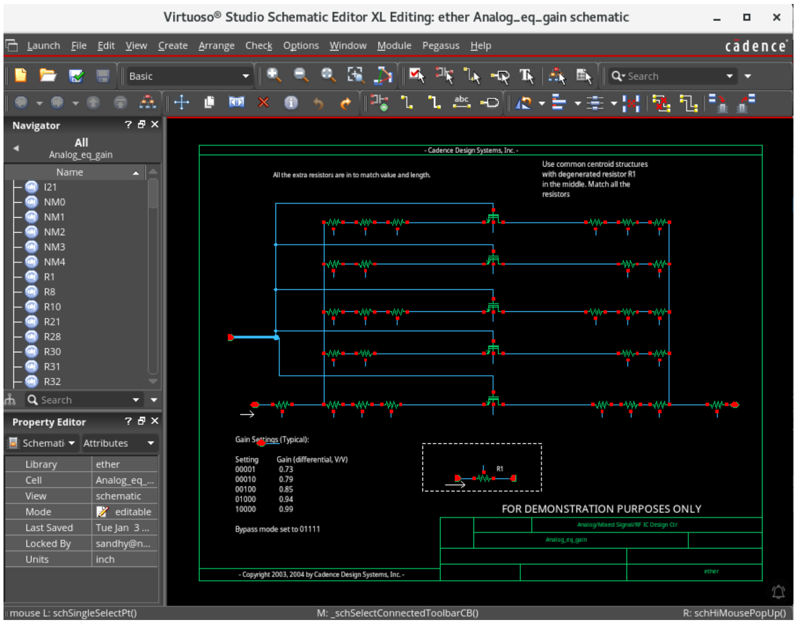
To place the components from your schematic quickly, you can use the Virtuoso Automatic Placement function. If some devices are rotated in the schematic, you can add a placement/orientation constraint to ensure all devices are oriented in a desired rotation.

1. Open the following schematic in XL:

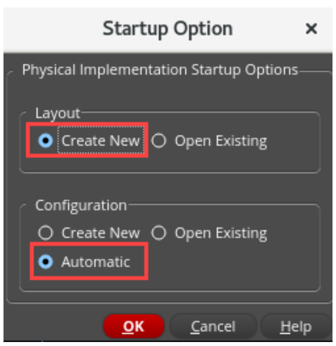
Library	Cell	View
`ether`	`Analog_eq_gain`	`schematic`



The *ether/Analog_eq_gain/schematic* window appears in XL mode.

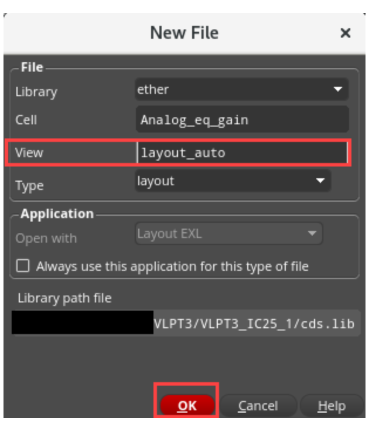


2. Choose **Launch – Layout EXL** and choose these options in the **Startup Option** form.



3. Click **OK**.

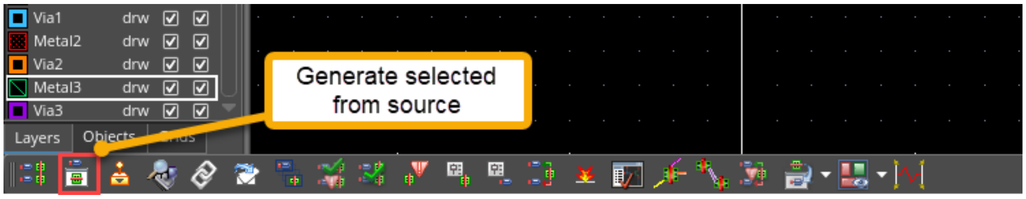
4. Name the new layout as ***layout_auto***.



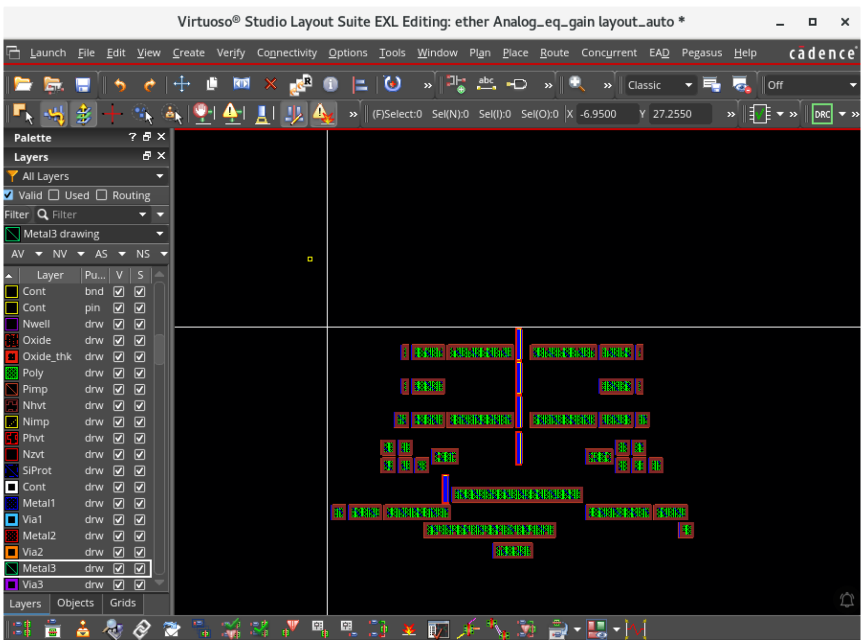
5. Click **OK**. The new layout palette appears.

6. In the schematic view, select all components by pressing **Control-A**.

7. In the layout view, click the **Generate Selected From Source** icon.

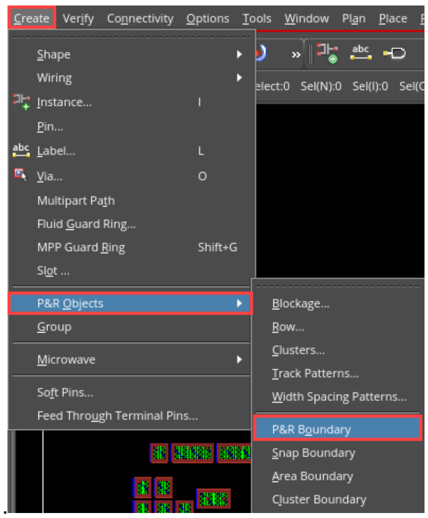


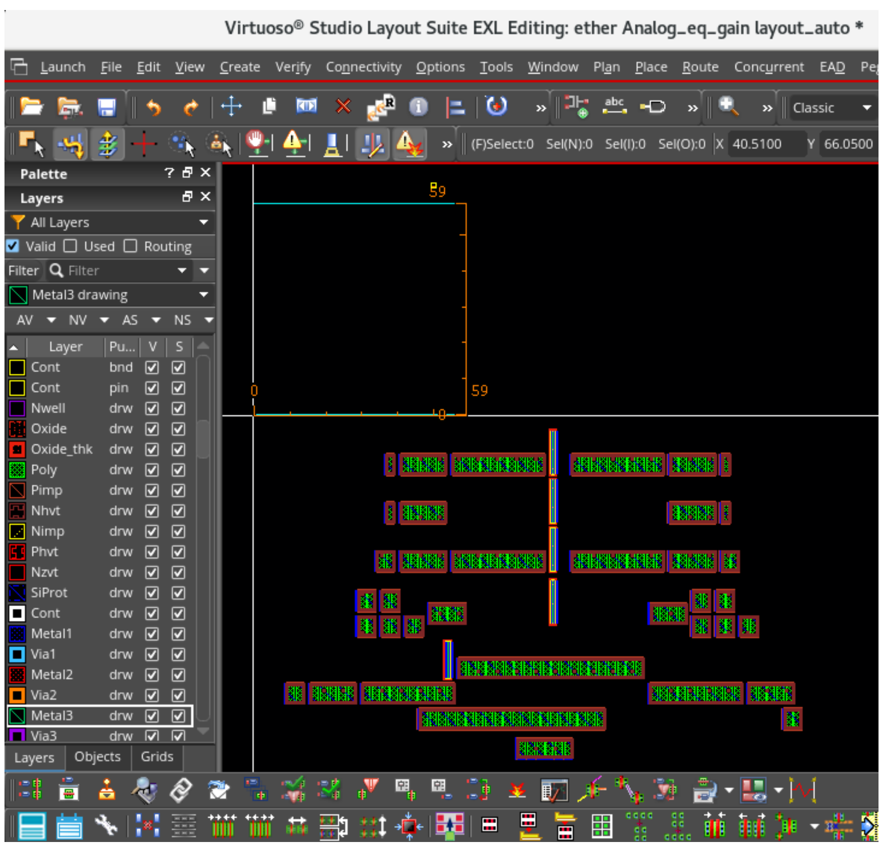
8. Keep the default values in the Generate Selected Components form and click in the layout to place the components, something like this one:



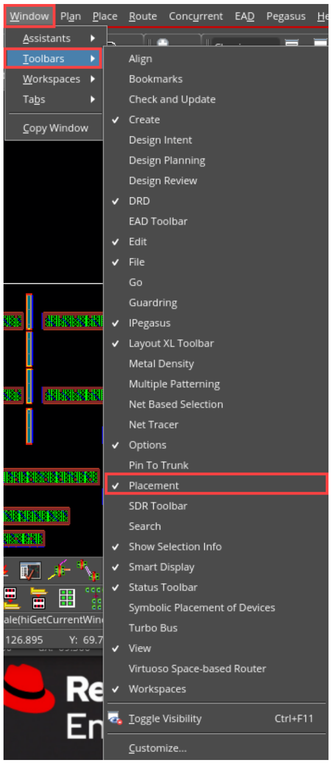
9. Create PR Boundary from the origin (**60 by 60** measurement) use **‘k’** bindkey to measure the Boundary accurately. Please follow the below steps to create PR Boundary.

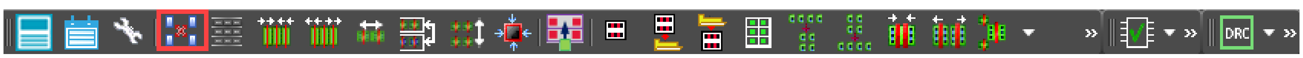
   Note: For Automatic Placement, a PR boundary is a must. So, when we generate selected from source, it won't generate a PR boundary. So, create the PR boundary manually.

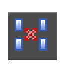


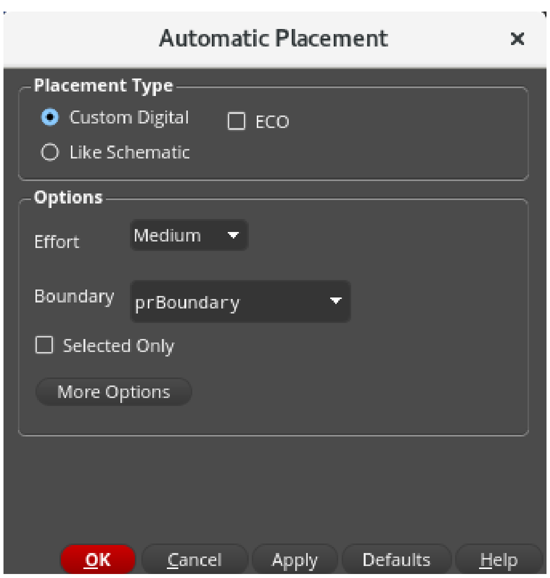


10. Activate placement Toolbar if it is not activated earlier, **Windows – Toolbars – Placement**.

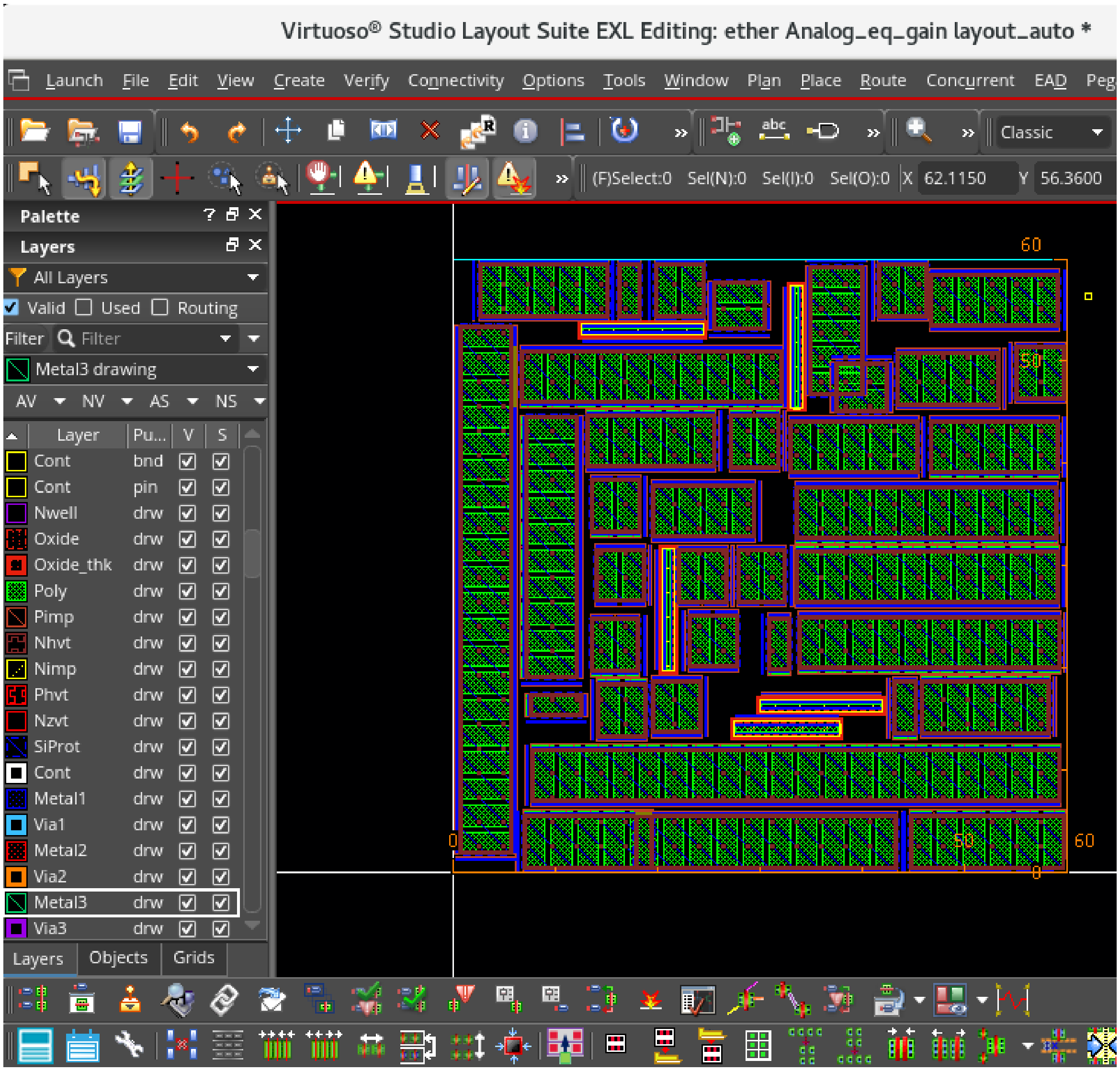




11. Choose the **Place – Auto Place**, or choose  from placement Toolbar. Automatic Placement form appears. Click '**OK**'.



12. Verify that your placement looks something like this one:



13. Close all the views. Do **NOT** save.

14. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned:

- How to use the Virtuoso Auto Place to place components.

End of Lab

Lab 2-4 Placing Components with the Custom Digital Placer

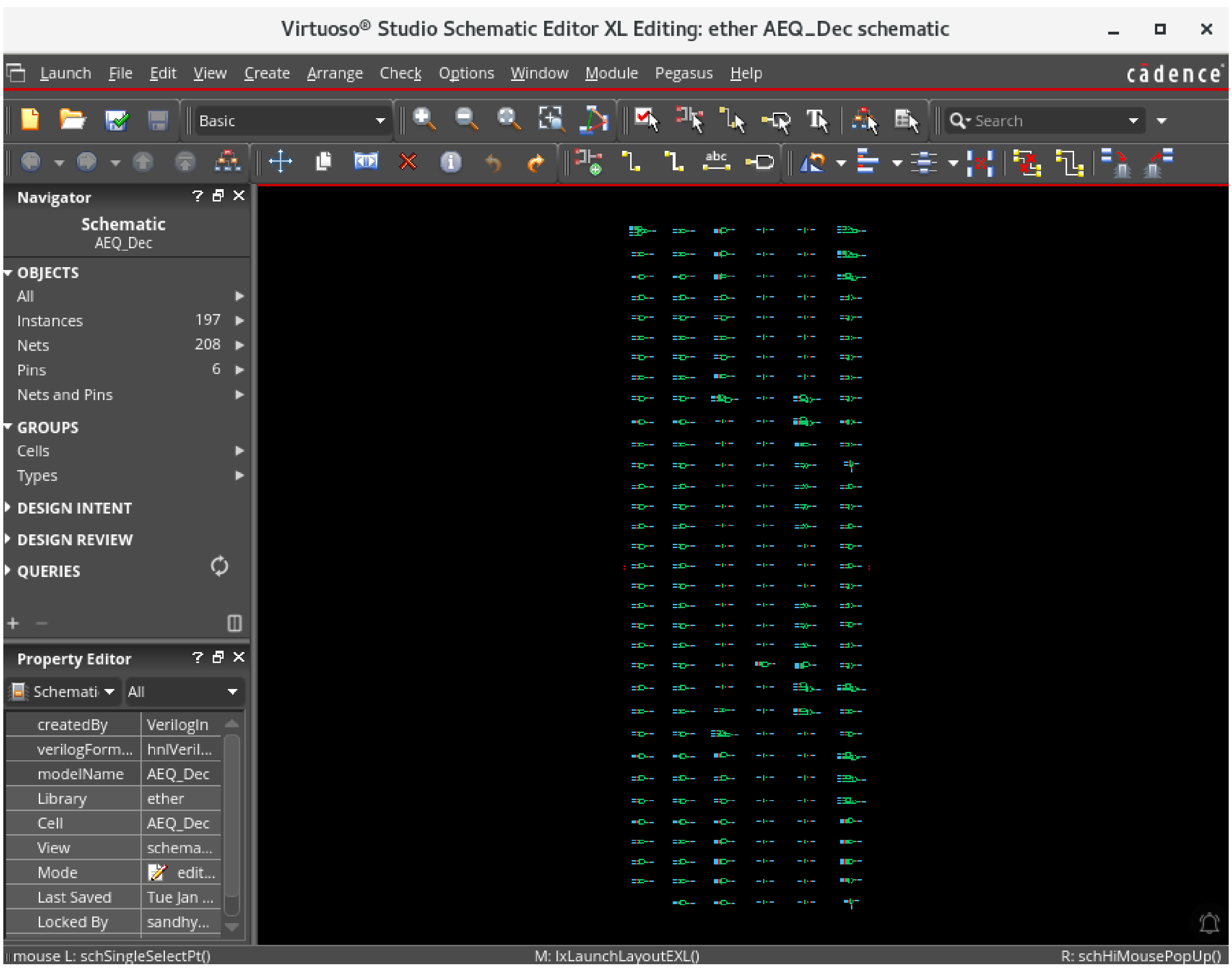
Objective: To use the Virtuoso Custom Digital Placer (VCP) to place components.

In this lab, you will use the Virtuoso Custom Digital Placer (VCP) to automatically place standard cells of the block within rows.

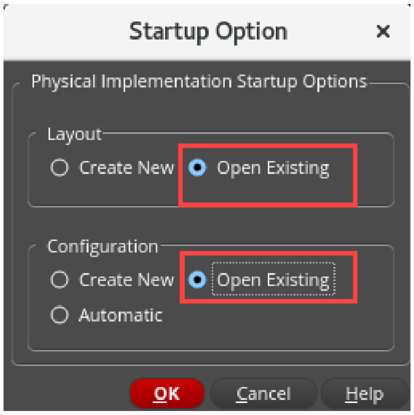
1. Open the following schematic in XL:

Library	Cell	View
`ether`	`AEQ_Dec`	`schematic`

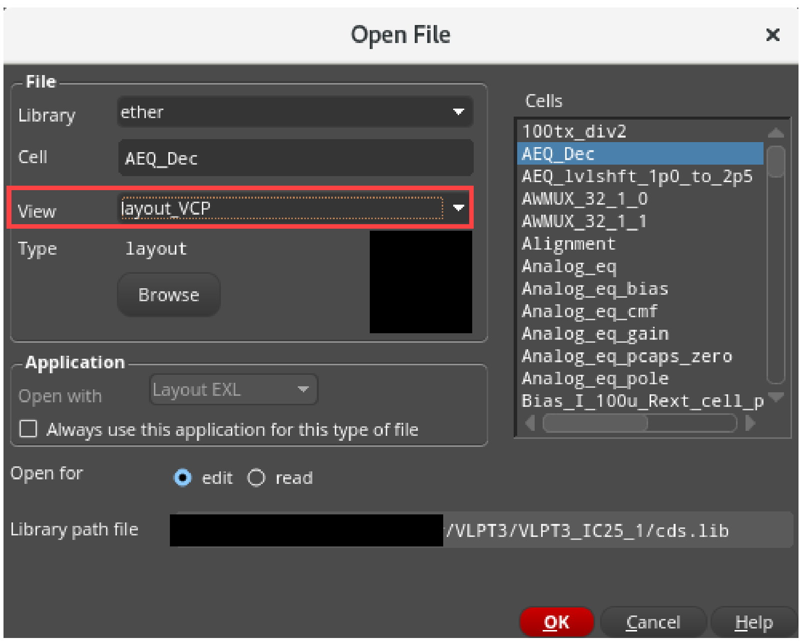
The *ether/AEQ_Dec/schematic* window appears in XL mode.



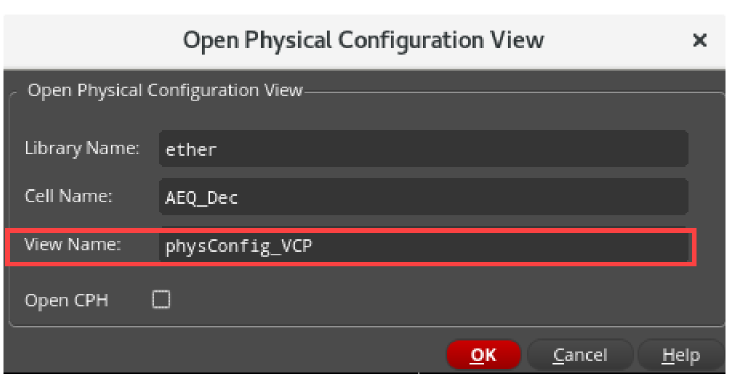
2. In the schematic, choose **Launch – Layout EXL** and select these options in the **Startup Option** form and click **OK.**



3. Fill out the Open File form, as shown here, and click **OK.**

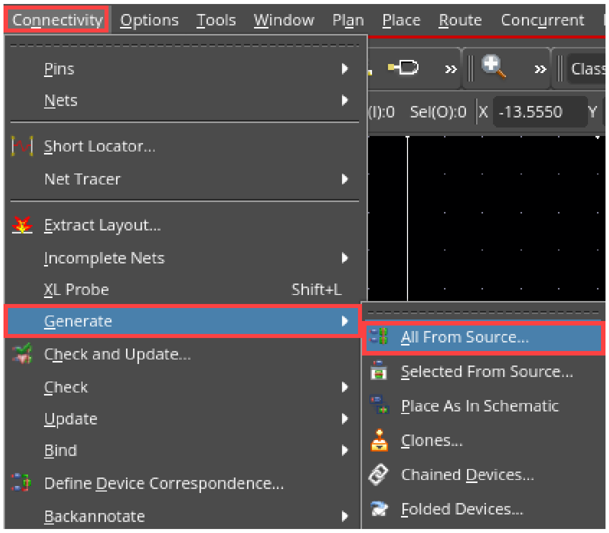


4. The Open Physical Configuration View form pops up. Fill out the form, as shown here, and click **OK.**

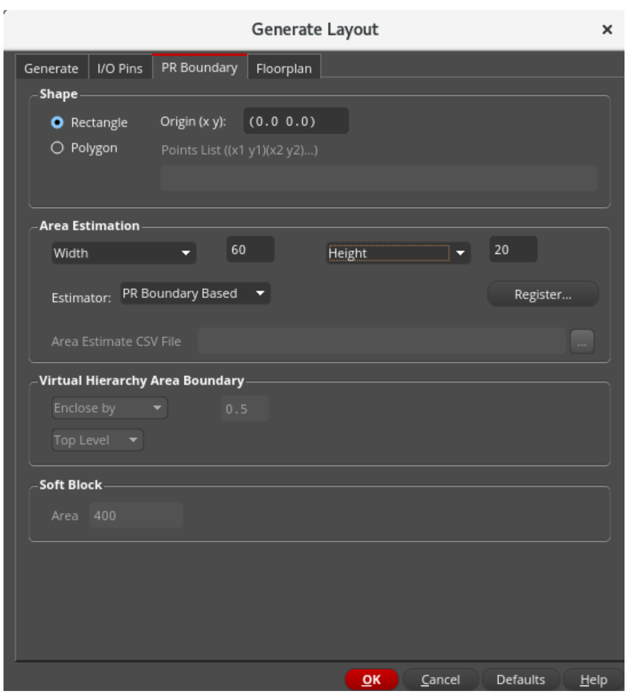


The Virtuoso Layout Suite EXL palette opens with the *ether/AEQ_Dec/layout_VCP* view.

5. Now, you will generate the layout components by choosing either the **Connectivity – Generate – All From Source** menu entry or clicking the **Generate All From Source (GFS)** icon.



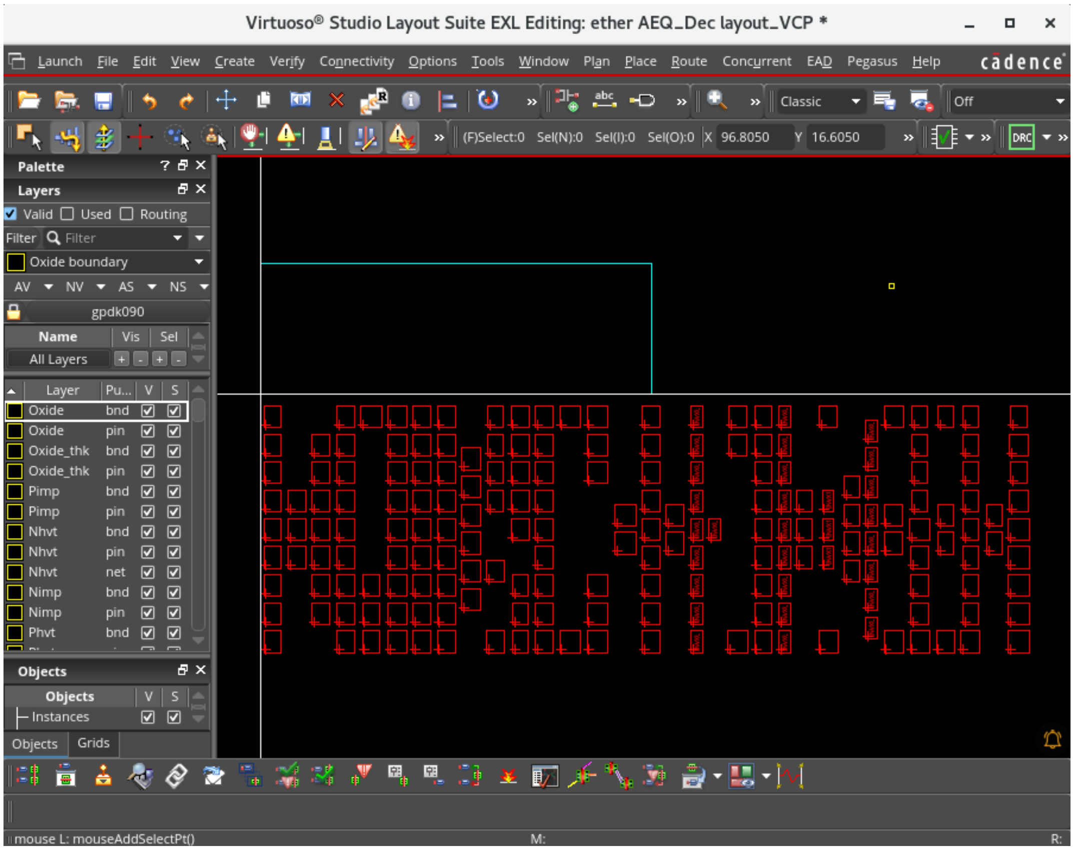
The Generate Layout form appears.



a. In the PR Boundary tab, in the Area Estimation field, select *Width* as **60** and *Height* as **20**.

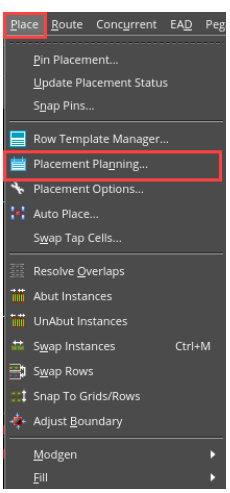
b. Keep the other tab values as default settings and click **OK**.

All the components from the source schematic are generated in the layout window, as shown here:



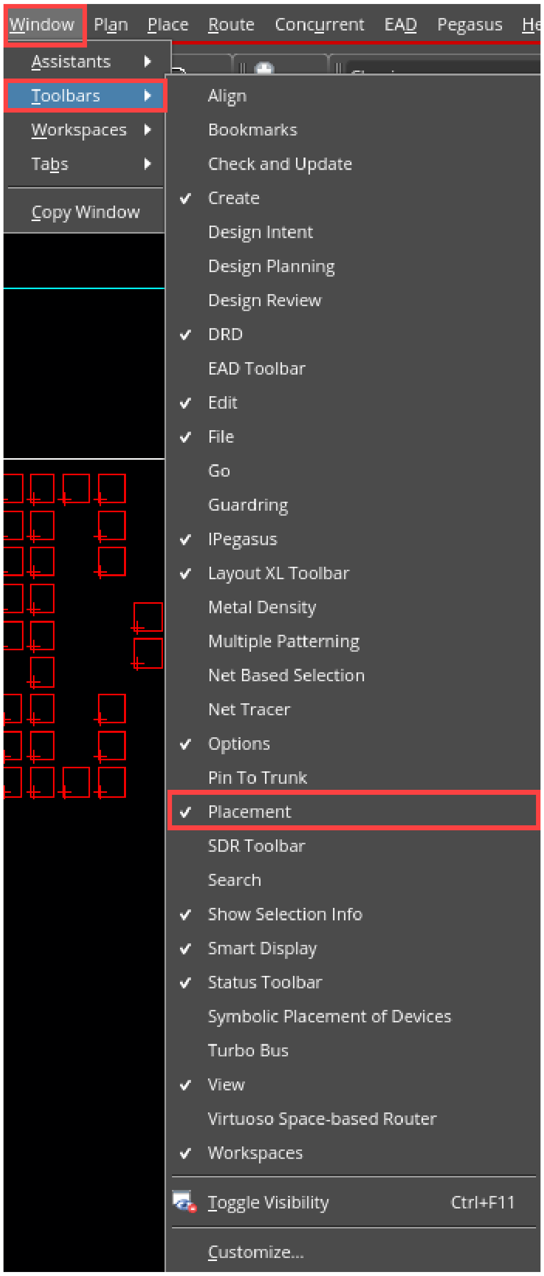
Now, you will create Rows using the Placement Planning GUI.

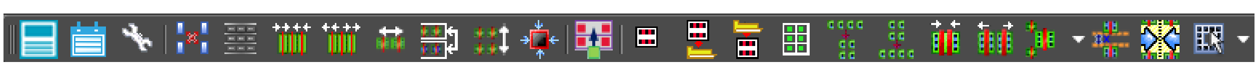
6. Run Placement Planning by choosing either the **Place – Placement Planning** menu command or clicking the **Placement Planning** icon (To enable placement Toolbar check the Note section below)



The Placement Planning form opens. Select the following options in the Placement Planning form.

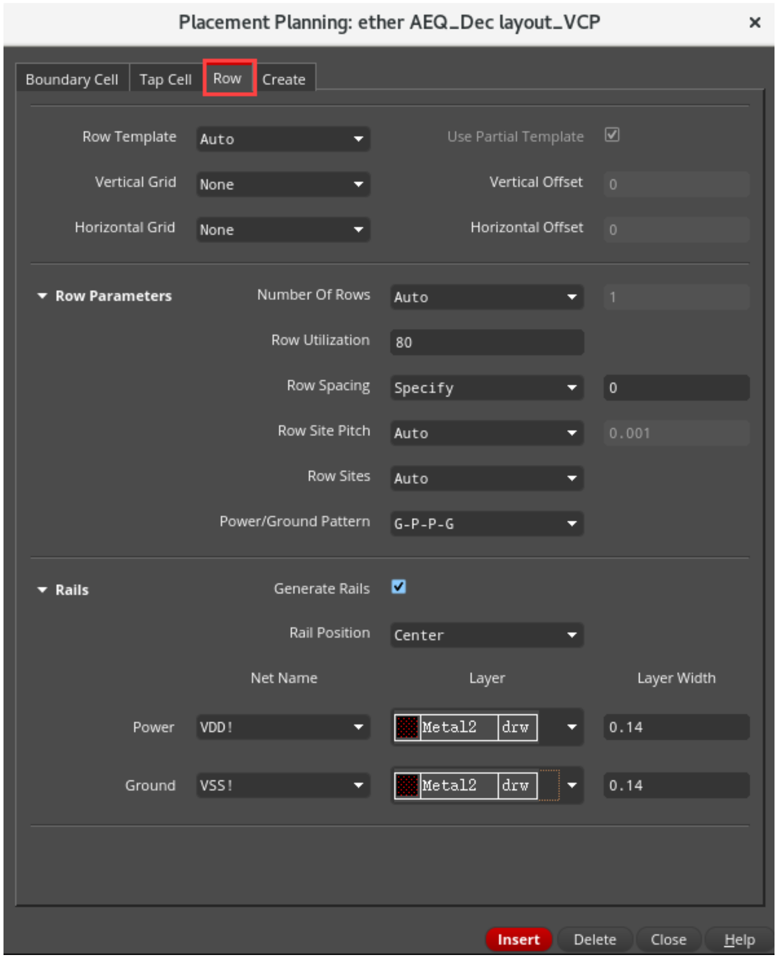
Note: To enable Placement Toolbar, go to **Windows** – **Toolbar** – **Placement**. This will enable the Placement Toolbar.





7. In the Placement Planning form, leave the **Tap Cell** tab and **Boundary Cell** with default values.

8. In the Placement Planning form, in the **Row** tab, select the following options:



a. **Number Of Rows**: *Auto*

b. **Row Utilization**: *80*

c. **Row Spacing**: **Specify** as *0*

d. **Row Site Pitch**: *Auto*

e. **Row Sites**: *Auto*

f. **Power/Ground Pattern**: *G-P-P-G*

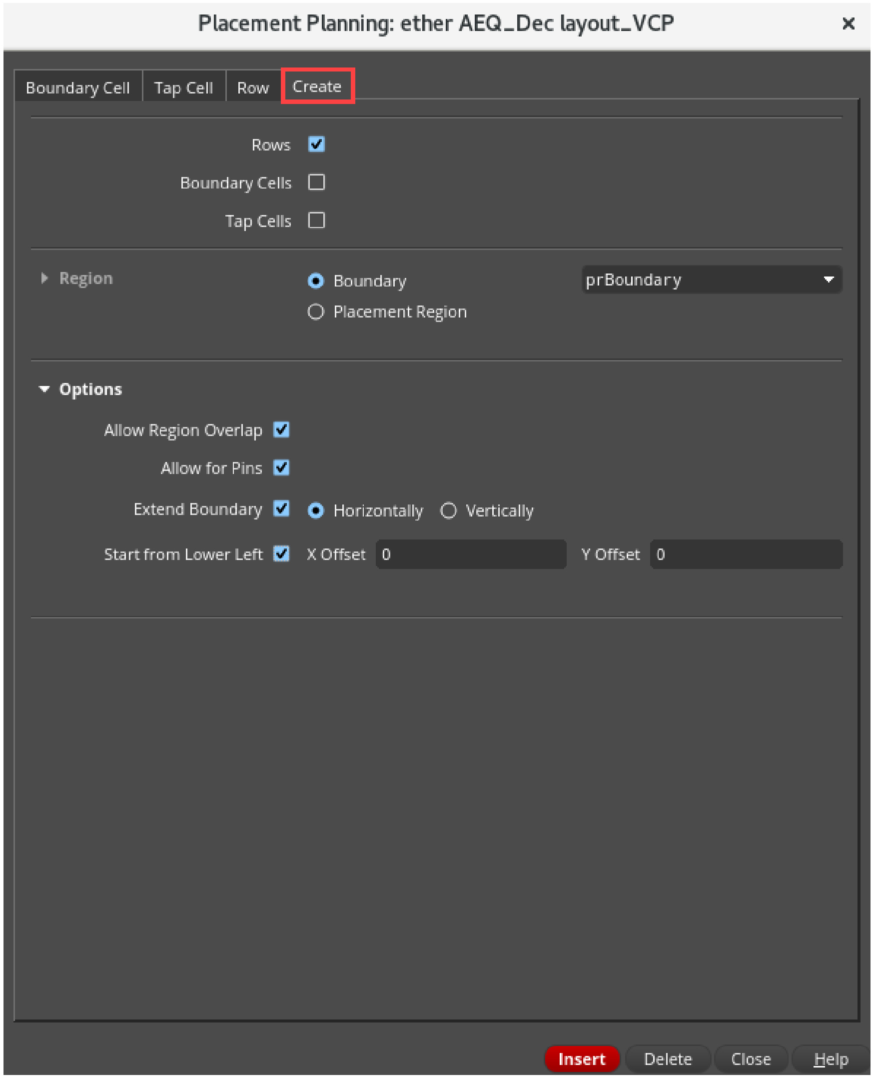
g. **Generate Rails:** *ON*

h. **Rail Position**: *Center*

i. **Power Net Name**: *VDD!*; **Layer**: *Metal2 drw*; **Layer Width**: *0.14*

j. **Ground Net Name**: *VSS!*; **Layer**: *Metal2 drw*; **Layer Width**: *0.14*

9. In the Placement Planning form, in the **Create** tab, select the following options:



a. **Rows**: *On*

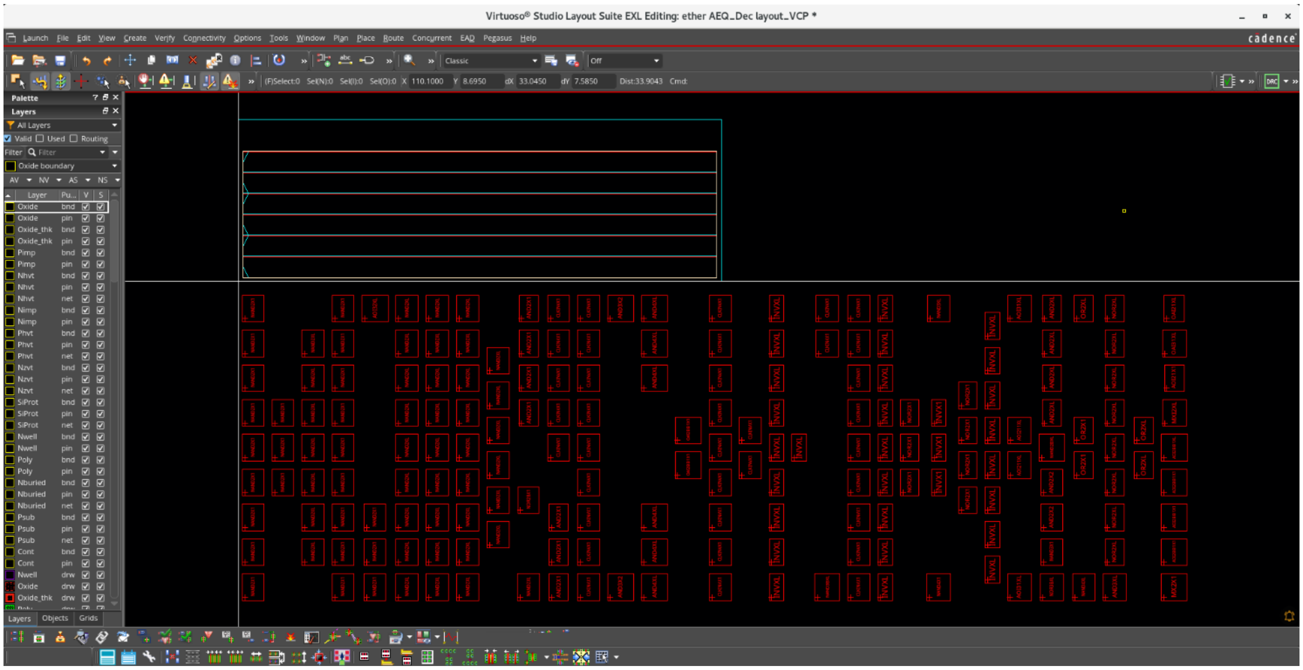
b. **Boundary**: *prBoundary*

c. **Allow for Pins**: *On*

d. **Extend Boundary**: *Horizontally*

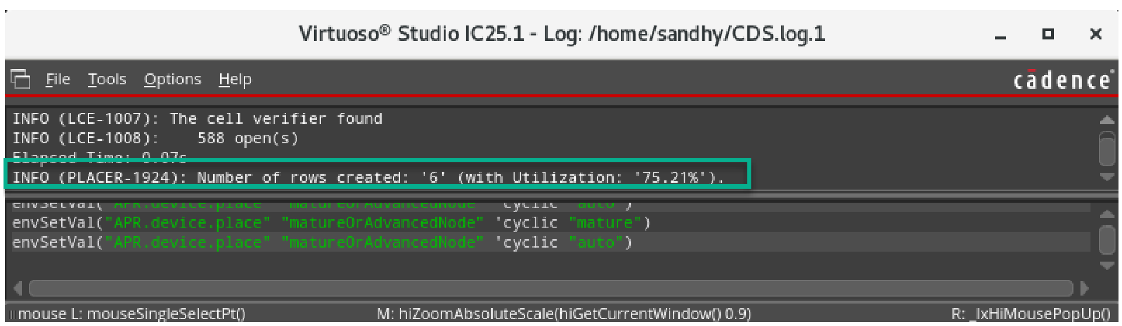
e. **Start from Lower Left**: *X Offset:0*; *Y Offset:0*

10. In the Placement Planning form, click the **Insert** button to generate the specified rows in the layout canvas, as shown here:

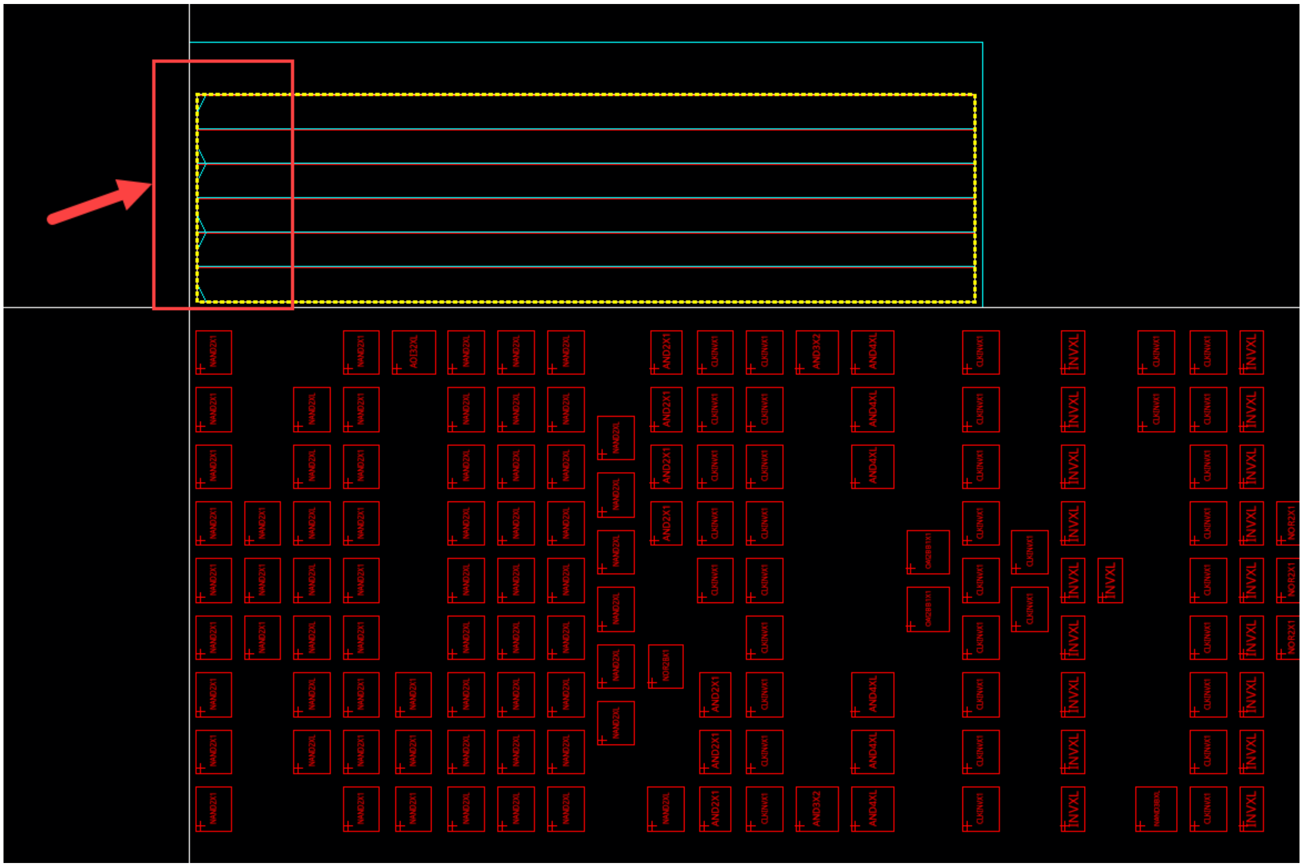


11. Click **Close** to close the Placement Planning form.

    Now, the estimated number of rows is created on the layout canvas. You can check it in the CIW.

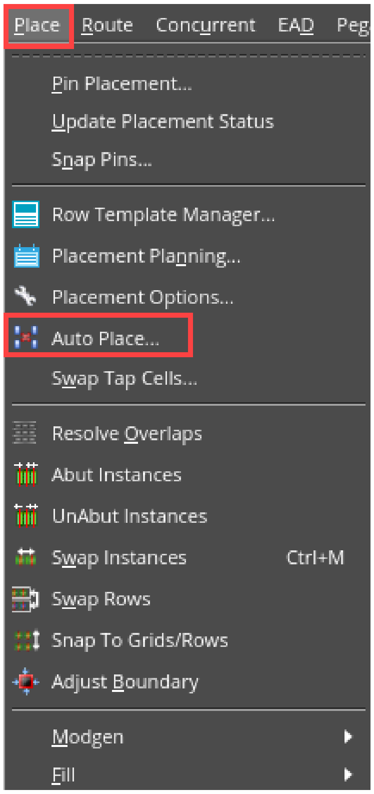


Also notice that some free space is available between the row edges and prBoundary for pin placement because you enabled the **Allow for Pins** option in the **Create** tab.

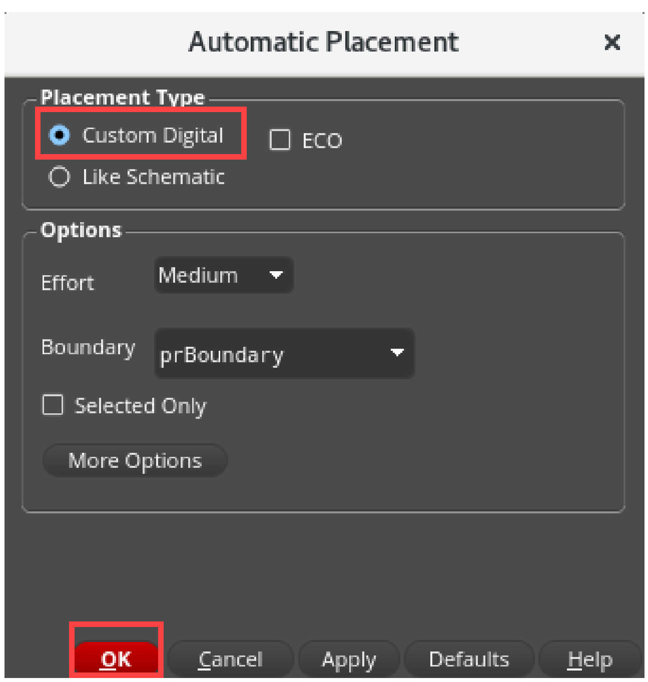


By changing the Tap Cell, Row or Create tab parameters as required, you can achieve the desired **Placement Planning** and proceed to place the components.

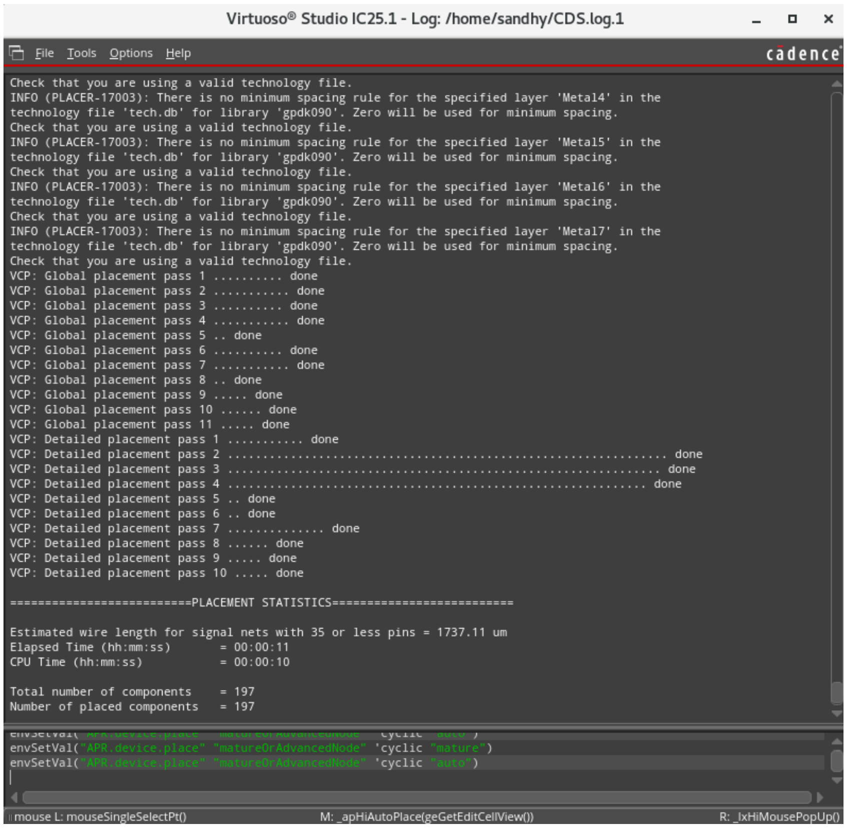
12. Now, you will run the Virtuoso Custom Digital Placer (VCP) to automatically place standard cells of the block within rows. Bring up the VCP by choosing either **Place – Auto Place** or clicking the **Auto Place** icon.



13. Automatic placement form appears: Choose the **Custom Digital** option and click **OK**.

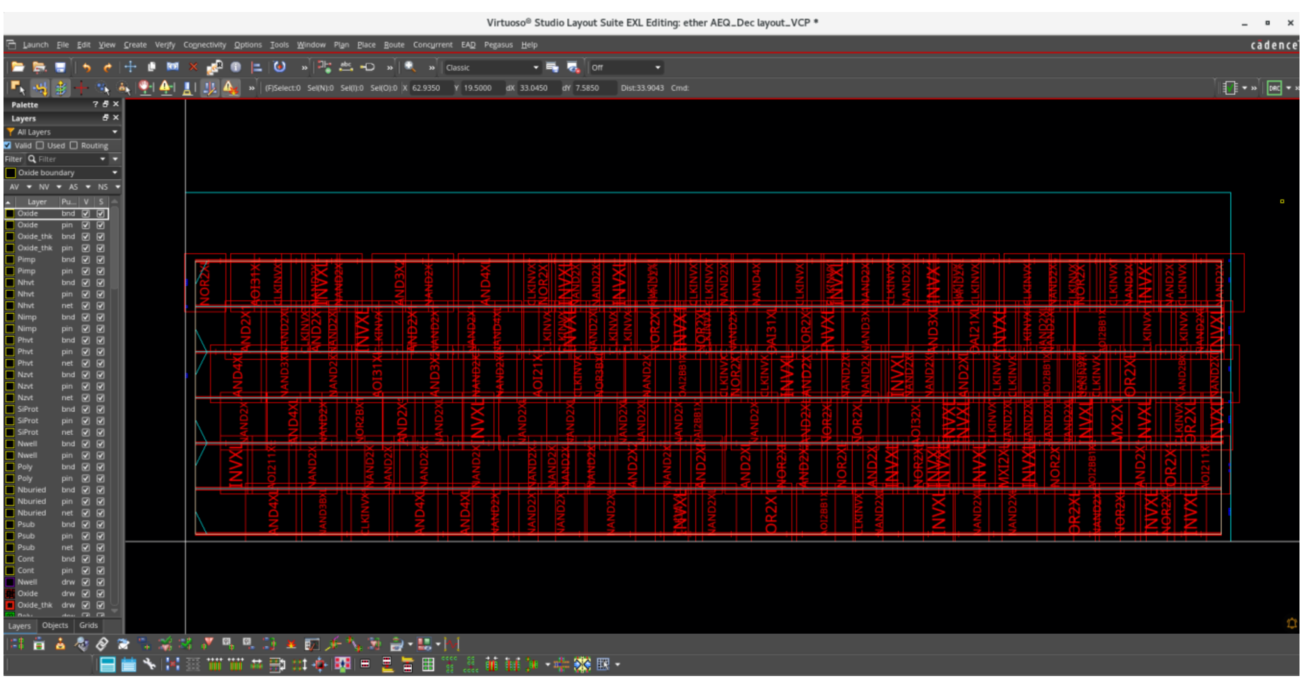


14. VCP runs and the VLS-EXL Placement Status opens in the CIW window, as shown here:

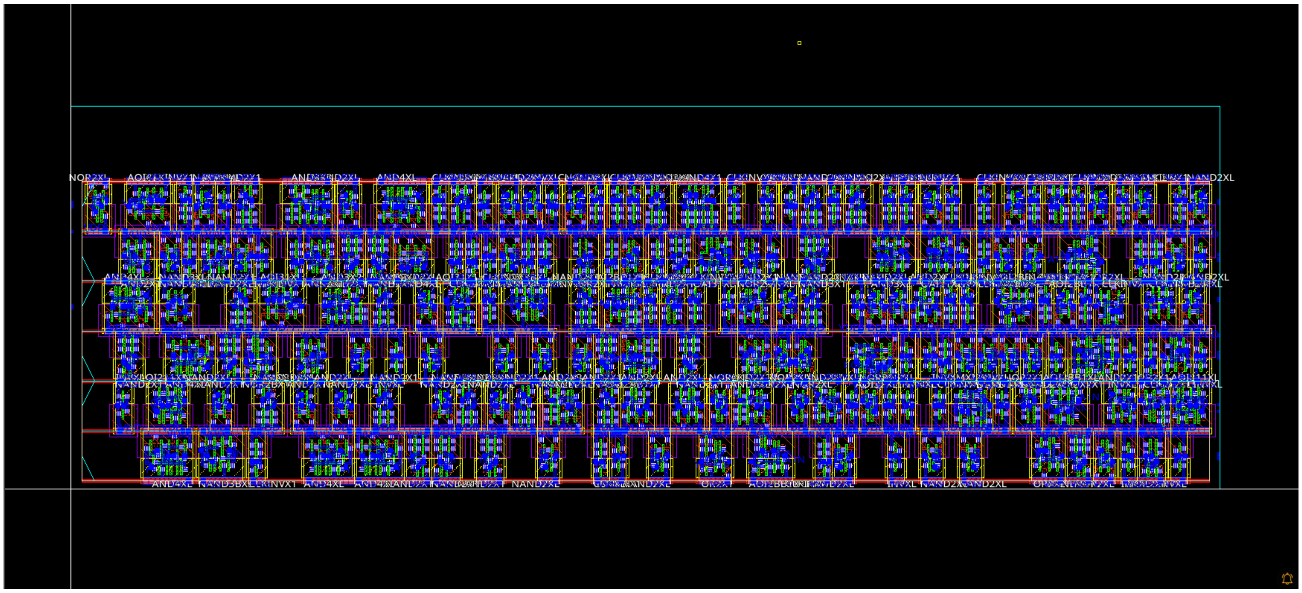


The Virtuoso Custom Digital Placer has placed the components within rows.

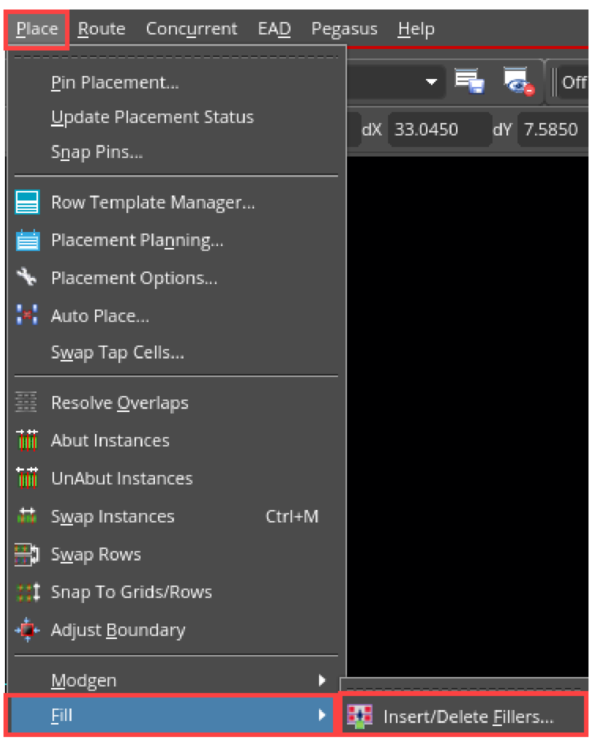
Notice that the Auto Placer has also placed the pins because you selected the **Allow for Pins** option in the Placement Planning form.

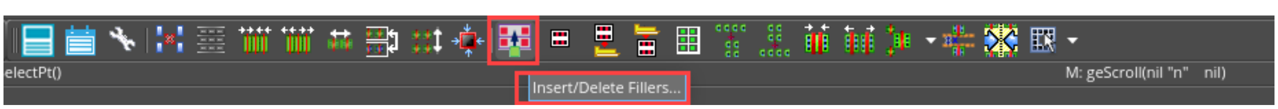


15. Press the bind key **Shift+F** to view all the levels of the hierarchy in the layout canvas.



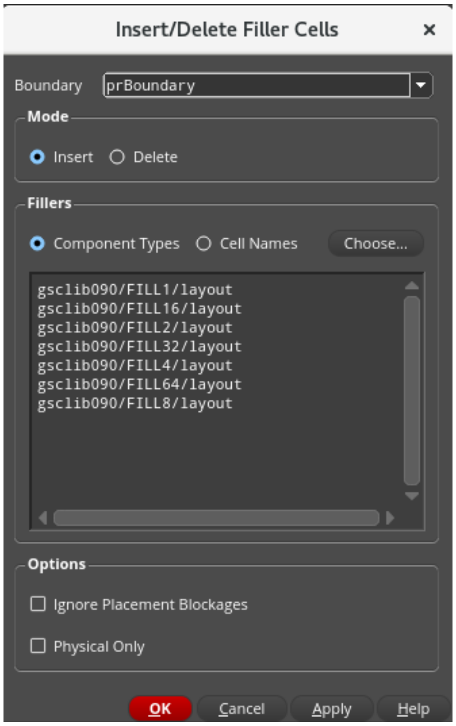
16. Now, you will run the **Insert/Delete Fillers** command to fill the empty spaces between the placed cells with filler cells. Choose the **Place – Fill –Insert/Delete Fillers** menu command or click the **Insert/Delete Fillers** icon.





The Insert/Delete Filler Cells form is displayed.

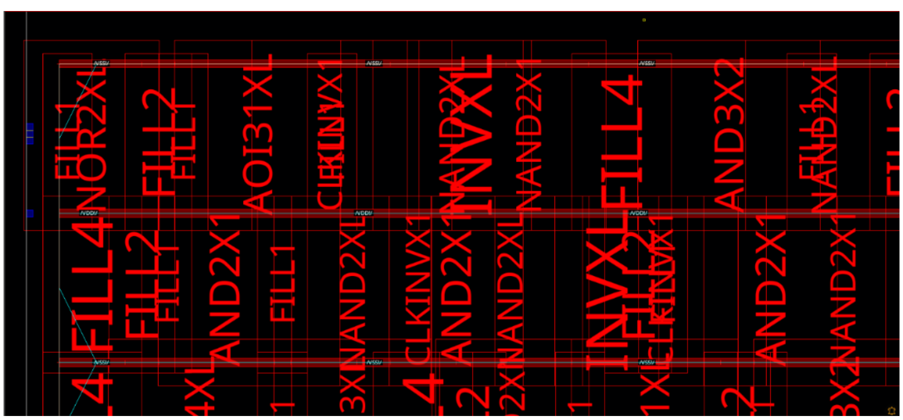
17. Allow the default settings in the Insert/Delete Filler Cells form.



18. Click **OK** to run Insert/Delete Filler Cells.

    Notice that all the empty spaces between the placed cells are now filled by filler cells.

    Here is a zoomed in view of the layout displaying filler cells occupying the empty spaces between the placed components.



19. Close all views. Do **NOT** save.

20. Close the Virtuoso session.

21. In the shell, enter the following command:

    cd ~

This command places you in your home directory.

Summary

In this lab, you learned:

- How to launch the connectivity-driven environment and generate components in the layout.
- How to run the Placement Planning form to plan and create rows and rails for placing the layout components.
- How to run the Virtuoso Custom Digital Placer (VCP) to run placement and select the appropriate placement options to get the desired placement outcome.
- How to run the Insert/Delete Filler Cells form to fill the empty spaces between the placed cells with filler cells.

End of Lab

Module 3: Binder and Extractor

Lab 3-1 Using Manual Binding

Objective: To explore the Manual Binding using the Define Device Correspondence (DDC) command.

In some situations, you might prefer to use the *Create Instance* or *Copy* commands when creating a layout. This is a supported flow in Layout XL, but not recommended since neither of these commands is connectivity aware. Therefore, making the generated layout **XL-compliant** requires additional effort. The unbound instances created by these commands are required to be bound to their schematic counterpart. One of the ways you can achieve this is by using Manual Binding using the Define Device Correspondence (DDC) command.

In this lab, you will learn how to bind layout instances with their schematic counterpart by **Manual Binding** using **DDC**.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. If you continue from the previous lab, move to the sub-heading: Starting the session Otherwise, proceed to the next step.

2. Update the following line in the /VLPT3_IC_25_1/. cshrc_VLPT3 file with the appropriate valid local tool paths for Virtuoso.

   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
   ```

3. Save and close the file.

4. Source the updated class setup file.

   ```
   Source .cshrc_VLPT3
   ```

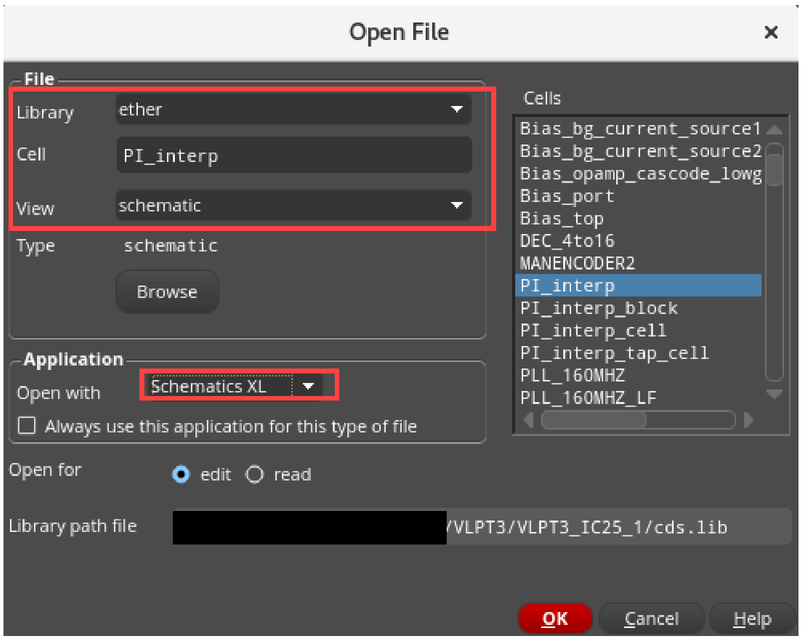
Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

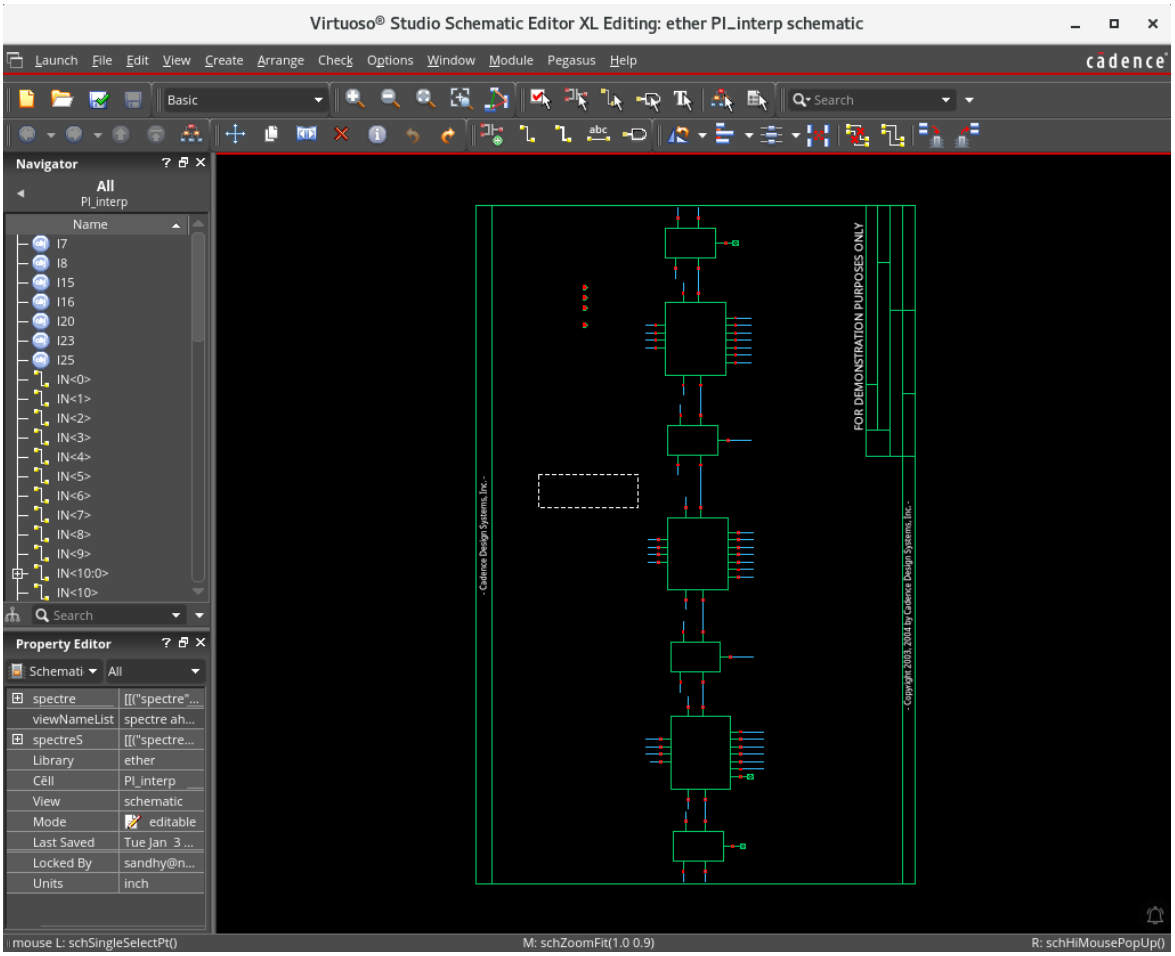
   ```
   cd ./VLPT3_IC_25_1/VLPT3_M03_BEX
   ```

2. Open the following schematic in XL with the **Library Manager** or choose **File – Open** from the CIW.

Library	ether
Cell	PI_interp
View	schematic

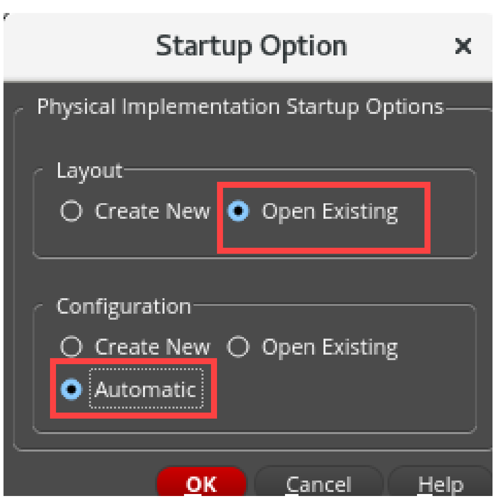


The *ether/PI_interp/schematic* window appears in XL mode, as shown here:

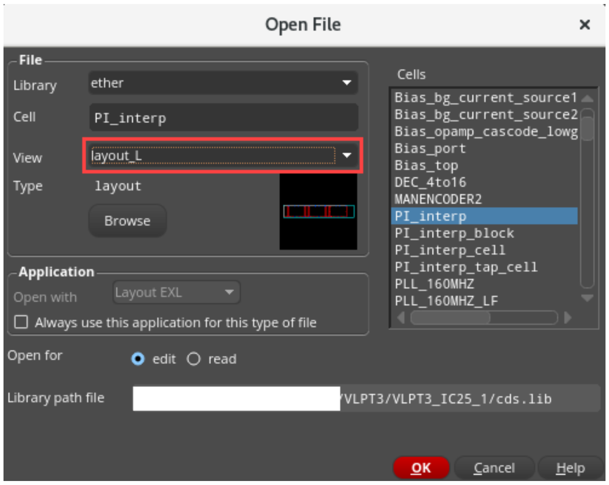


3. In the schematic, choose **Launch – Layout EXL** to bring up the **Startup Option** form.

4. Select the options as shown here to open an existing layout view and click **OK**.

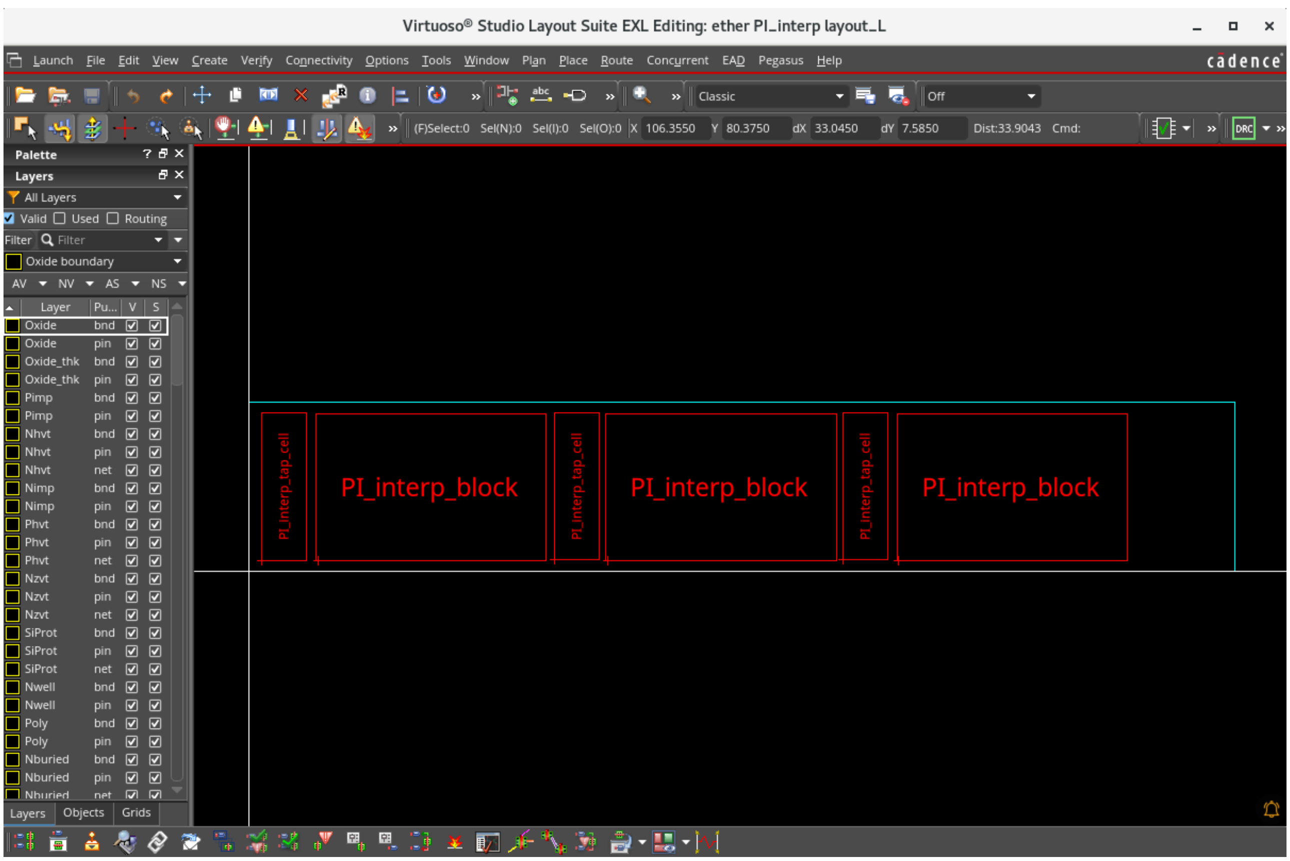


5. Fill out the Open File form, as shown here:



6. Click **OK** to open the layout view.

The Virtuoso Layout Suite EXL palette opens with the *ether/PI_interp/layout_L* view.

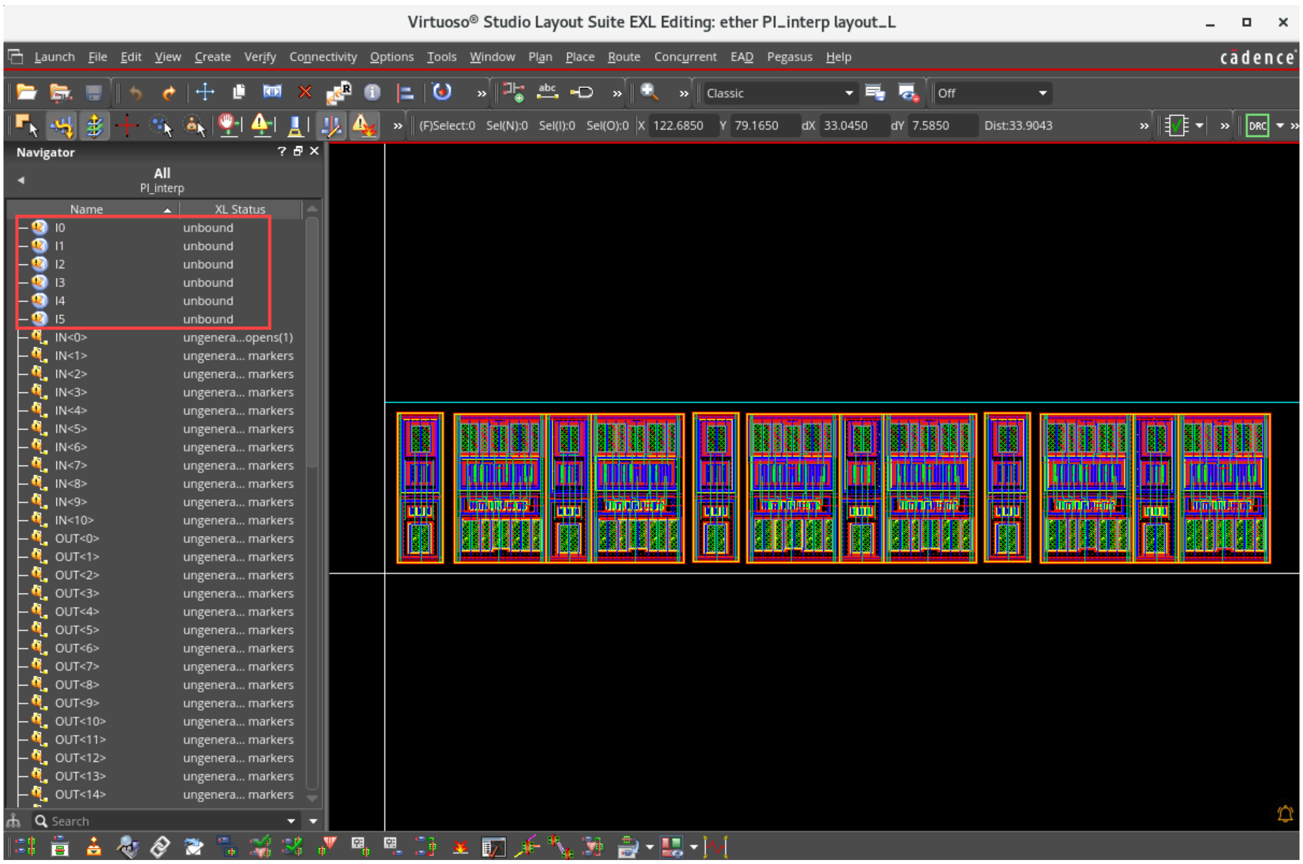


Note: In the layout canvas, from the **Window – Toolbars** menu, you can make the toolbars appear/disappear as per your requirement.

Note: In the layout canvas, from the **Window – Assistants** menu, you can enable/disable the assistants as per your requirement.

Note: In the layout canvas, from the Workspace Configuration cyclic field, you can choose the required workspace.

This layout has been created using *Create Instance* and *Copy* commands, and therefore, all the instances are **unbound**.

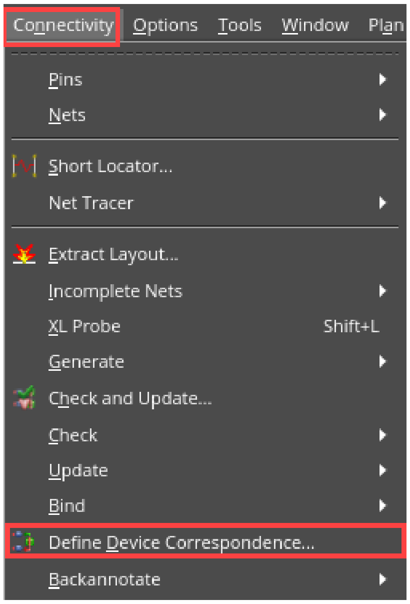


I0, **I2**, and **I4** are instances of **PI_interp_tap_cell**.

I1, **I3**, and **I5** are instances of **PI_interp_block**.

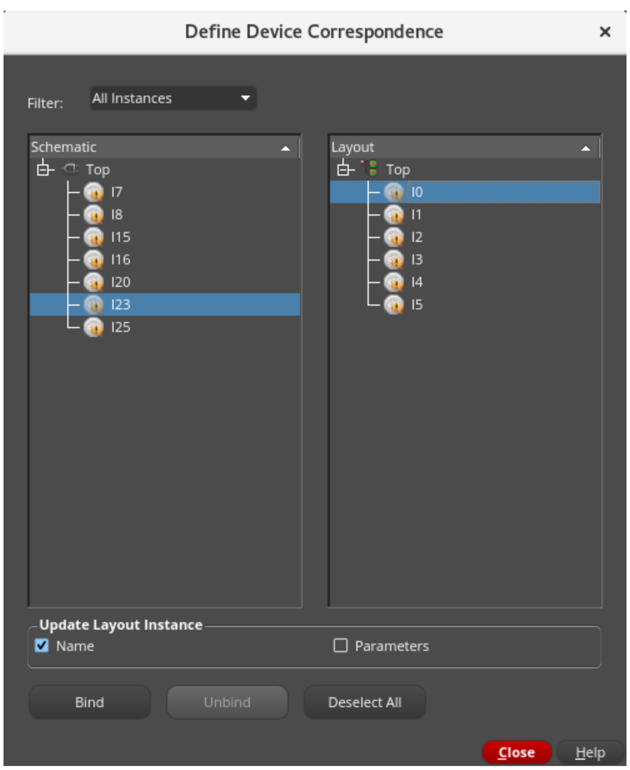
Now, you can manually bind the layout instances to their schematic counterparts using the *Define Device Correspondence (DDC)* form.

7. Run Define Device Correspondence by choosing either the **Connectivity – Define Device Correspondence** menu command or clicking the **Define Device Correspondence** icon on the Layout EXL toolbar, as shown here:



The Define Device Correspondence form appears.

8. In the DDC form, select **All Instances** in the Filter cyclic field to show the already mapped ones.

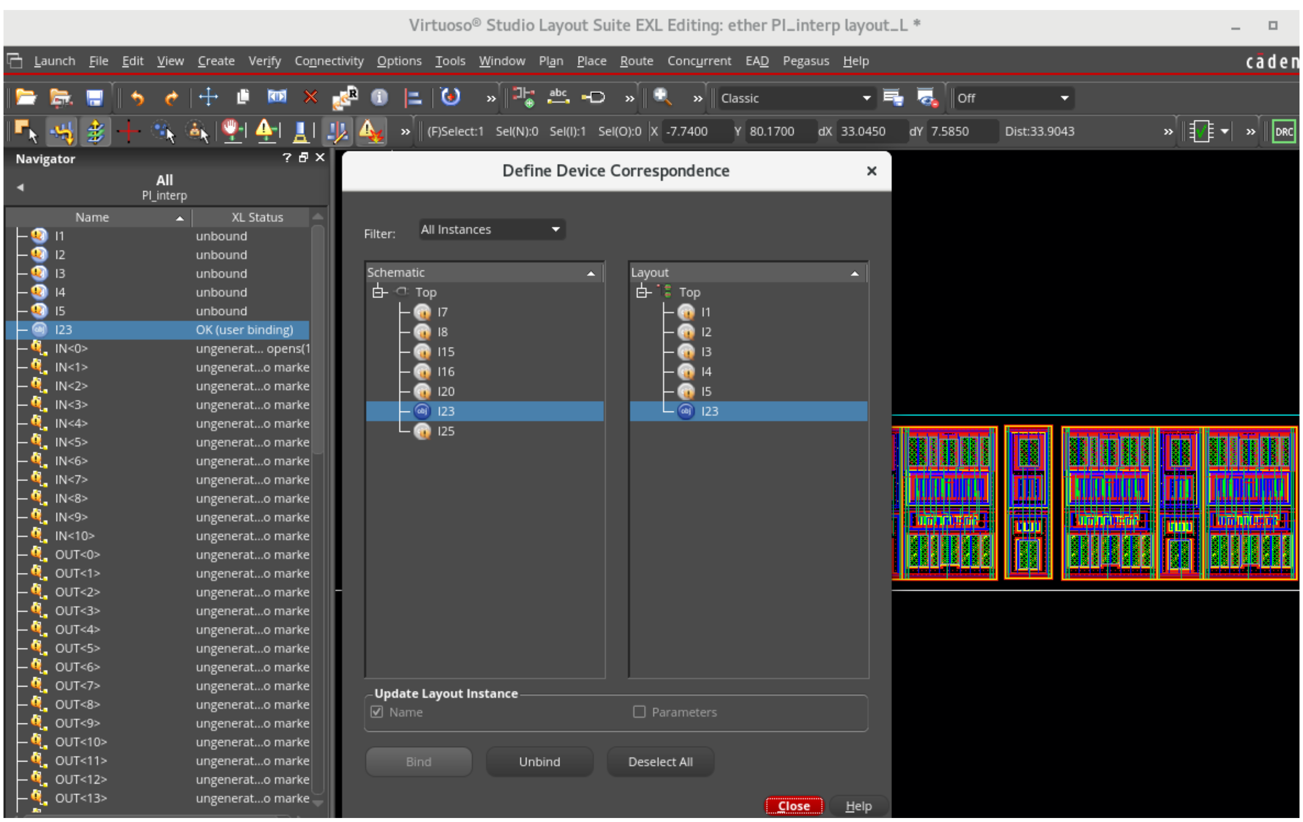


Schematic Instance	**Layout Instance**
`I23`	`I0`
`I7`	`I1`
`I16`	`I2`
`I8`	`I3`
`I15`	`I4`
`I20`	`I5`

For the design in this example, the table details the bindings that should be made to achieve XL-compliance.

Important: It is recommended to check the *Name* option in the *Update Layout Instance* section of the form. This will update the layout instance name to match that of the schematic.

9. In the *Define Device Correspondence* form, select an instance in the schematic pane and the corresponding instance in the layout pane that it should bind to as per the table given here and click the **Bind** button.

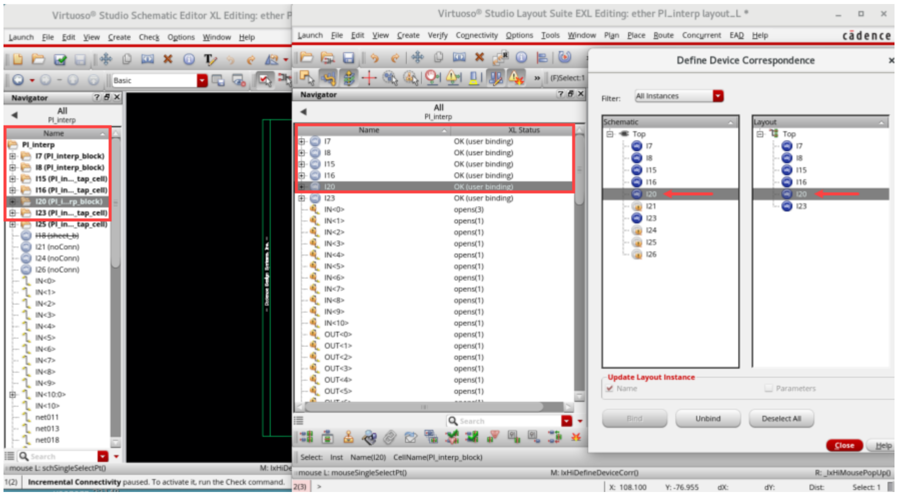


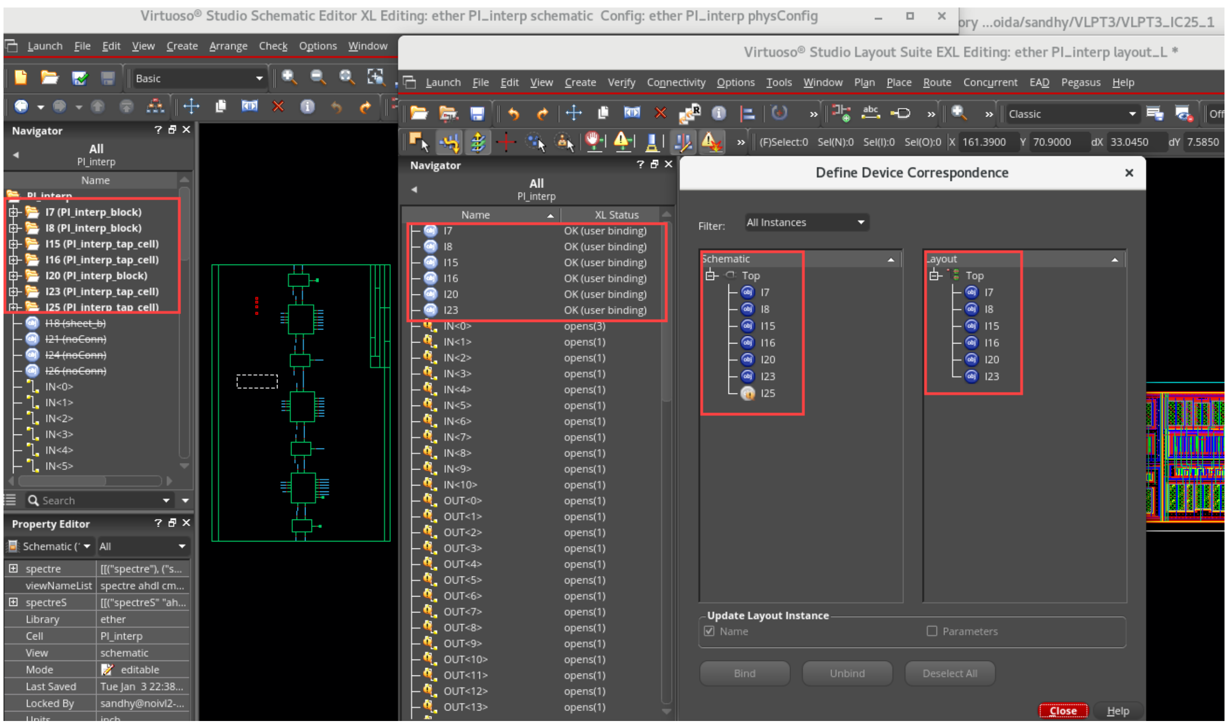
As shown in the *DDC* form, the binding is done between ***I23*** of Schematic Instance and ***I0*** of Layout Instance and the binding between schematic and layout is shown here.

As the bindings are made, you will notice a number of things happen that indicate the design is now XL-compliant.

- The icon in the Define Device Correspondence form changes to a **dark blue** to signify this is a user binding.
- The Navigator XL Status changes to **OK** (user binding).
- The correct connectivity is added to the instance's terminal, as can be seen in the **Connectivity** tab of the Property Editor assistant.

10. Complete the binding for all the schematic and layout instances as per the given table. When all the bindings are made, your schematic/layout will look as shown here:





11. Close the schematic and layout views. Do **NOT** save.

12. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned:

- How to bind layout instances with their schematic counterpart by Manual Binding using Define Device Correspondence (DDC).

End of Lab

Lab 3-2 Using Incremental Binding

Objective: To explore the Automatic Incremental Binding of "unknown" instances.

In some situations, you might prefer to use the *Create Instance* or *Copy* commands when creating a layout. This is a supported flow in Layout XL, but not recommended since neither of these commands is connectivity aware. Therefore, making the generated layout **XL-compliant** requires additional effort. The unbound instances created by these commands are required to be bound to their schematic counterpart. One of the ways you can achieve this is by using **Automatic Incremental Binding** when connectivity is added to the layout.

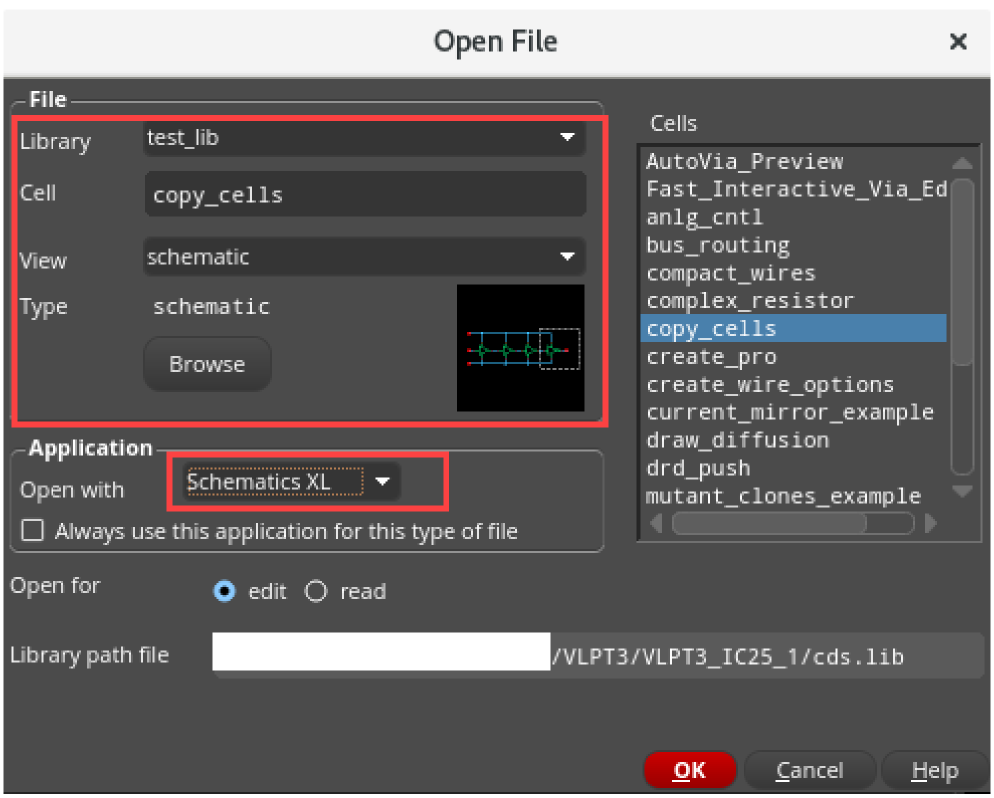
You add connectivity to the layout with routing and when there is sufficient connectivity information to consider the instance unique, the Incremental Binder automatically binds it to its schematic counterpart.

In this lab, you will learn how to bind layout instances with their schematic counterpart using the Automatic Incremental Binder and Update Components and Nets (UCN) commands.

1. In the CIW, choose **File – Open**.

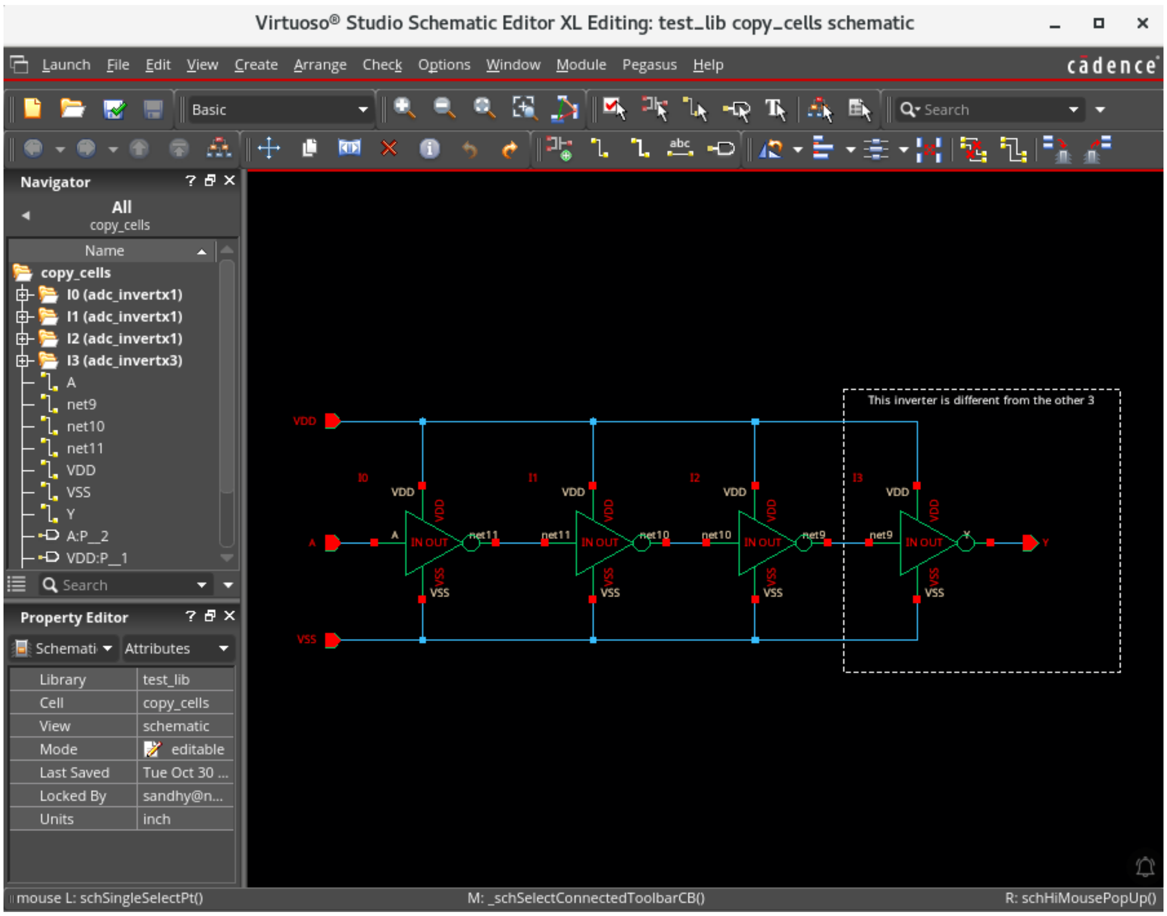
   The Open File form appears.

2. Fill out the form, as shown here:



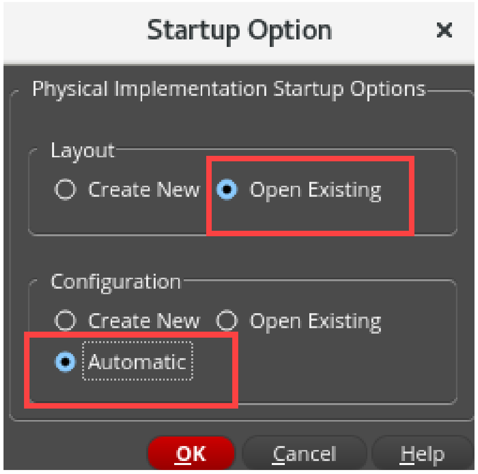
3. Click **OK**.

   The schematic cellview *test_lib/copy_cells/schematic* opens for editing. The circuit is simply built by four inverters chained together. The first three instances are identical. The 4th cell is the only different one.

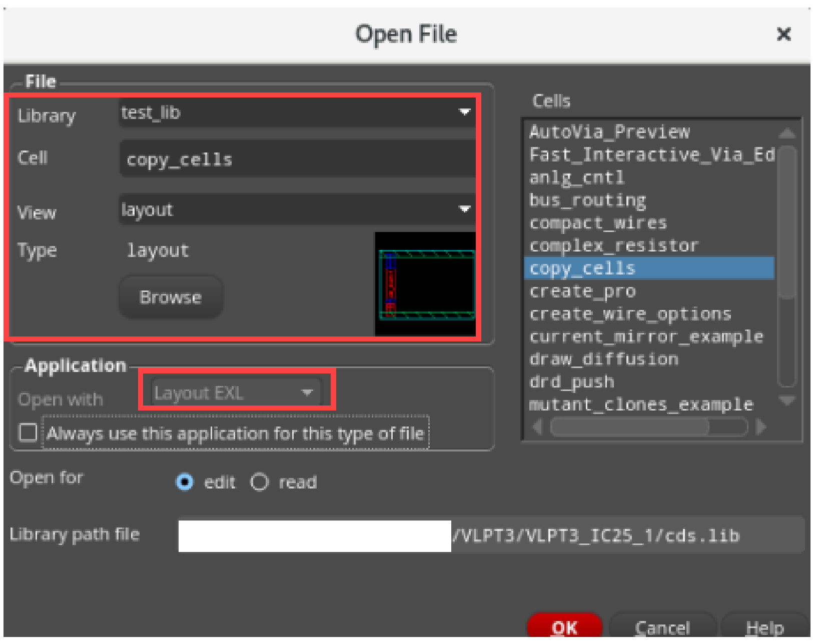


4. In the schematic, choose **Launch – Layout EXL** to bring up the **Startup Option** form.

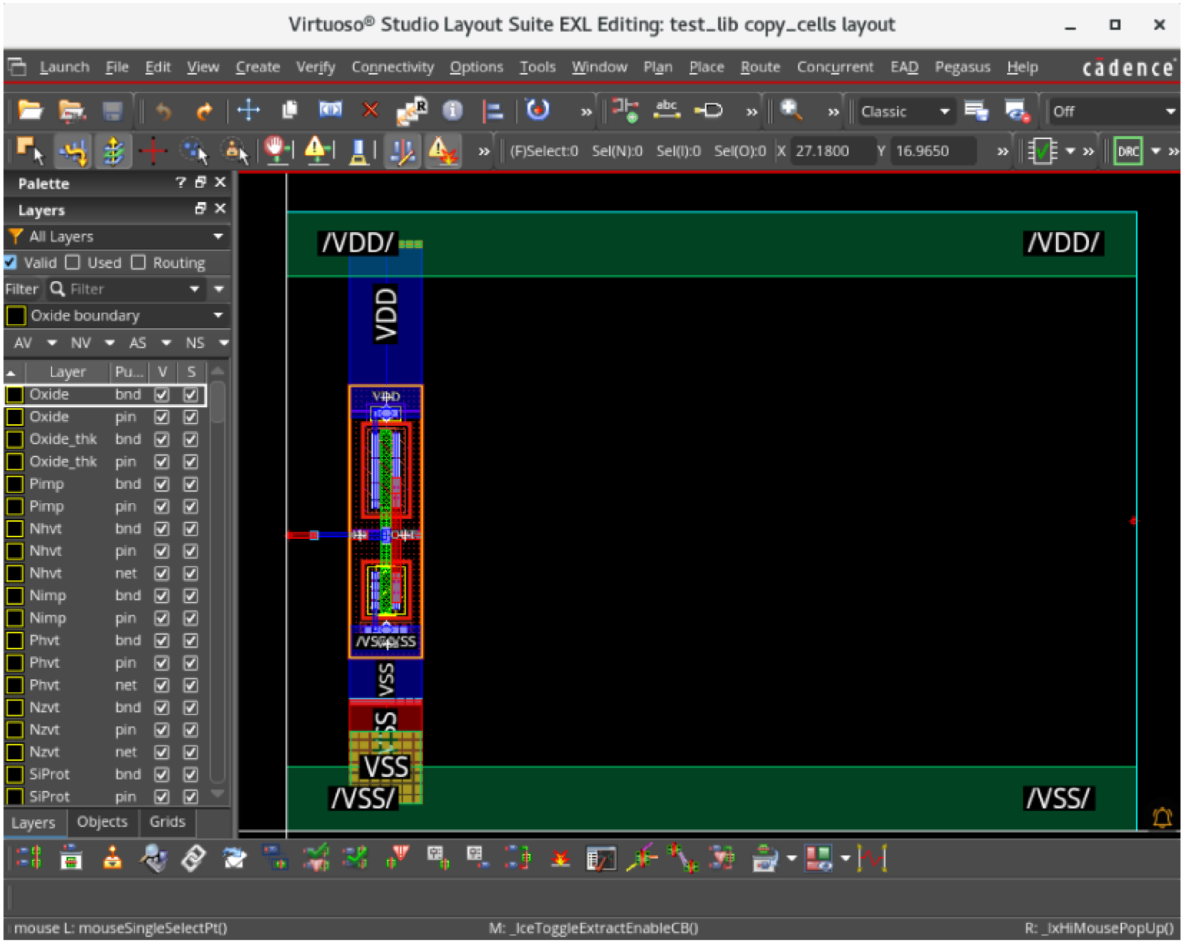
5. Select the options as shown here to open an existing layout view and click **OK**.



6. Fill out the Open File form, as shown here:



7. Click **OK** to open the *test_lib/copy_cells/layout* view.



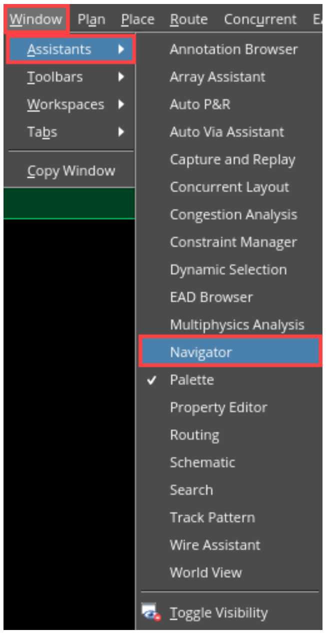
Note: If you do not get “Ungenerated Instances” in the Navigator assistant in the layout canvas, you can click on the **Ungenerated** section in the OBJECTS category to view the ungenerated instances.

8. Activate the **Navigator** assistant and enable the **XL Status** column if it is not already turned on.

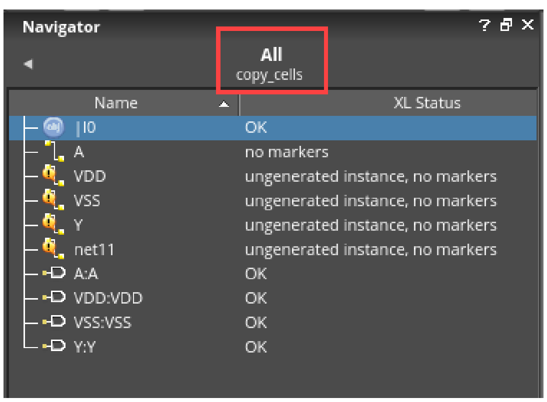
   a. Choose **Window** – **Assistants** – **Navigator** on the layout menu banner to turn on the Navigator assistant if it is not already turned on.

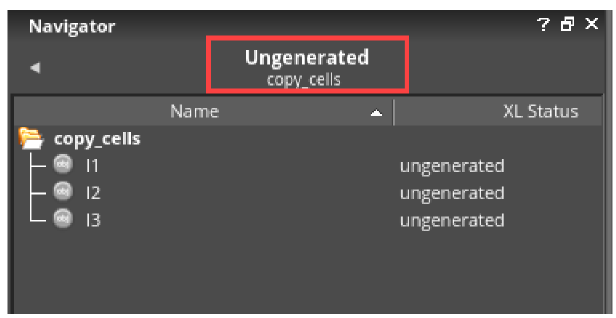
   b. Use the RMB on the **Navigator** column header (where **Name** is present) and enable the XL Status column if it is not already turned on.

c. The new column is not immediately visible; stretch the column handle to enlarge it to see the XL Status column.



9. Choose **Window – Assistants – Navigator**, click on **All** and **Ungenerated,** as shown here:



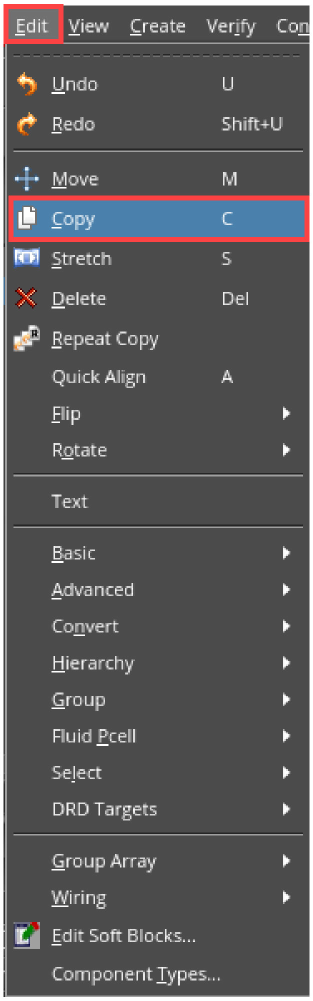


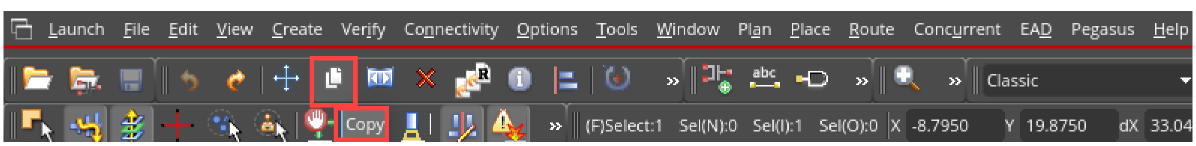
Note: In the Navigator, **XL Status** of the instances are:

- **I0 – OK**
- **I1 – ungenerated**
- **I2 – ungenerated**
- **I3 – ungenerated**

You will now create two copies of the inverter.

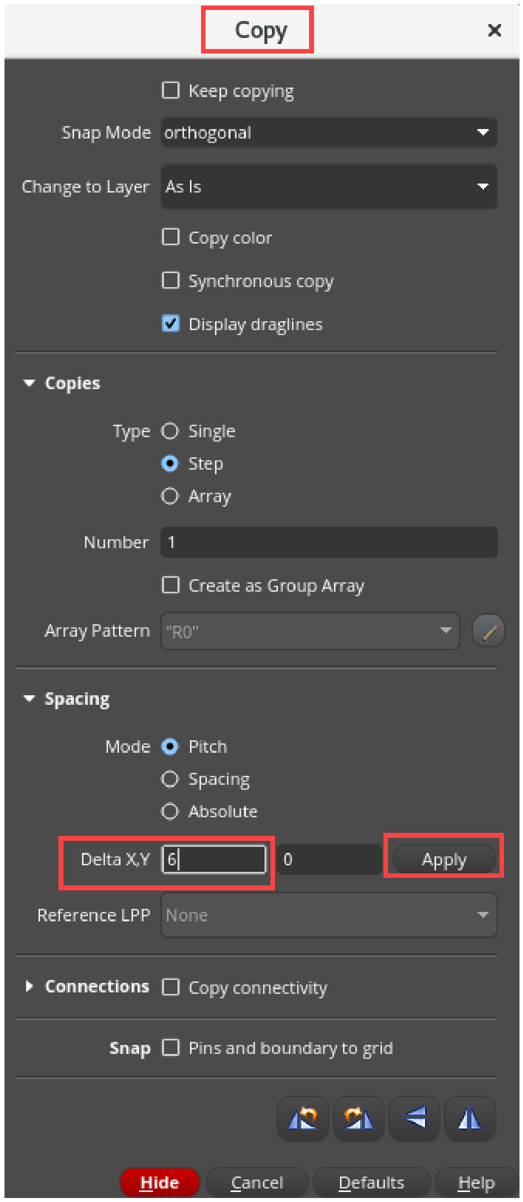
10. Invoke the Copy form by choosing either the **Edit – Copy** menu command or pressing the bindkey **C** or clicking the **Copy** icon on the Edit toolbar. If the Copy form does not pop up, press **F3** to bring up the form.





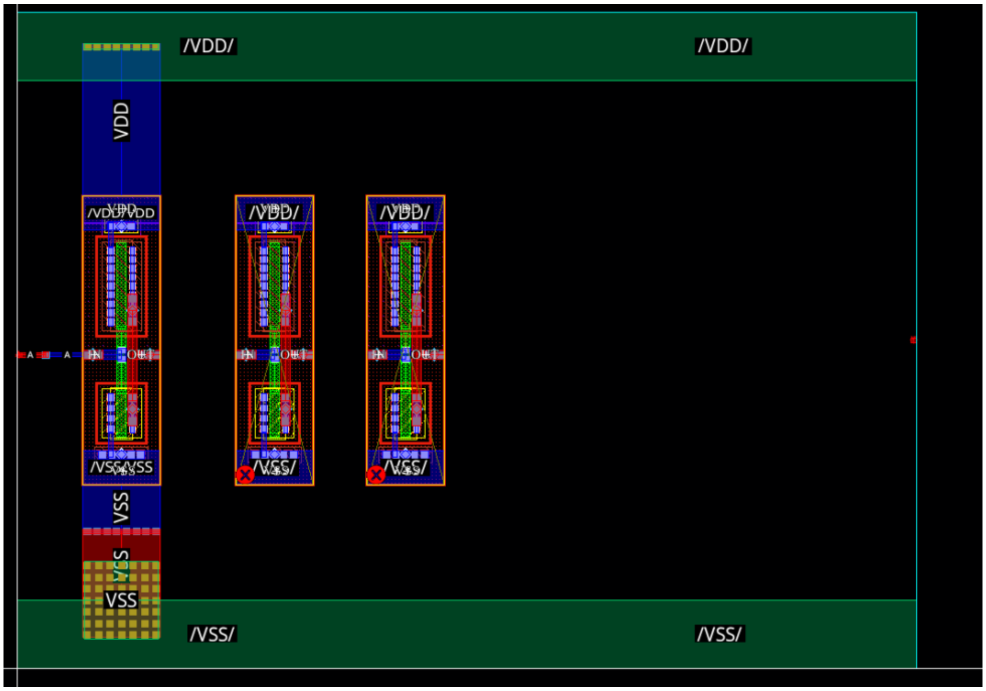
11. Fill out the Copy form, as shown here:

    a. In the **Delta X Y** field, enter **X=6**.

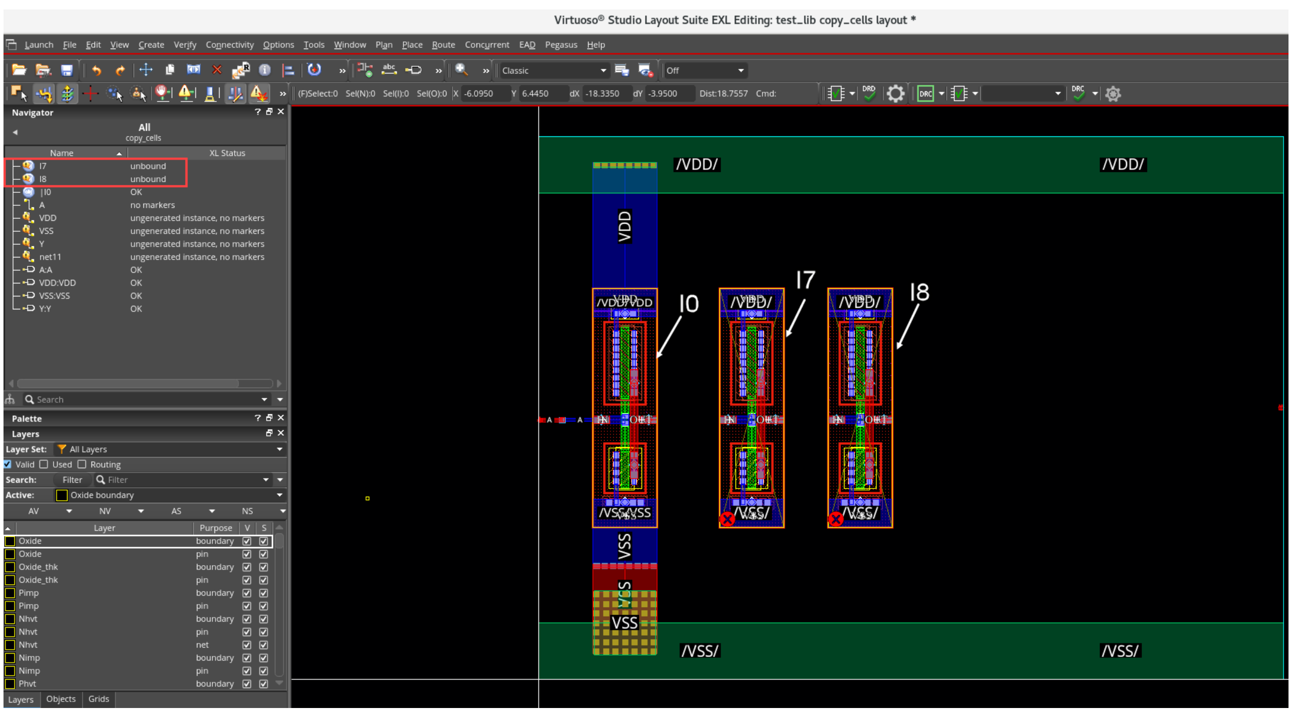


12. Select the instance **I0** and click the **Apply** button to make a copy (**I7**) of the instance.

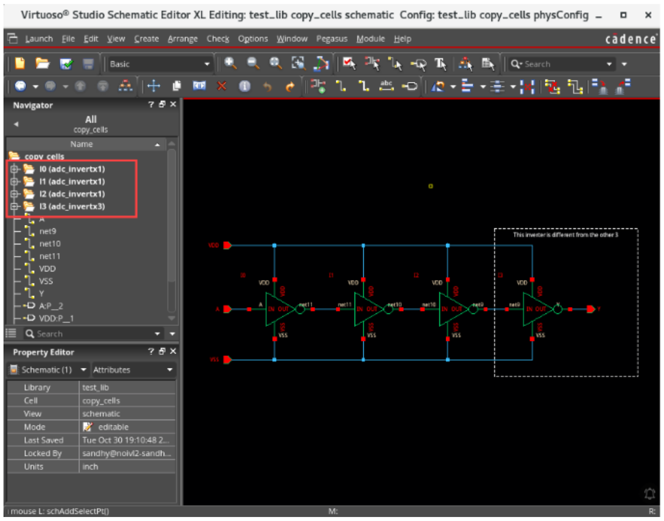
13. Select the newly copied instance (**I7**) and repeat the previous step to make another copy (**I8**).

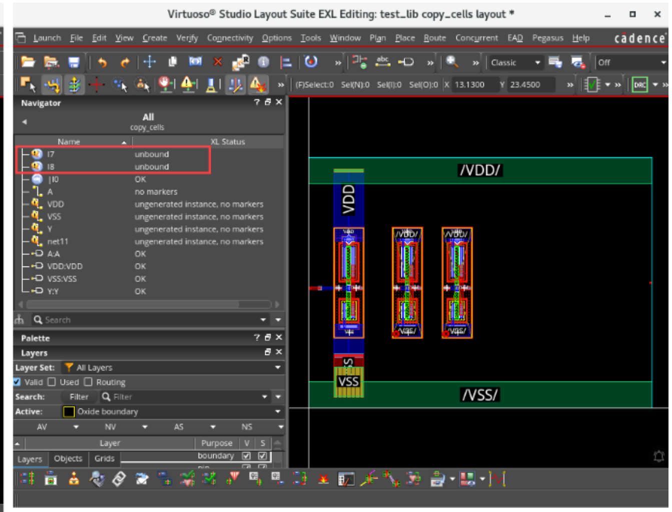


You have now created **two** copies of the inverter (**I7** and **I8**) at an interval of **6 units** along the X-axis. Since the Copy command is used, the new copies will initially be **unbound,** as shown here.



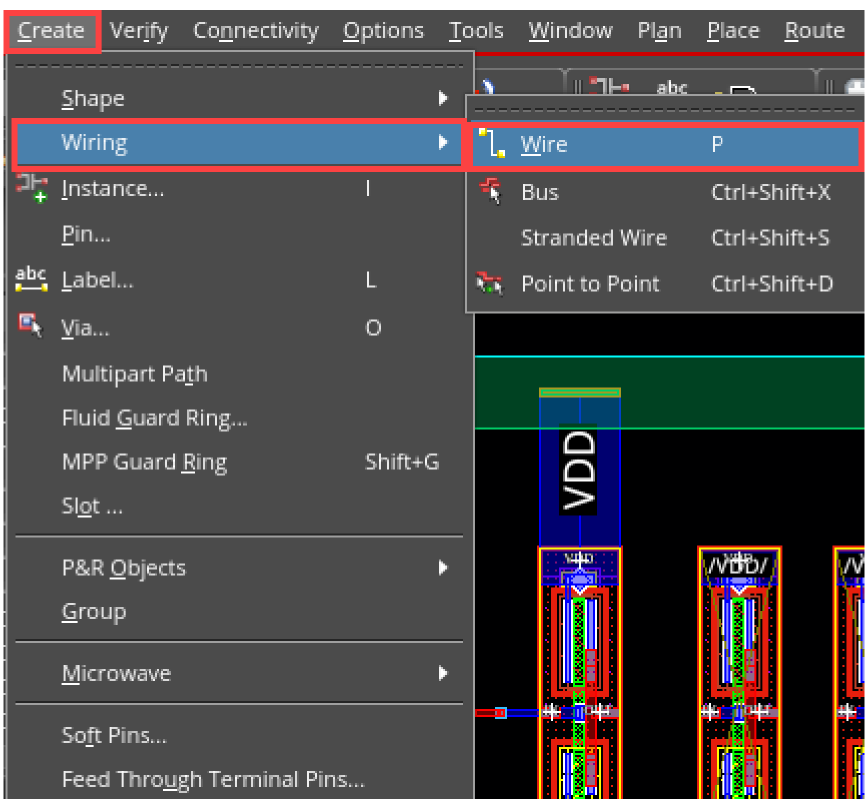
Notice how the instance names of the two copies of the layout (**I7** and **I8**) do not reflect the names in the schematic. Also, the cross-probing between the schematic/layout is not working as expected for these two copies of the layout because they are **NOT bound** yet.

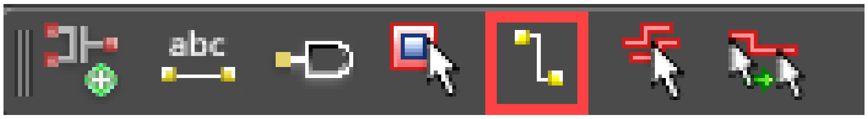




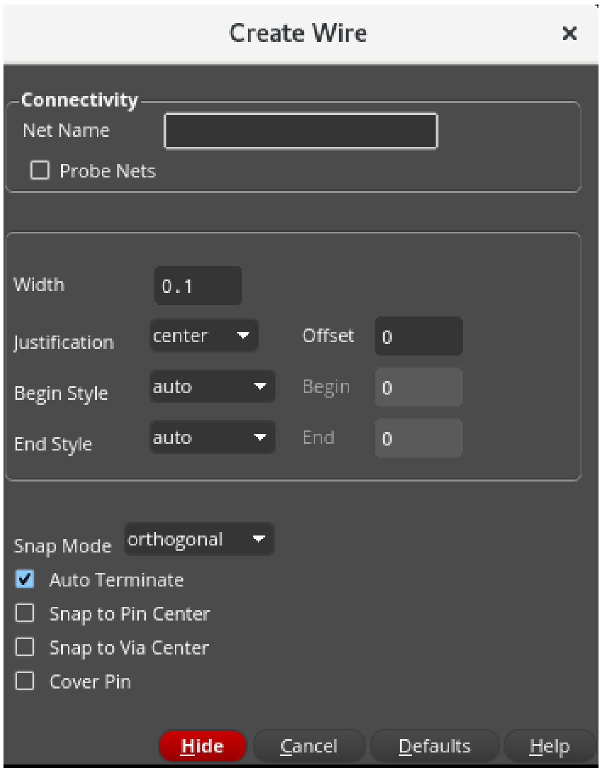
Now, let us connect the output of the first inverter (**I0**) to the input of the second one (**I7**). The incremental binder will immediately recognize the second inverter (I7) as **I1**.

14. Invoke the Create Wire form by choosing either **Create** – **Wiring** – **Wire** pull-down menu or pressing the bindkey **P** or clicking the **Create Wire** icon on the Create toolbar. If the Create Wire form does not pop up, press **F3** to bring up the form.

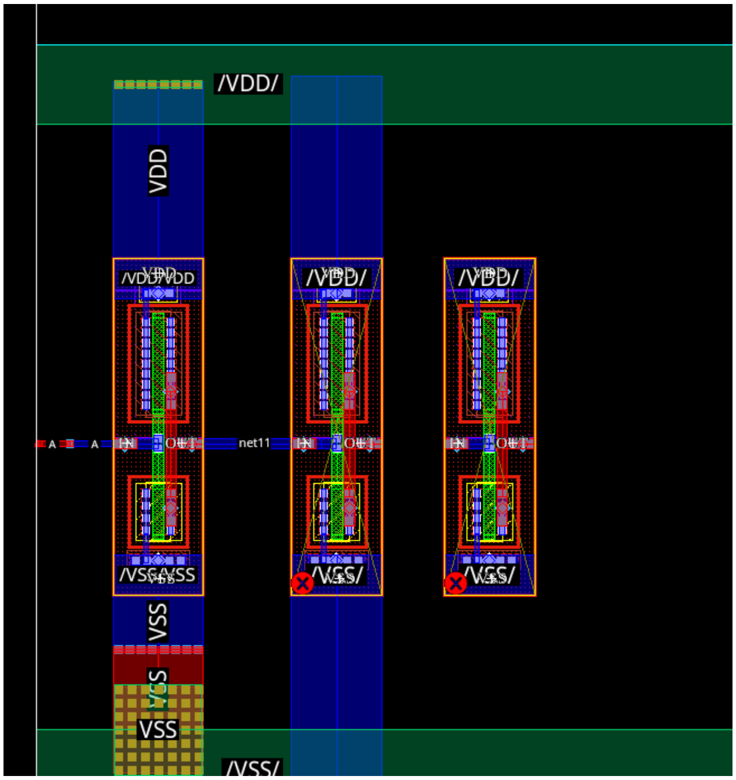




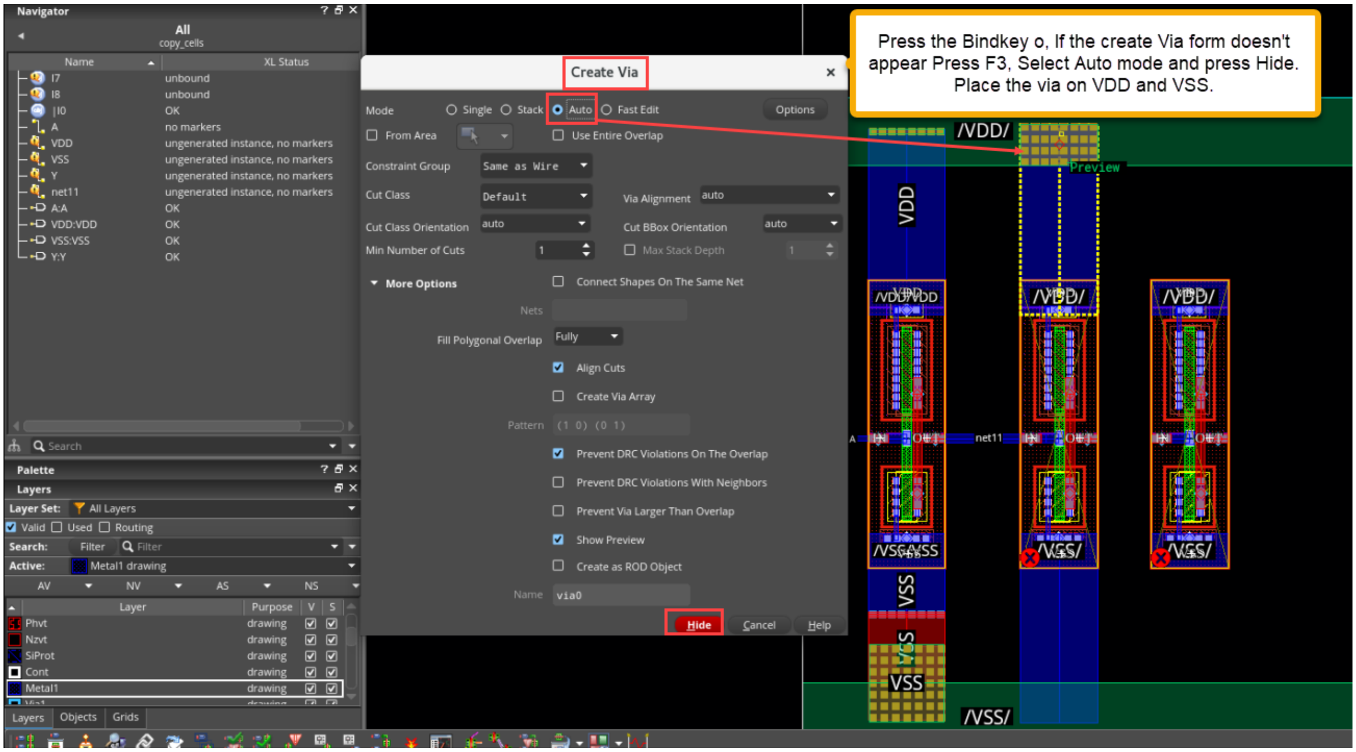
The Create Wire form appears, as shown here:



15. Press Bindkey P, Start interactive routing with **Metal1** layer from the inverter **I0** to the inverter **I7** and complete the connection as shown here.

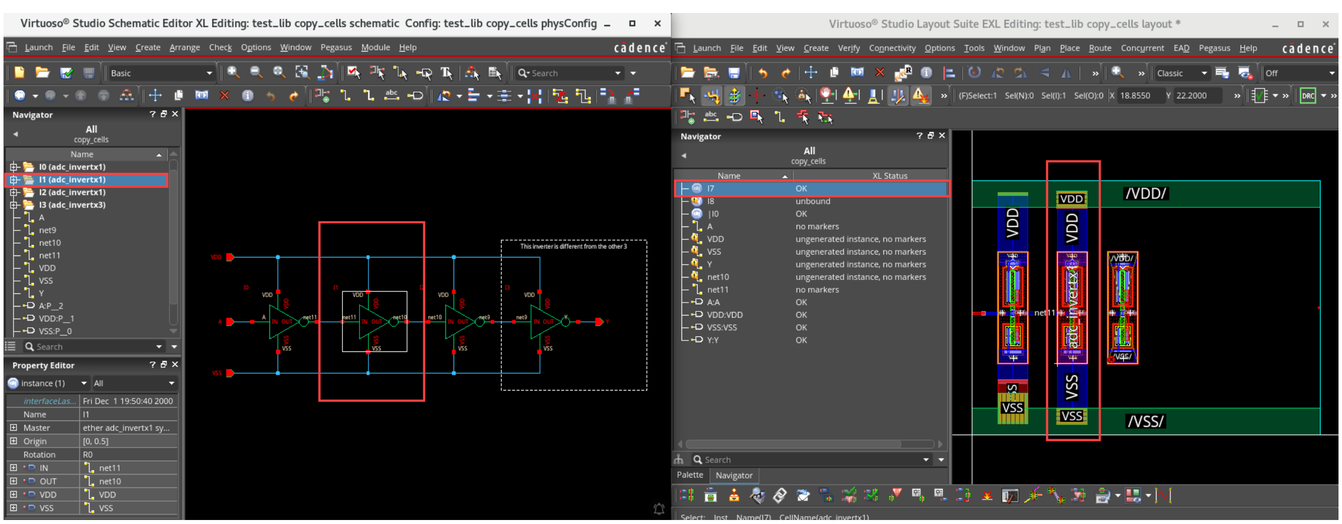


16. To connect to pins **VDD** and **VSS** which are on Metal3, Press the bindkey ‘o’ if the create via form doesn’t appear press F3, choose auto via and press hide then place the mouse pointer on VDD and VSS. Via get placed perfectly.



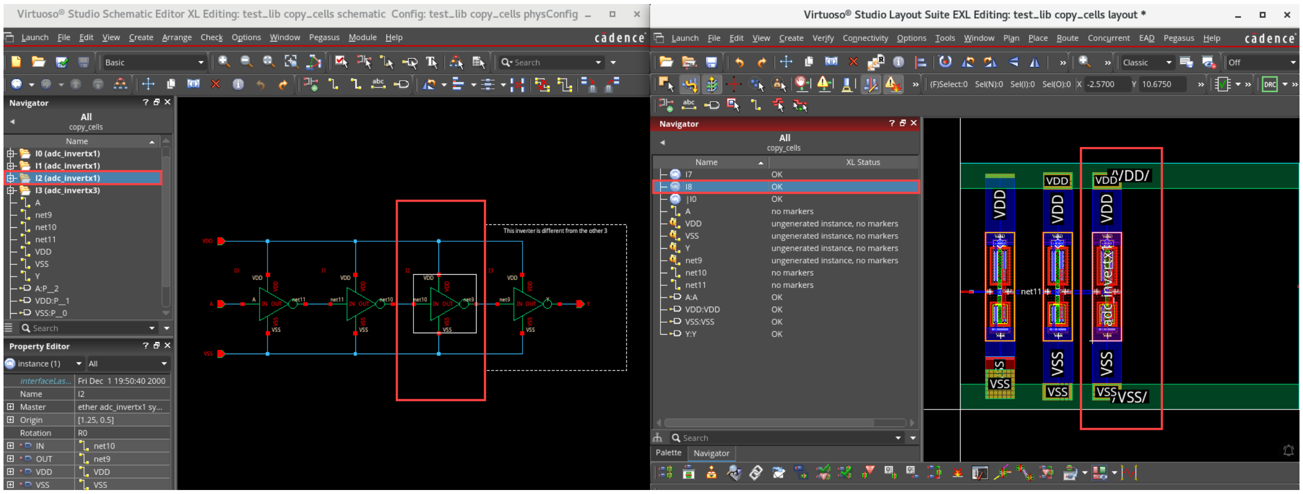
Important: As soon as the connection is completed, the binder identifies the instance (**I7**); the cross-probing between schematic(**I1**)/layout(**I7**) is now working as expected; the instance(**I7**) is marked as **OK** in **XL Status,** as shown here.

Note: Cross-probing is **NOT** available between schematic(**I2**)/layout(**I8**), because the third inverter(**I8**) is **NOT bound** yet.



17. You can now complete the connection between the 2nd inverter (**I7**) and the 3rd inverter (**I8**) as described in **Steps 13** and **14**.

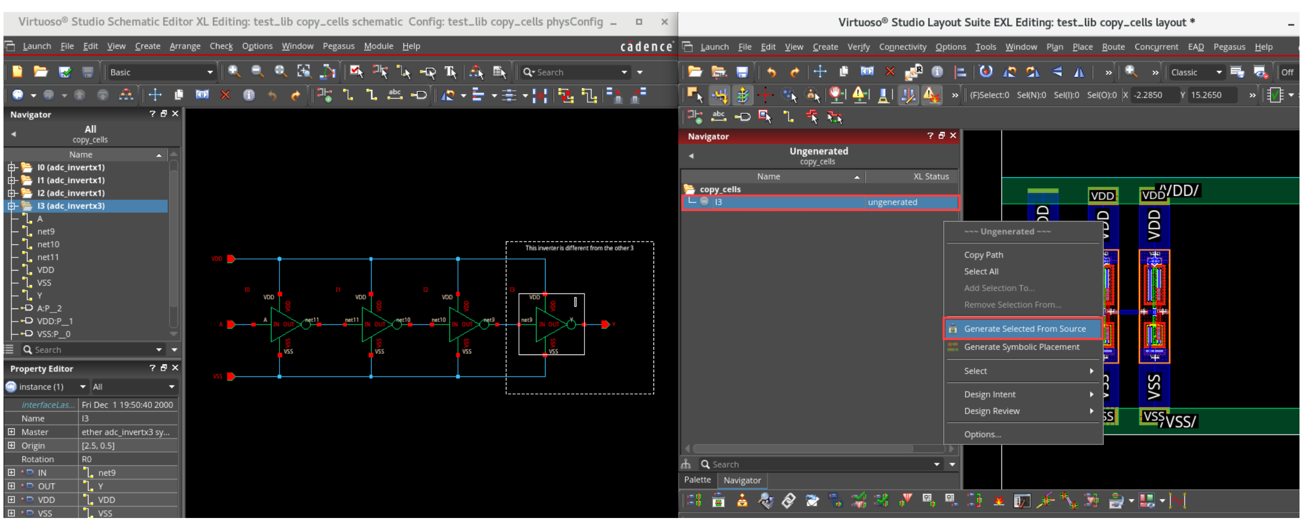
    Important: As soon as the connection is completed, the binder identifies the instance (**I8**); the cross-probing between schematic(**I2**)/layout(**I8**) is now working as expected; the instance (**I8**) is marked as **OK** in XL Status, as shown here:

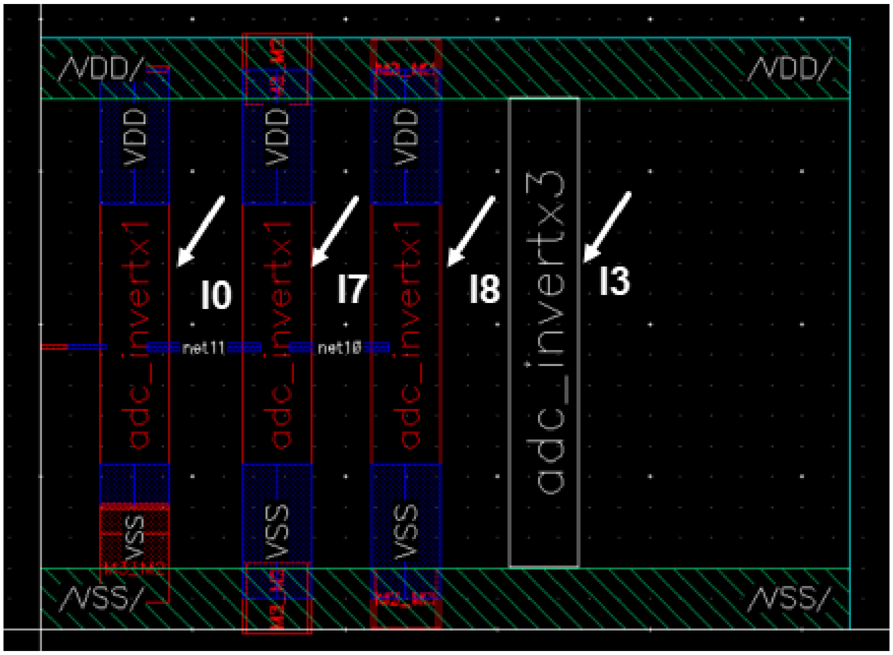


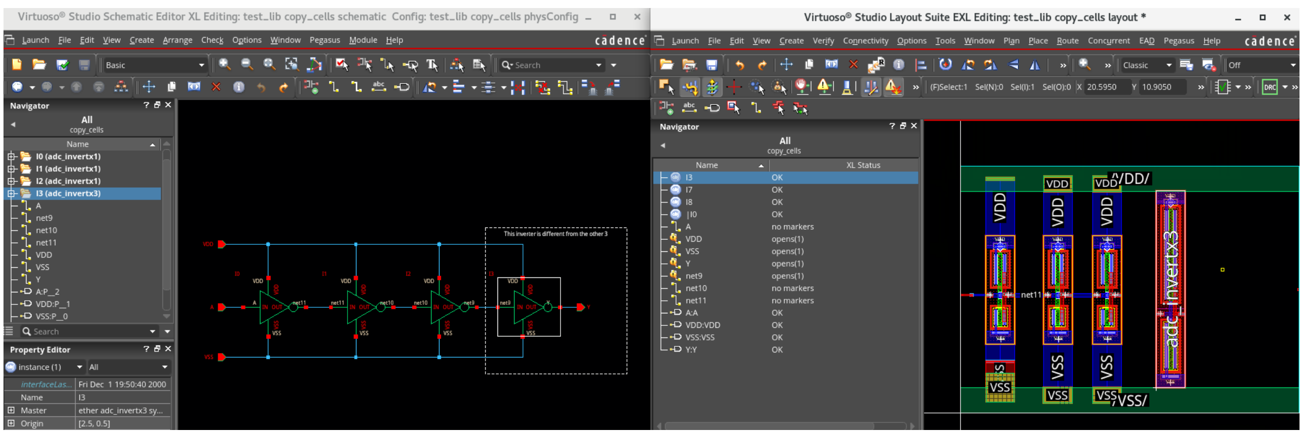
You will now generate the 4th inverter using the Generate Selected From Source command.

18. In the Navigator, **right**-click on the instance (**I3**) and select the **Generate Selected From Source** (**GSFS**) command option.

    The device is selected and remains glued to your cursor until you place it in the layout. You can now place it in the layout, as shown in the next graphic.



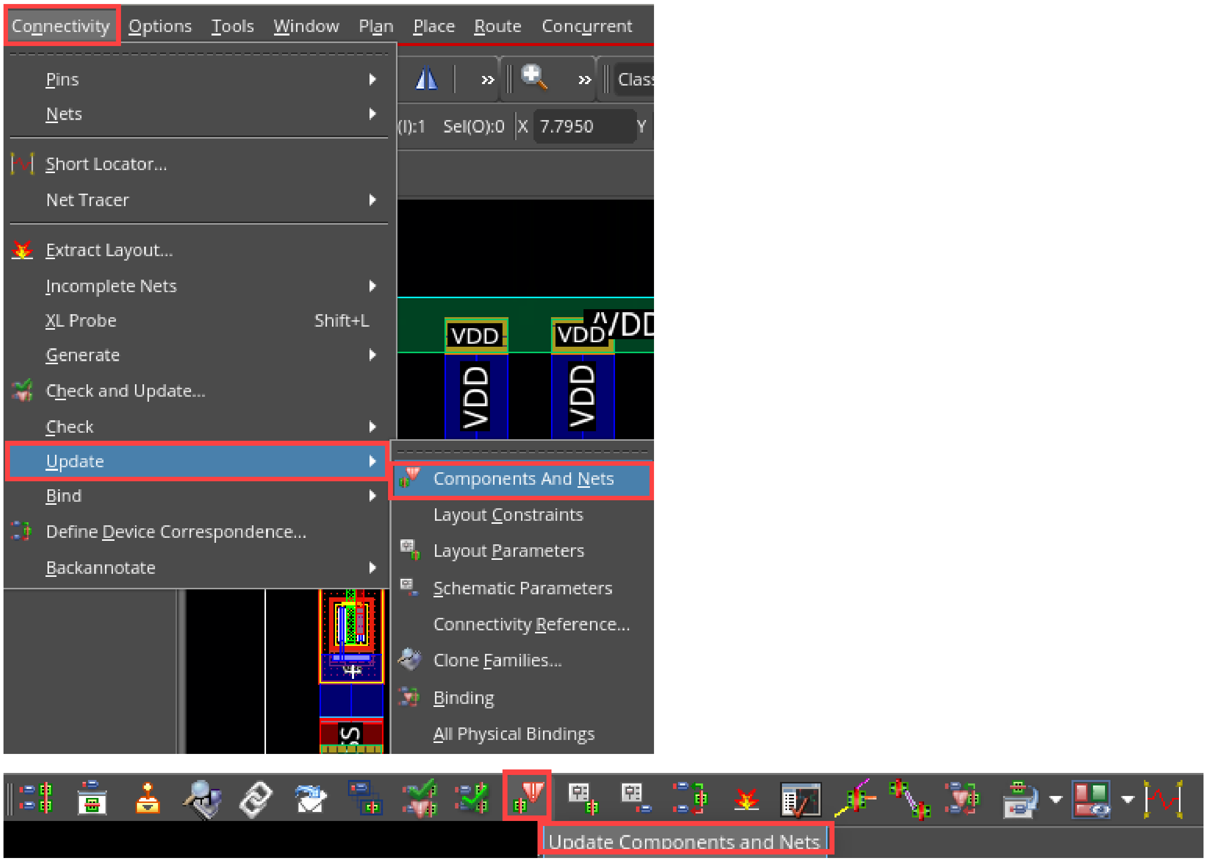




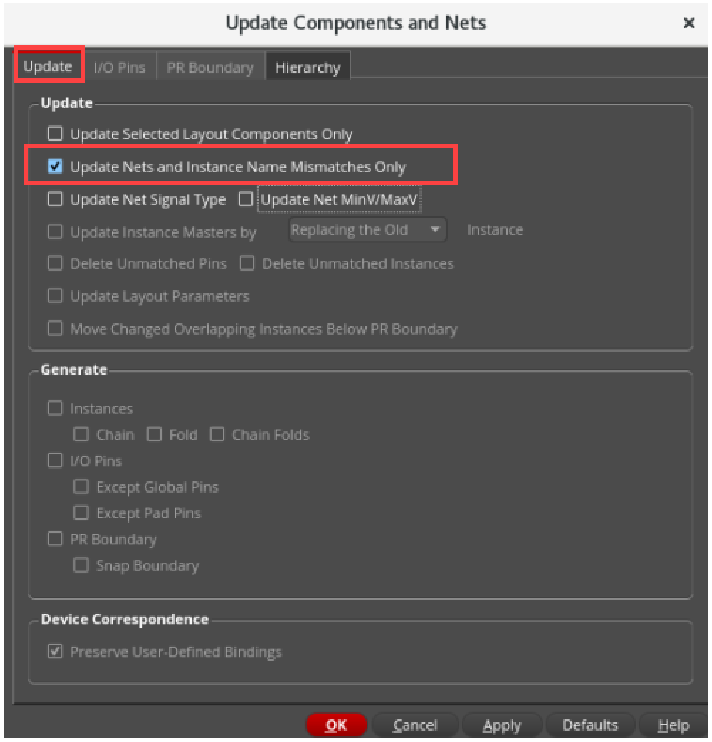
The layout placement is now complete; all instances are in the layout view, and they are all **bound**, as you can easily verify by cross-probing them.

In the next step, you will make the naming consistent between the schematic/layout.

19. Invoke the Update Components and Nets form by choosing either **Connectivity** – **Update** – **Components And Nets** menu entry or clicking the **Update Components and Nets** icon on the Layout toolbar, as shown here:

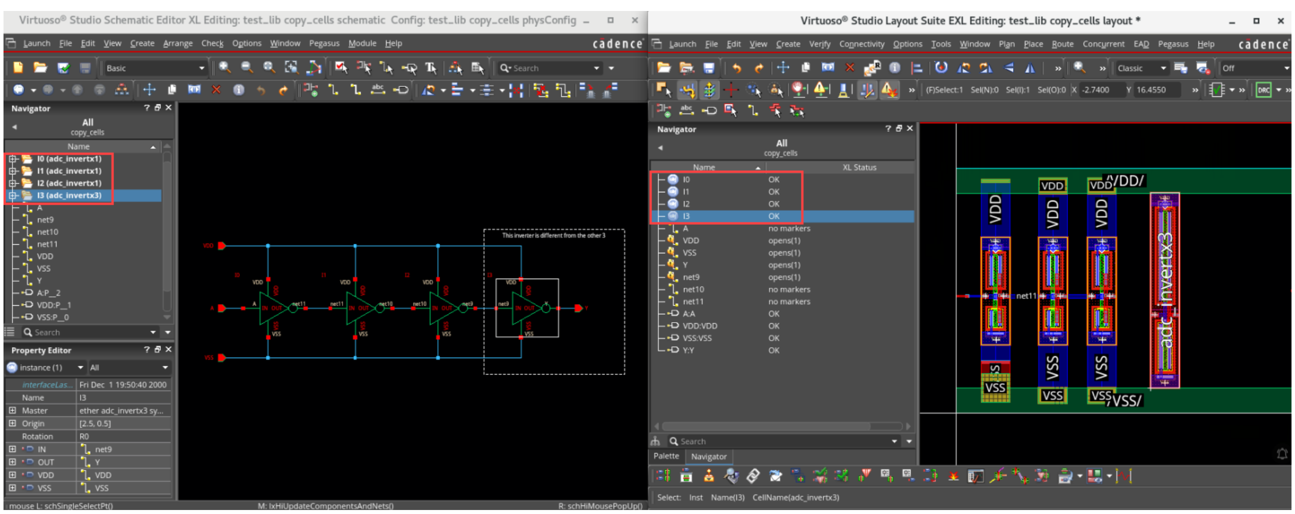


20. In the Update Components and Nets form, activate the **Update Nets and Instance Name Mismatches Only** checkbox. Your form should look like this, as shown here:

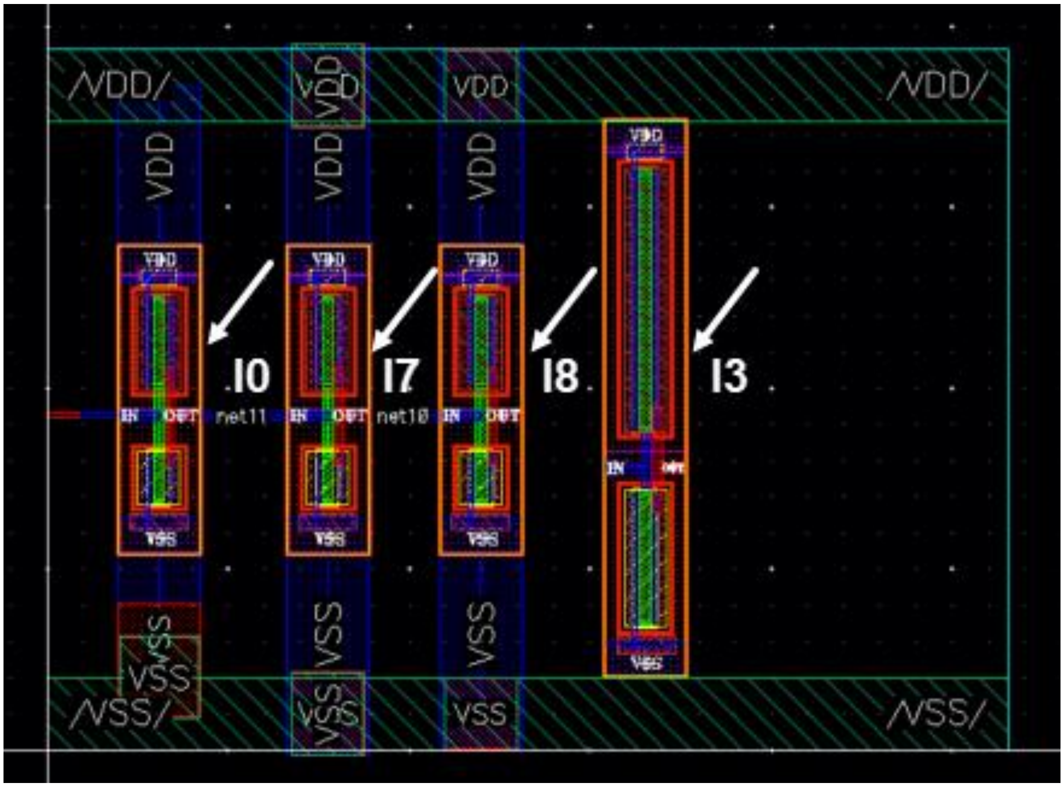


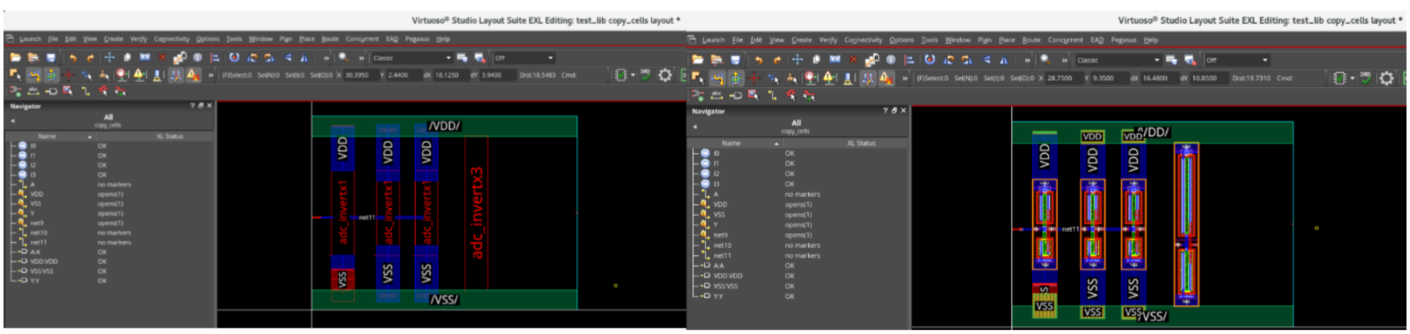
21. Click **OK**.

This will change the name of the layout instances to match the ones used in the schematic, as shown here:



22. Press the **Shift+F** bindkey to view all the levels of the hierarchy in the layout design, as shown here:





23. Close the schematic and layout cellviews without saving the changes.

24. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned:

- How to generate instances in VLS-EXL using the Copy command.
- How to add connectivity to the layout with routing and trigger the Incremental Binder.
- How to generate instances using the Generate Selected From Source (GSFS) command.
- How to correct the name mismatches between schematic/layout using the Update Components and Nets (UCN) command.

End of Lab

Lab 3-3 Using Update Binding

Objective: To explore the Update Binding features.

Layout productivity and design turnaround time are increasingly important. To use the productivity-enhancing functionality in VLS XL/EXL/MXL, the layout must be XL compliant.

Legacy designs or those completed outside of Layout XL can be LVS clean but may not be XL-compliant. Previously, it was difficult and error-prone to make such a layout XL-compliant, which would negate the VLS XL productivity advantage. You can use the *Update Binding* feature to make this process quicker and easy.

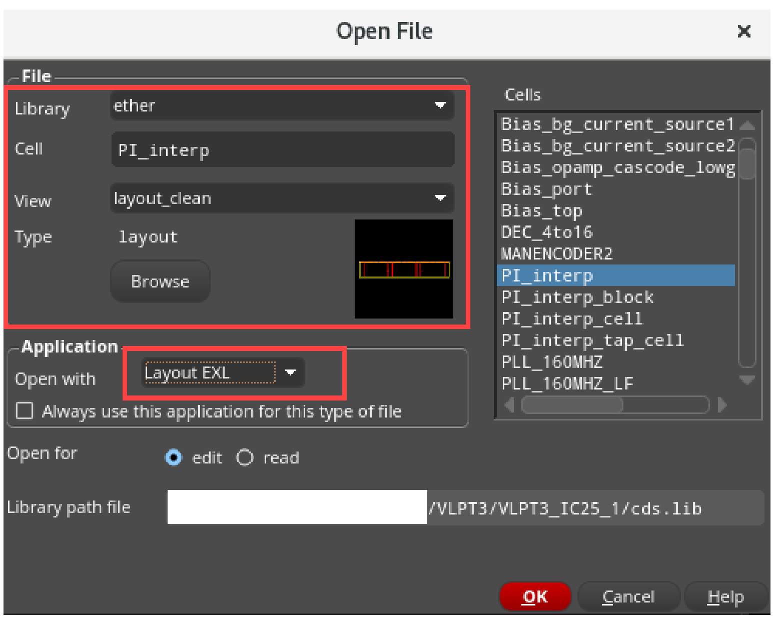
Update Binding uses the physical connectivity in the layout, such as routing, to trace connectivity through the layout and make bindings based on that. This means it is important that the design is LVS clean. Incorrect physical connectivity can very likely prevent the Update Binding from making the layout XL-compliant.

In this lab, you will consider a design that was completed outside of Layout EXL and make it XL-compliant using Update Binding.

1. In the CIW, choose **File – Open**.

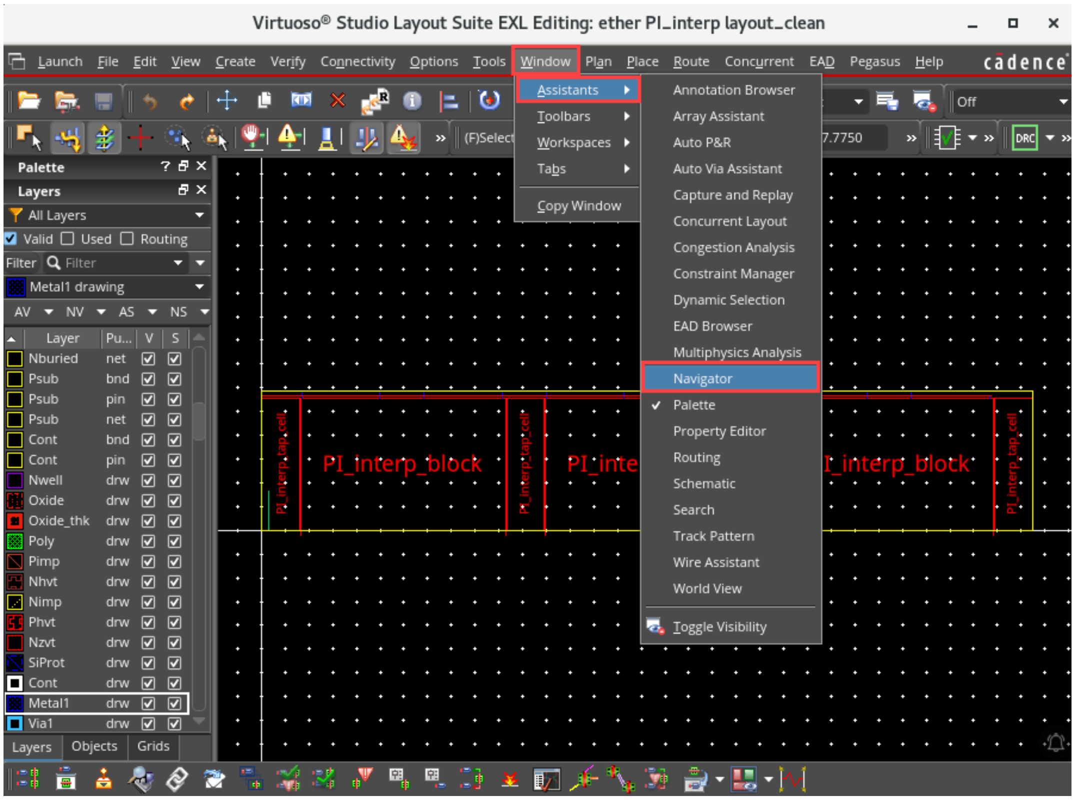
   The Open File form appears.

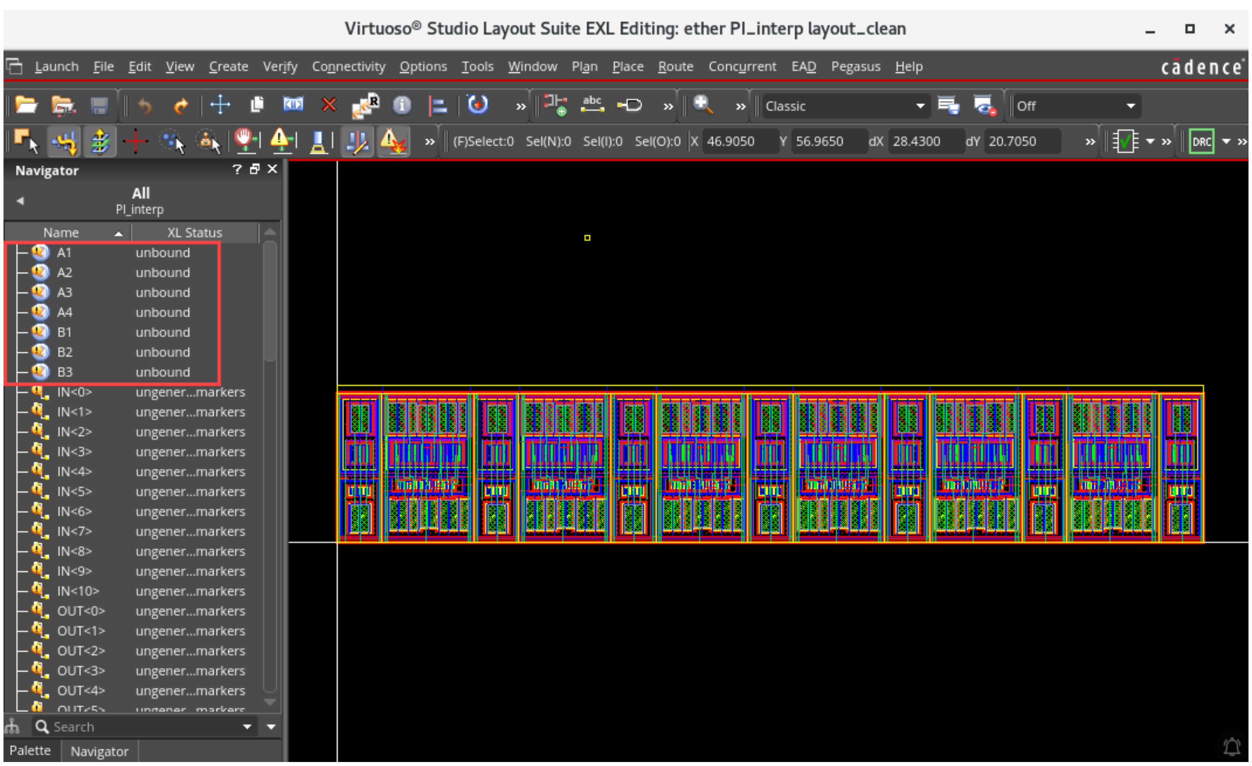
2. Fill out the Open File form, as shown here:



3. Click **OK**.

The *ether/PI_interp/schematic* and *ether/PI_interp/layout_clean* cellviews open. Go to **Windows – Assistants – Navigator.** Minimize the schematic cellview and maximize the layout cellview.

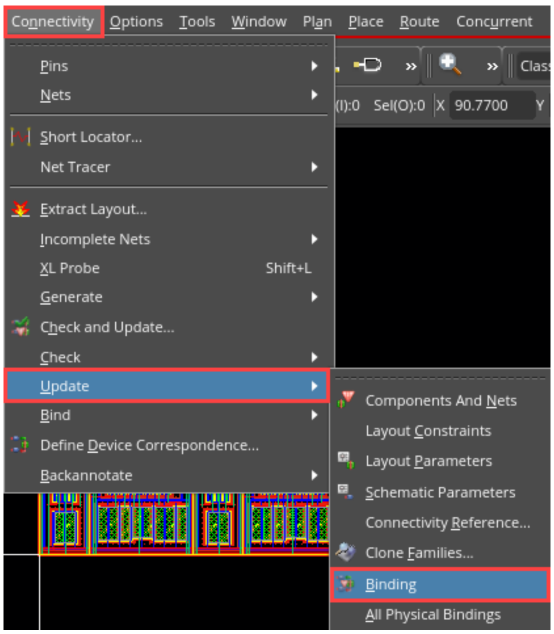


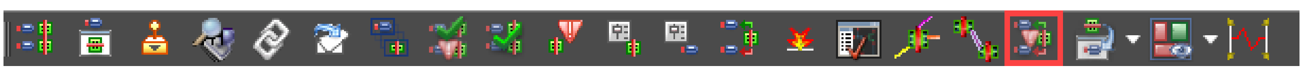


As the design was completed outside of Layout EXL, the first time it is opened in Layout EXL, the binder will attempt to bind instances by name.

In this case, none of the instances binds, indicating the design is **NOT** XL-compliant. Therefore, let us run Update Binding to resolve this.

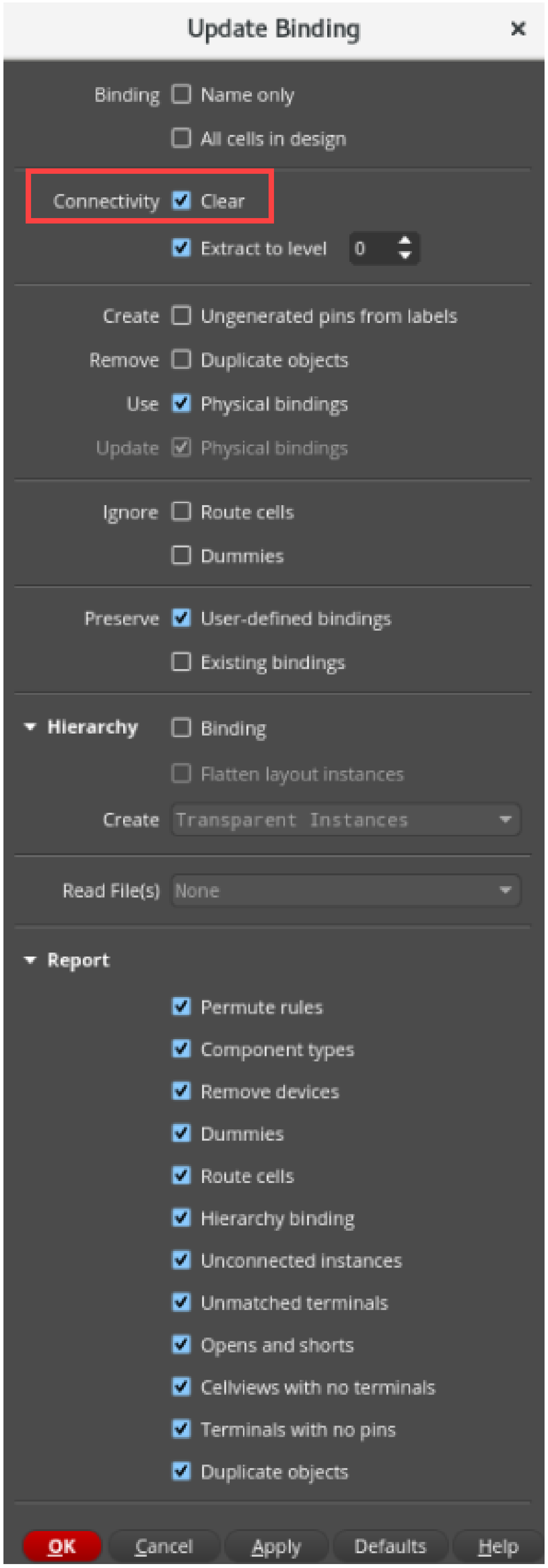
4. Run Update Binding either from the Update Binding icon on the Layout EXL toolbar or by selecting the **Connectivity** – **Update** – **Binding** menu command, as shown here:





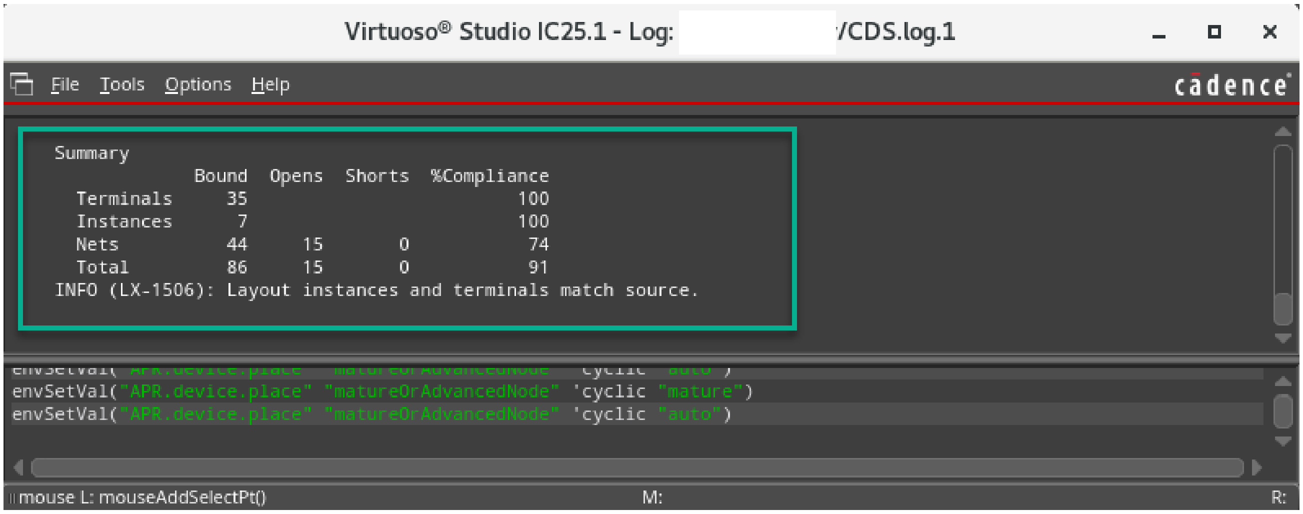
The Update Binding form appears.

5. Ensure that the options are set as shown here. The only option changed from the default is **Clear Connectivity**, which is now selected. This is done to clear any incorrect or sticky connectivity in the layout.

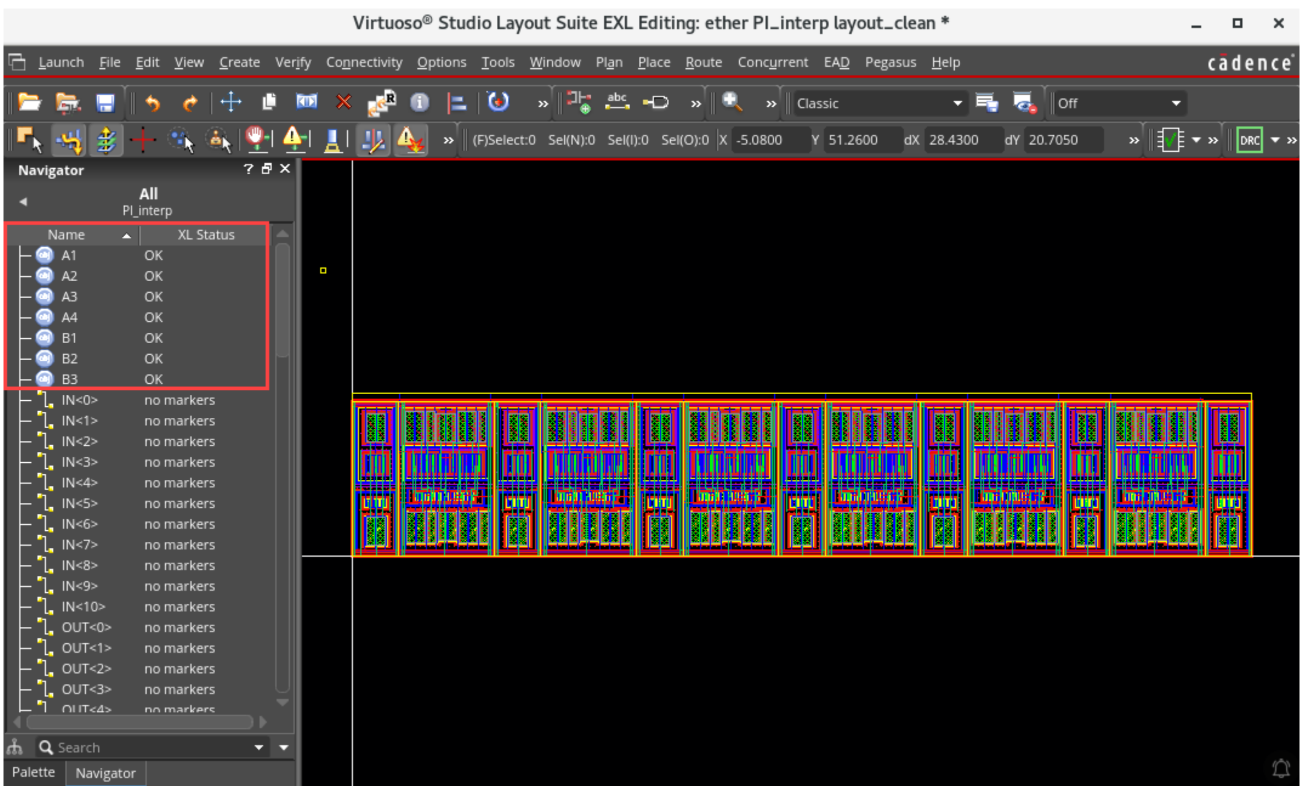


6. Click **OK** to proceed with the update.

   Update Binding runs and reports the binding status in the CIW window.

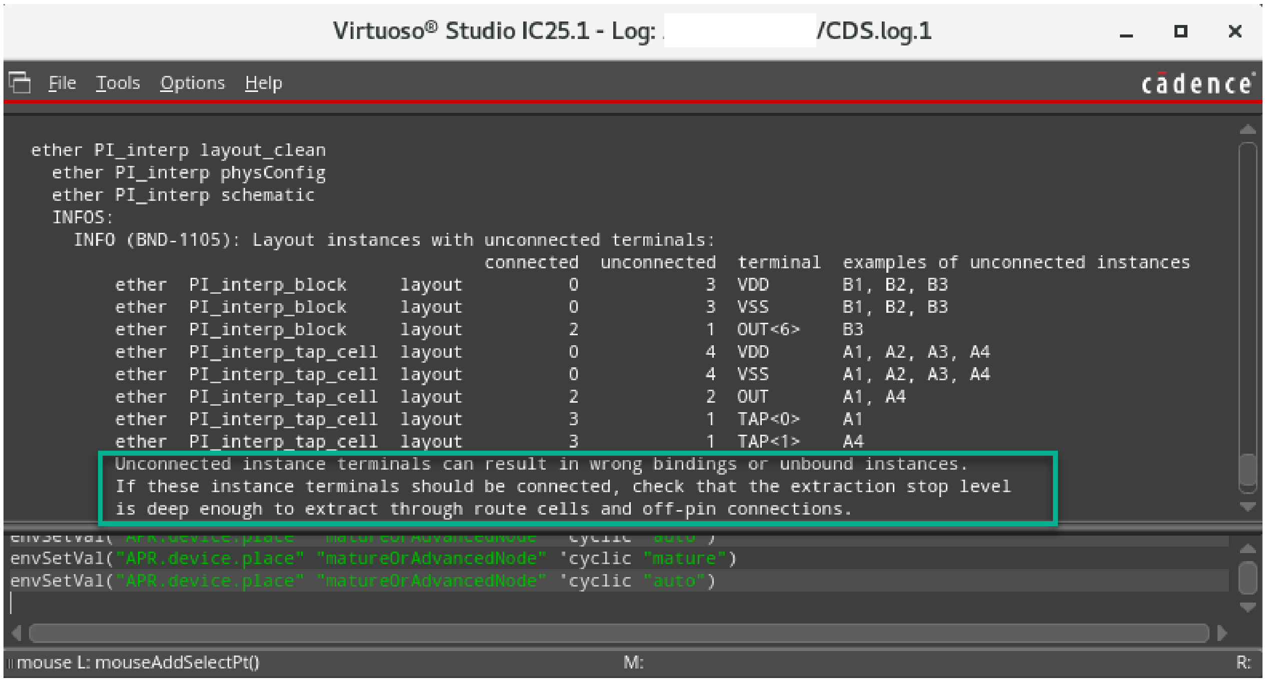


   You can see that all the instances, terminals, and nets are bound. This can be confirmed by checking the XL Status in the Navigator, as shown here:

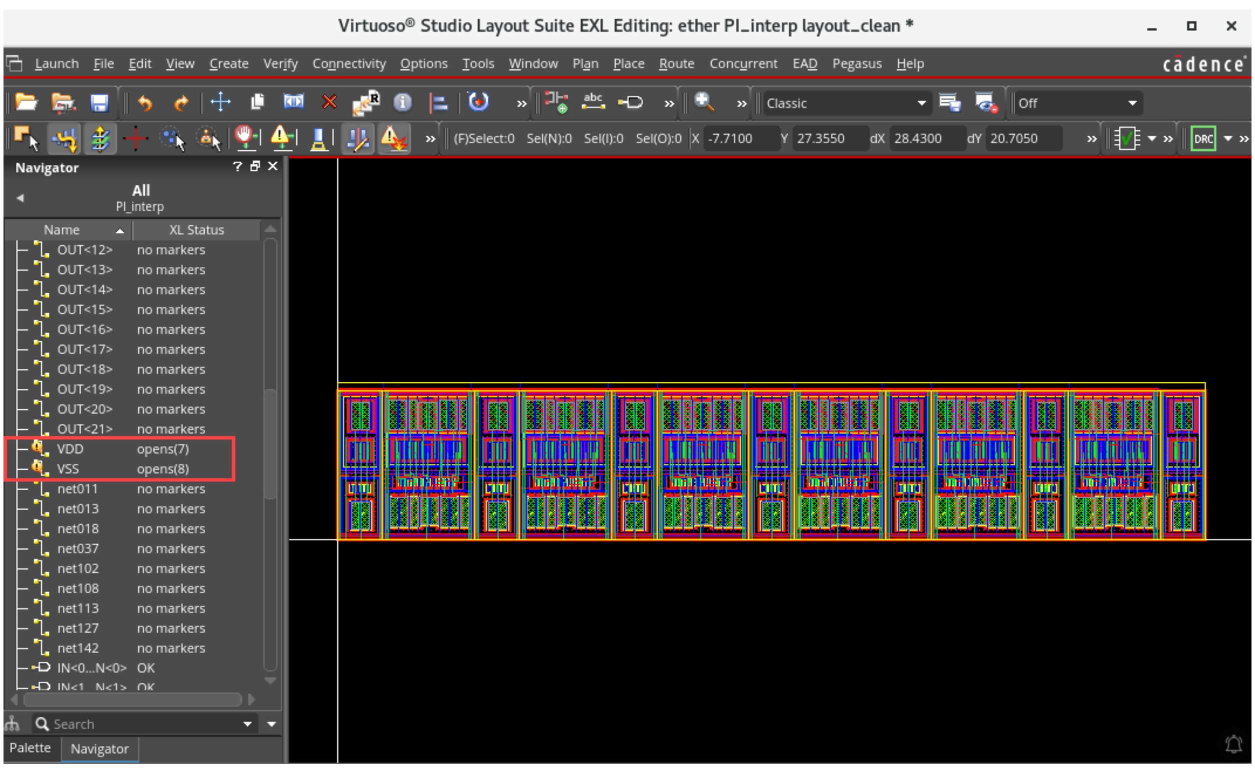


   Notice that the CIW window reports **15 Opens** in the layout that need to be investigated. Scrolling up the CIW window, you can see that several INFO messages have been output.

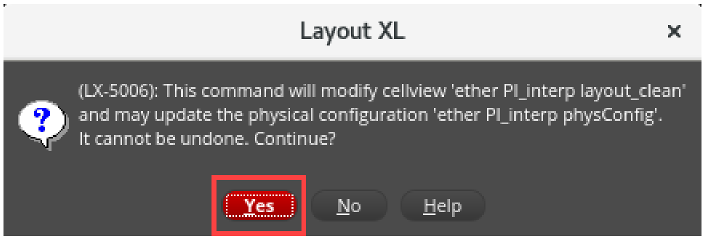
Update Binding issues INFO messages to assist you with suggested actions to make the design XL-compliant, as shown here:



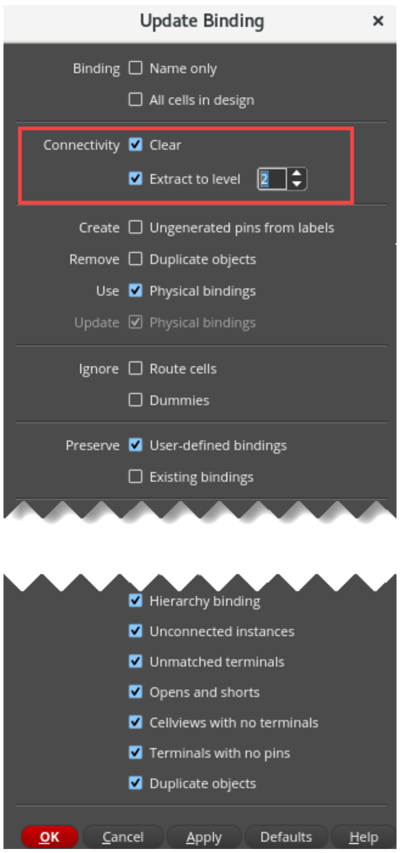
The opens are on the **VDD** and **VSS** nets as seen here in the Navigator window and are caused because there are off-pin connections in the cells' lower hierarchy. To resolve the opens, you need to increase the extraction level.



7. Invoke the Update Binding form again, **Connectivity – Update – Binding**. You get a pop-up message, as shown here:



8. Click **Yes** to **Continue,** and the Update Binding form opens.

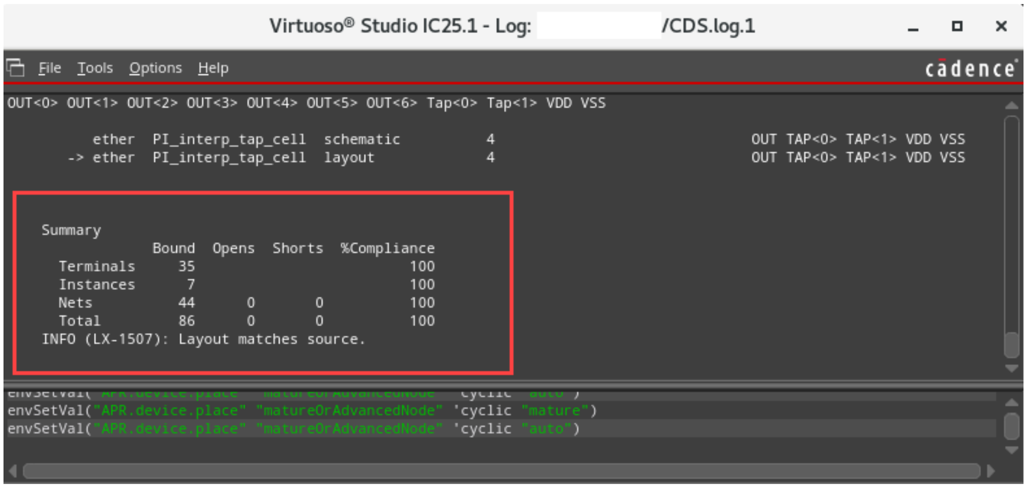


9. Change the value in the *Extract to level* field to **2** to increase the extraction depth.

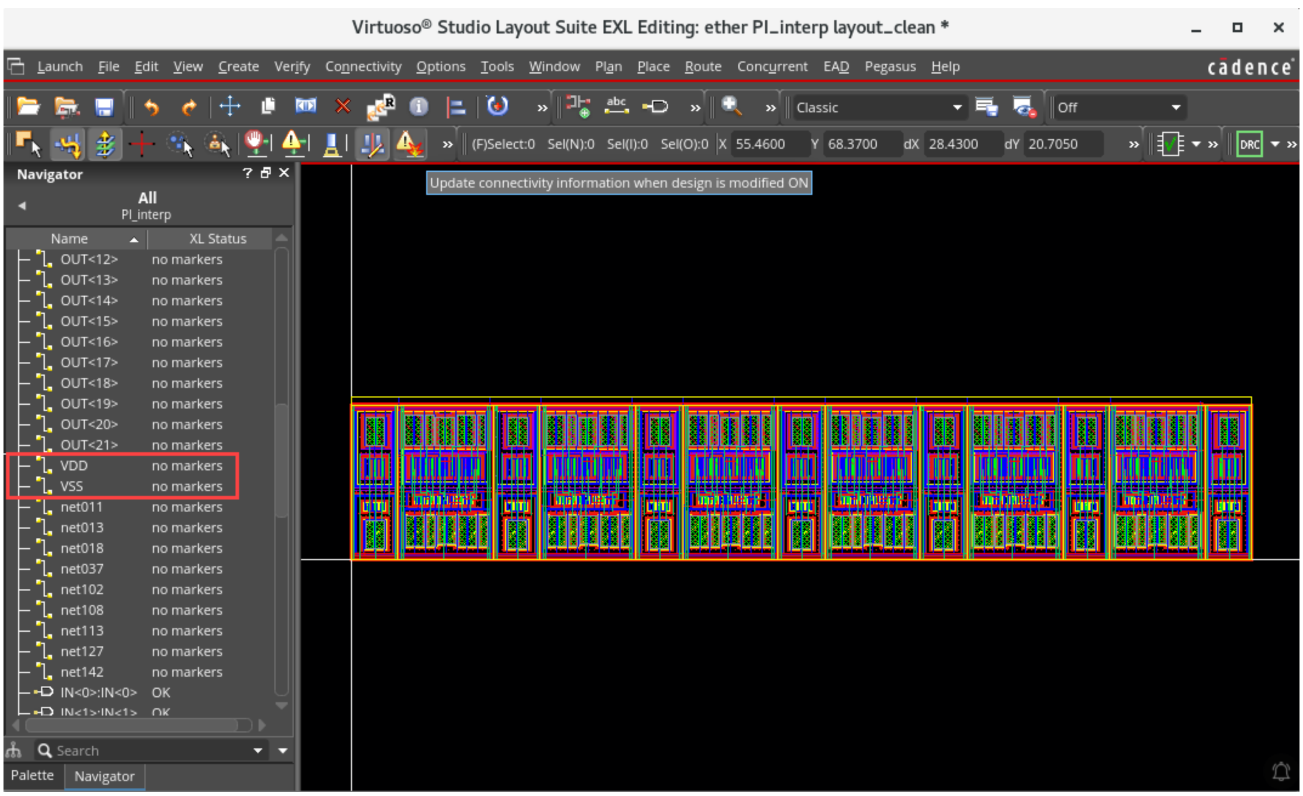
10. Run **Update Binding** again by clicking **OK**.

Important: As the bindings were correct, it was not strictly necessary to re-run the Update Binding. Running **Extract Layout** to a depth of level **2** would also resolve the opens.

After Update Binding completes, you can see the layout is fully bound with **NO Opens** and **Shorts** and therefore, fully XL compliant. You can verify this in the CIW window.



You can verify that the opens on the VDD and VSS nets are removed after Update Binding, as shown here by checking the XL Status in the Navigator window.



11. Close the schematic and layout views. Do **NOT** save.

12. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned:

- How to use the Update Binding features.

End of Lab

Lab 3-4 Using Complex Binding

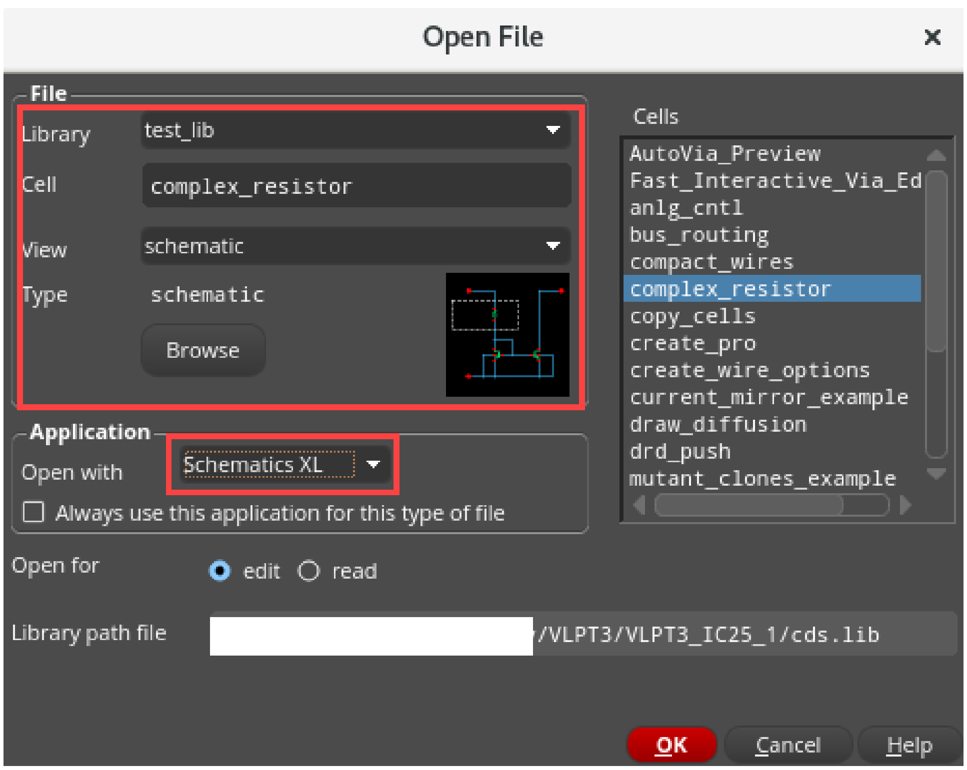
Objective: To explore series/parallel mapping and the Complex Binding feature.

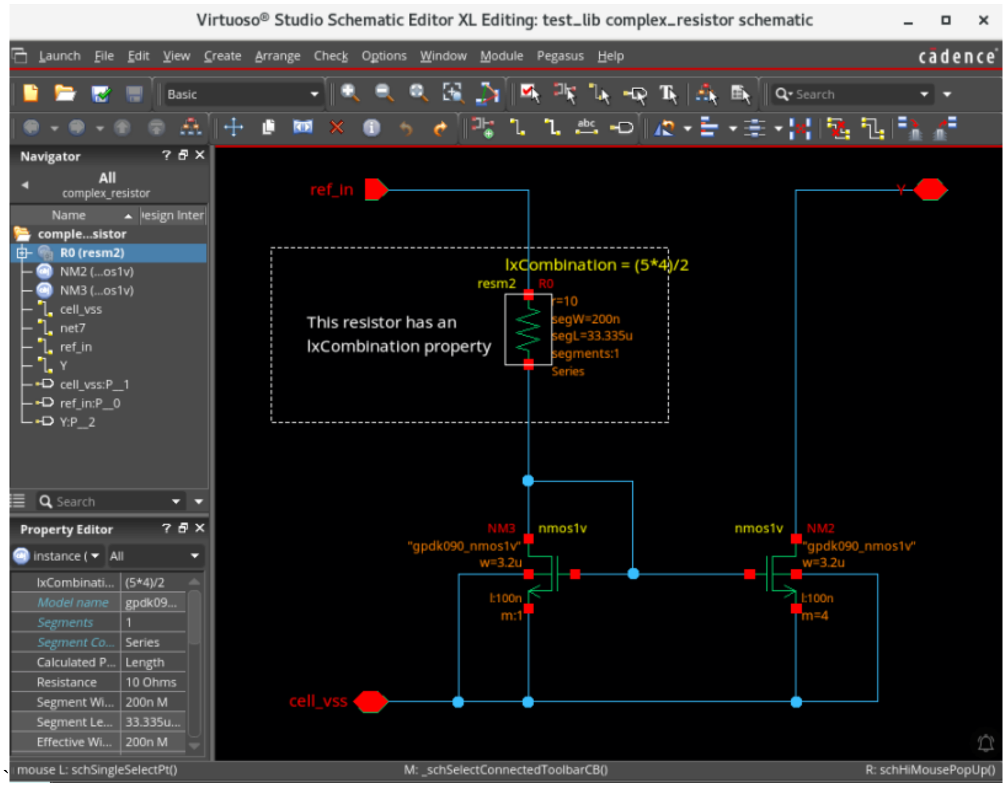
You typically use lxCombination to split a single resistor into multiple smaller devices to achieve maximum density in the layout by placing devices in spaces otherwise not utilized. This lets you approximate the resistance value defined in the schematic without violating the manufacturing grid.

If the lxCombination property is defined on the schematic symbol, the **r** parameter is ignored by the Generate All From Source (GFS) and Update Components And Nets (UCN) commands. The layout instances will have the correct parameters based on the lxCombination factor setting.

1. Open the following schematic in XL.

Library	`test_lib`
Cell	`complex_resistor`
View	`schematic`



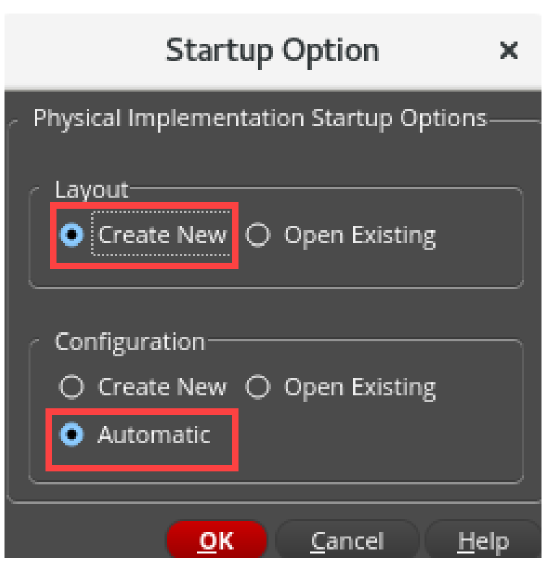


The schematic opens. You can check the properties on the resistor instance.

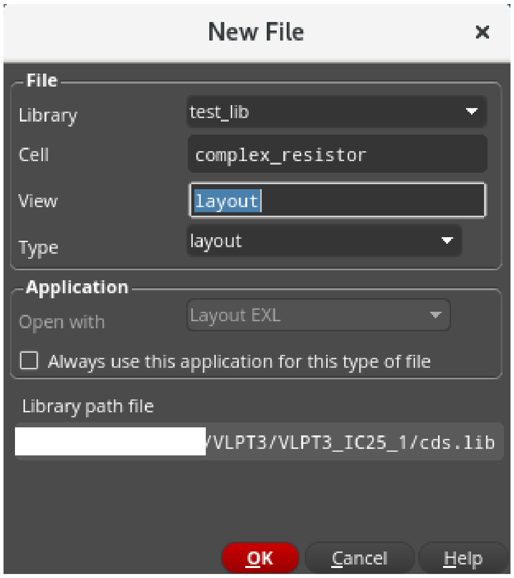
You can notice the *lxCombination* property. The total resistance must be 10 Ohms. In this case, it is obtained by using “2 parallel branches of 4 resistors in series each”. In this way, only 5 Ohm resistors are used.

2. In the schematic, choose **Launch – Layout EXL** to bring up the **Startup Option** menu.

3. Select the options as shown here to create a new layout view and click **OK**.



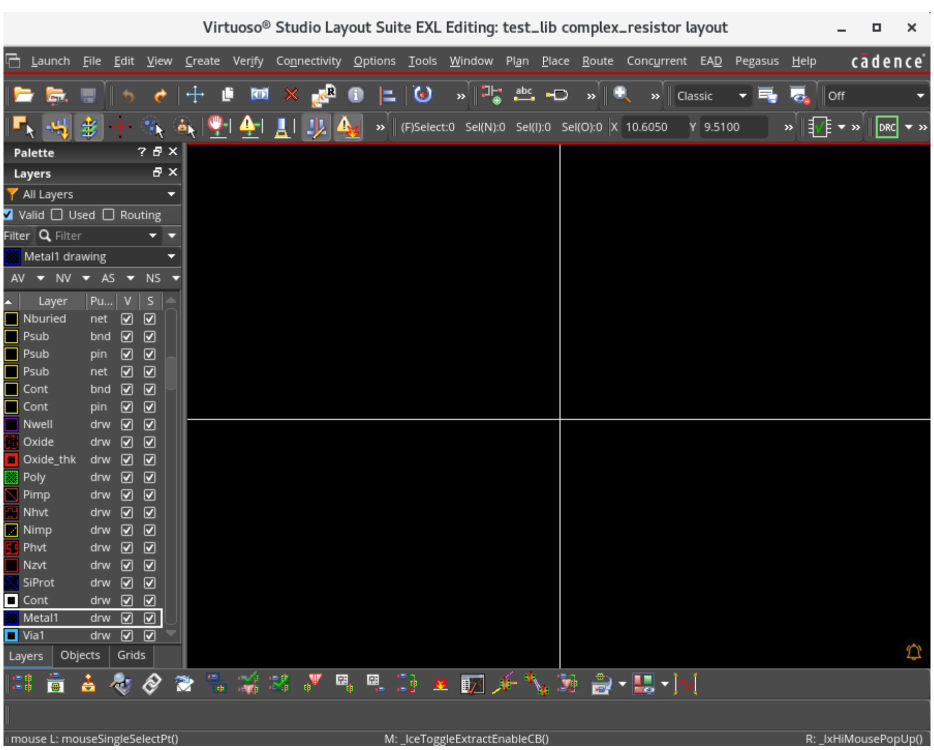
4. Leave the default settings on the New File form to create a new layout view called **layout**.



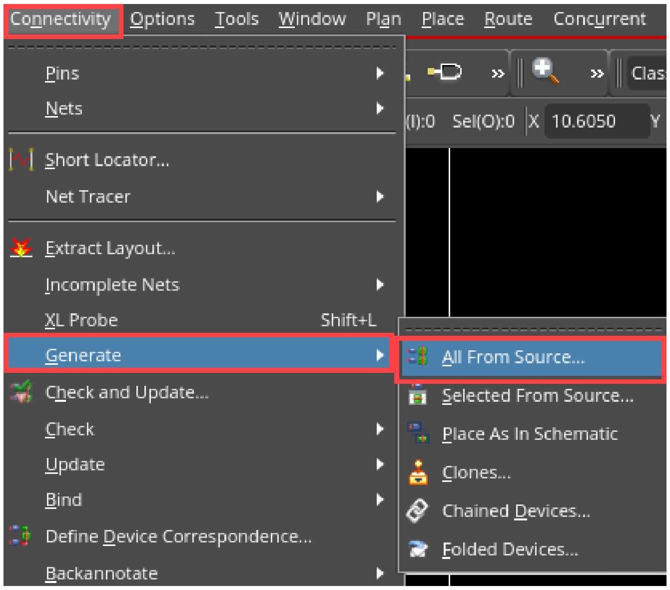
5. Click **OK**.

   The Virtuoso Layout Suite EXL palette opens with the *test_lib/complex_resistor/layout* cellview.

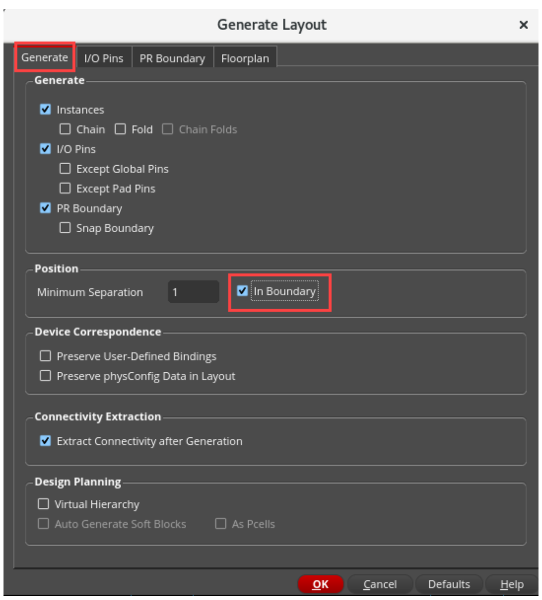
   Note: If you do not see “ungenerated” in the XL Status in the Navigator assistant in the layout canvas, you can click on the Ungenerated section in the OBJECTS category to view the ungenerated instances.

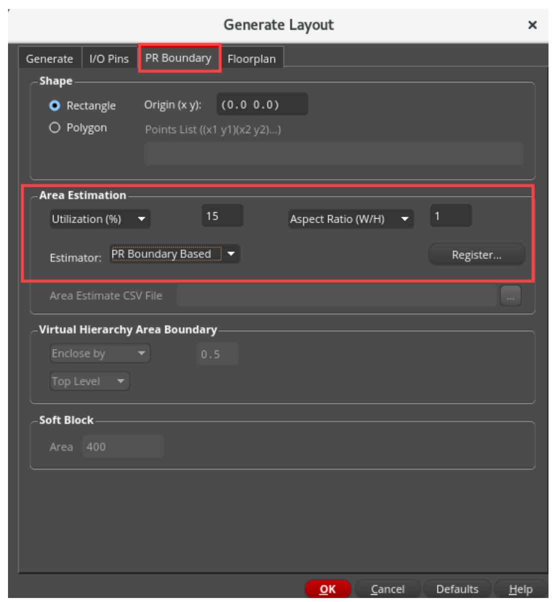


6. Generate All From Source by choosing either **Connectivity – Generate – All From Source** pull-down menu or clicking the **Generate All From Source (GFS)** icon on the Layout EXL toolbar.



The Generate Layout form appears.





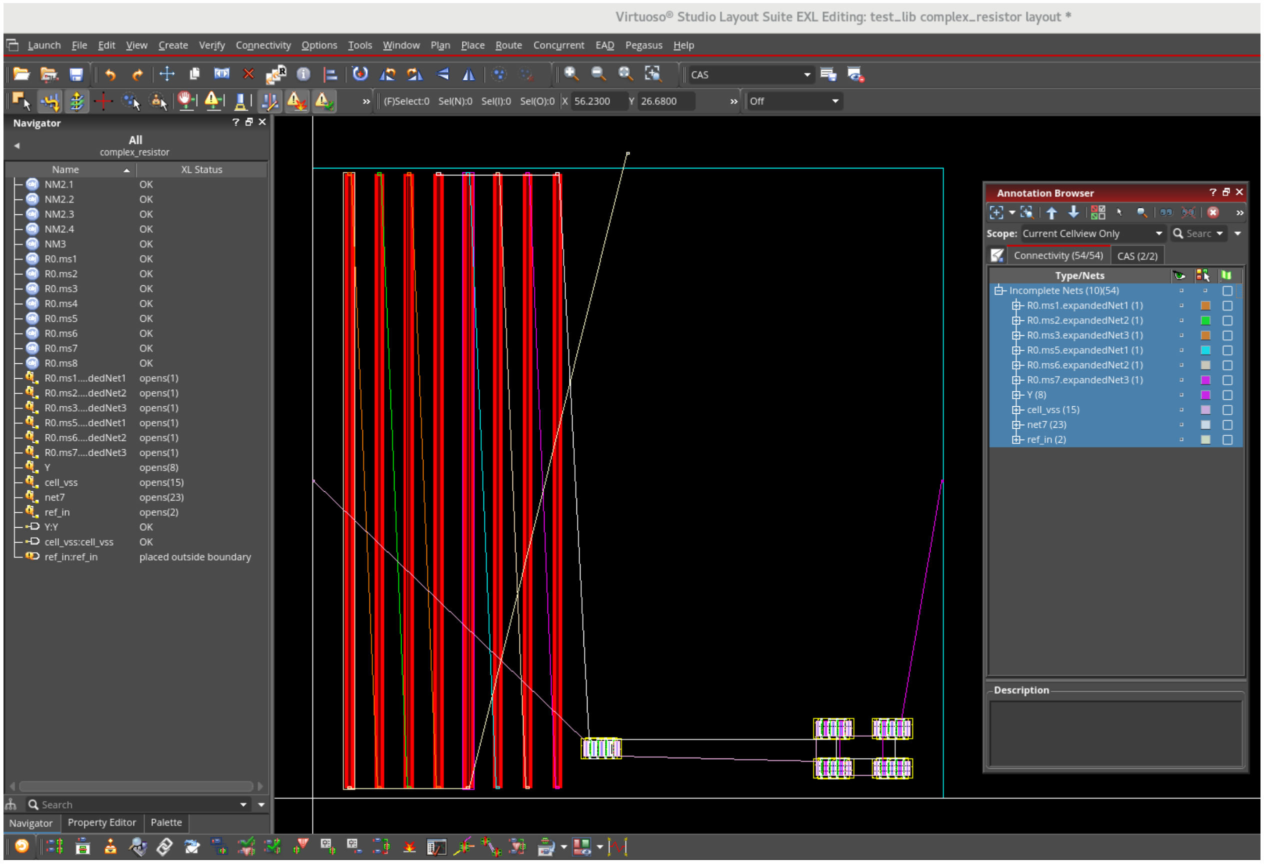
7. Select the settings, as shown here:

   a. In the **Generate** tab, keep the *In Boundary* option **ON**.

   b. In the **PR Boundary** tab, keep *Utilization (%)* as **15** and *Aspect Ratio (W/H)* as **1**.

8. Click **OK.**

Your placement looks like this:

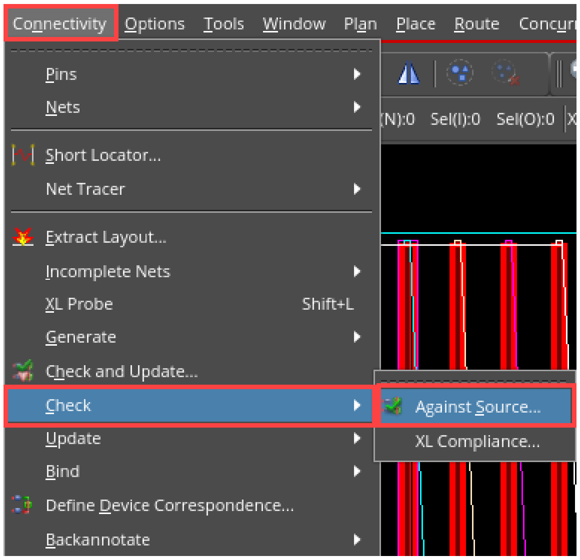


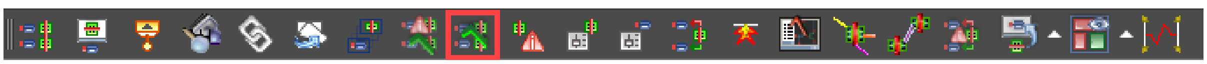
As expected, eight resistors of 5 Ohms each have been generated. You can display the flight lines using the Annotation Browser assistant to see how they are connected; there are two groups in parallel, each group including four resistors in series.

Note: You can invoke the Annotation Browser assistant by choosing **Window – Assistants – Annotation Browser**.

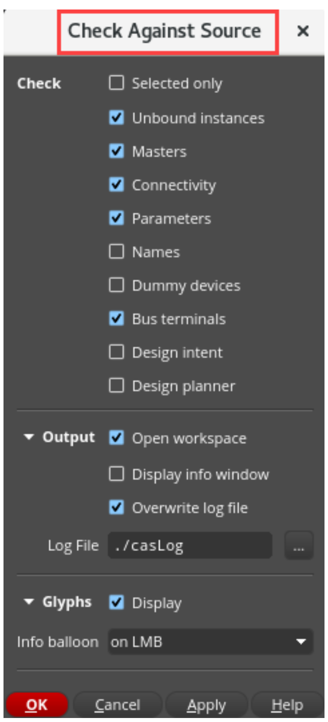
Note: Even if the flight lines are not showing the “optimal” connectivity, you can simply “ignore” them and connect the resistors. Layout EXL will then use the “permutable-pins” property to automatically swap the pins according to the connections you are creating.

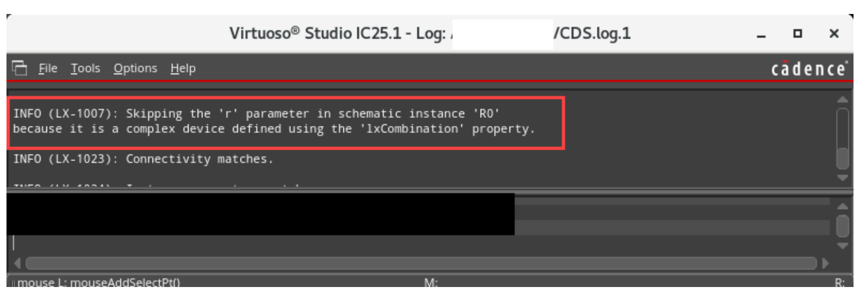
9. Execute **Check Against Source** by choosing either **Connectivity** – **Check** – **Against Source** or clicking the **Check Against Source** (CAS) icon.





The Check Against Source (CAS) form appears.





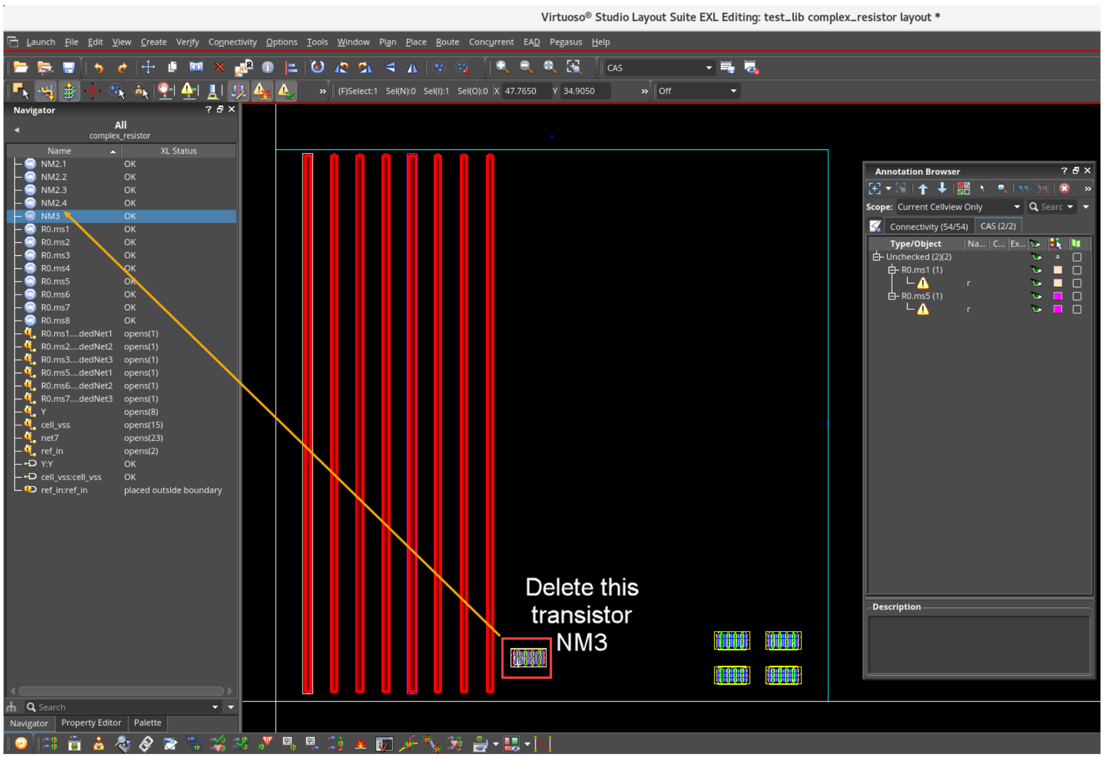
10. Have the settings, as shown here, in the CAS form and click **OK**.

CAS generates a summary report as shown here. You can see that no problem is reported, and the “total resistance” value is **NOT** compared between schematic/layout since the lxCombination value is used instead.

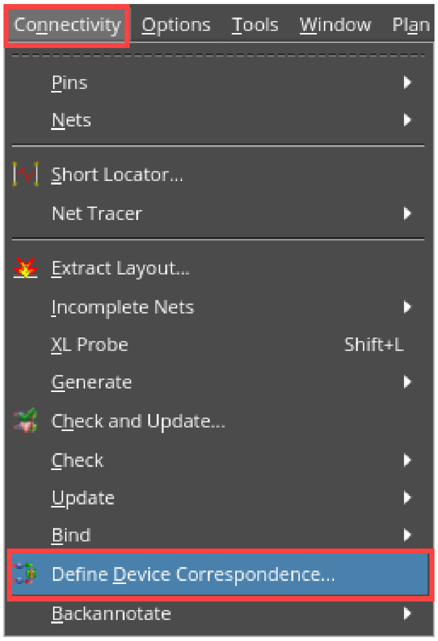
Now, you are going to use the Define Device Correspondence (DDC) form to investigate the schematic/layout instance mapping and to define a new device correspondence.

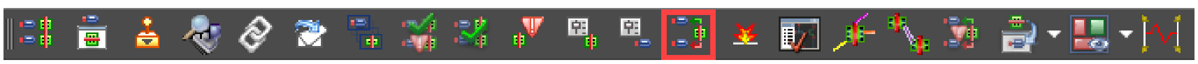
11. Close the Info window.

12. Delete the layout instance called **NM3**, as shown here:



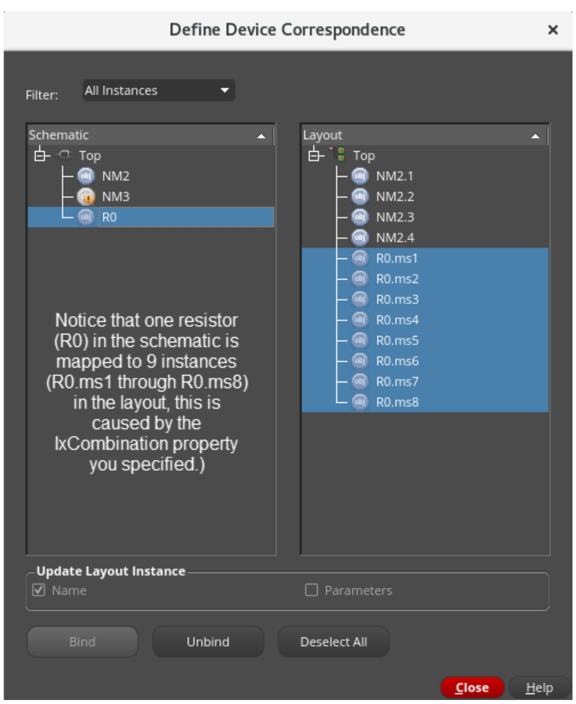
13. Use the menu entry **Connectivity** – **Define Device Correspondence** or the **Define Device Correspondence** icon on the Layout XL toolbar.





This will bring up the DDC form, which can be used to define the mapping between schematic and layout instances.

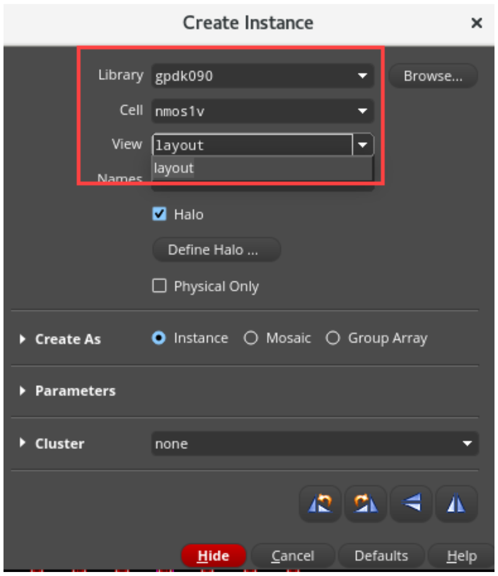
14. In the DDC form, in the *Filter* cyclic field, choose **All Instances** to show the already mapped ones.



The schematic instance **NM3** is now unmapped, hence you see the flag close to its name.

15. Close the Define Device Correspondence form.

    Now, you will create a new layout instance: ***library: gpdk090; cell: nmos1v; view: layout.***

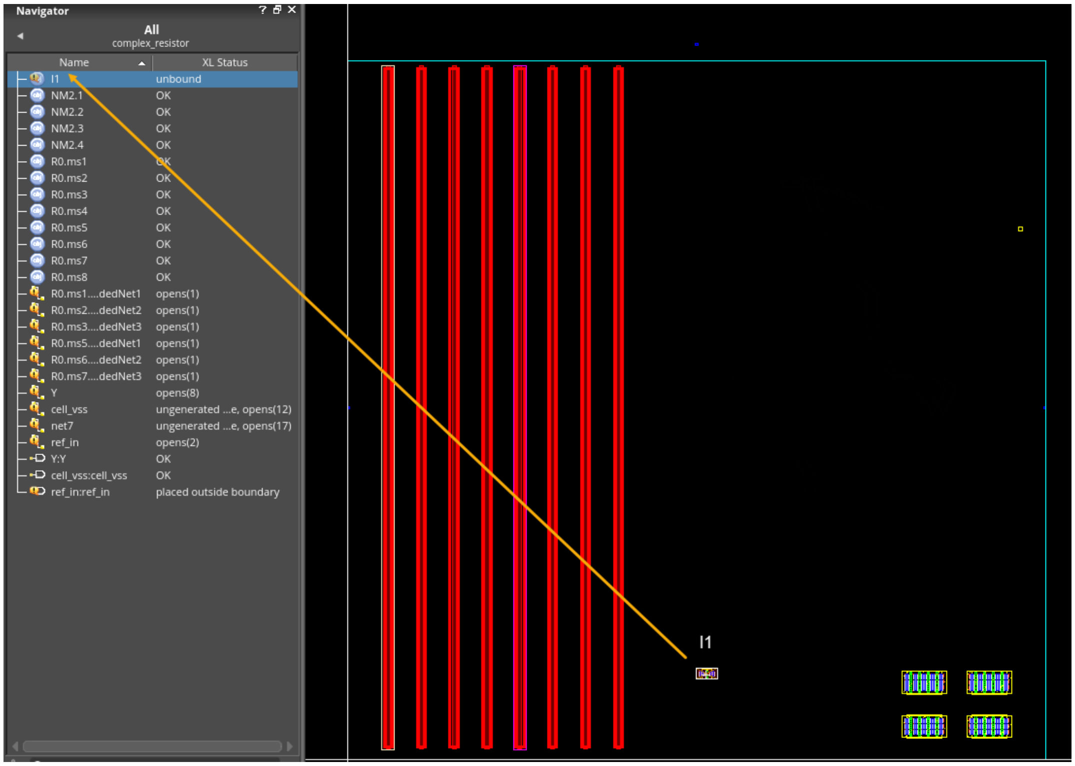
    

16. Invoke the Create Instance form by choosing either the **Create – Instance** menu command or pressing the bindkey **I** or by clicking the **Create Instance** icon on the Create toolbar.

    a. Click on the **Defaults** button to ensure that the remaining parameters are initialized with their default values.

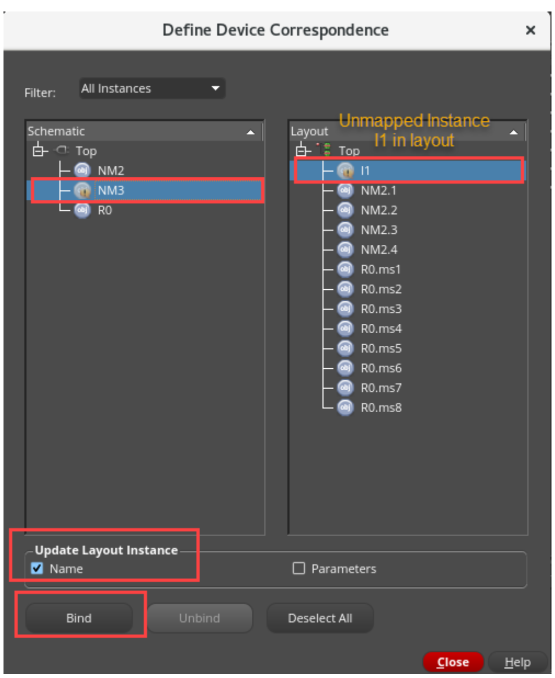
b. Place the instance *I1:gpdk090/nmos1v/layout* in the layout canvas.

This new instance **I1** is **unbound**.



17. Press the **ESC** key to cancel the Create Instance form.

18. Invoke **Connectivity** – **Define Device Correspondence** to bring up the mapping form again.

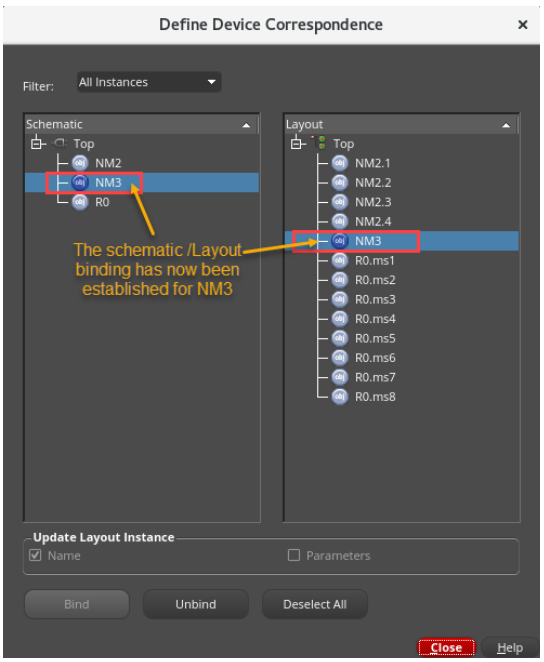


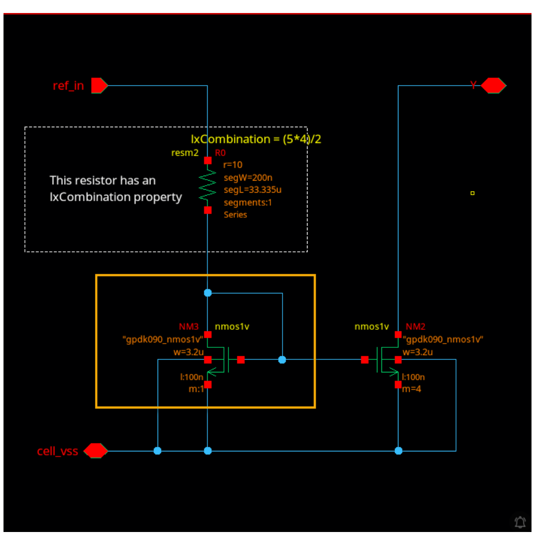
You can see that there is one unmapped instance (**NM3)** in the schematic and one in the layout (**I1**).

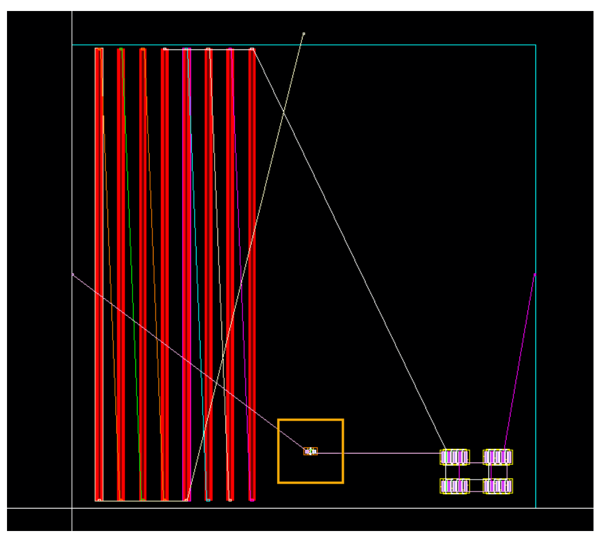
Important: It is recommended to check the *Name* option in the *Update Layout Instance* section of the form. This will update the layout instance name to match that of the schematic.

19. Select both the instances **NM3** and **I1** using **CTRL+Click** and then click the **Bind** button.

    The schematic/layout binding has now been established as seen in DDC form, and you can cross-select the instances in schematic/layout, as shown here:







20. Close the Define Device Correspondence (DDC) form and close the schematic and layout cellview without saving the changes.

21. Leave the Virtuoso session.

22. In the shell, enter the following command:

    ```
    cd ~
    ```

    This command places you in your home directory.

Summary

In this lab, you learned:

- How to define a complex series/parallel mapping.

End of Lab

Module 4: Using Constraint Groups in Virtuoso

Lab 4-1 Using Constraint Groups

Objective: To take advantage of the Constraint Groups during editing.

In this lab, you will learn how to define Constraint Groups and how to select the proper Constraint Group for your design.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. If you continue from the previous lab, move to the sub-heading: Starting the session Otherwise, proceed to the next step.

2. Update the following line in the /VLPT3_IC_25_1/. cshrc_VLPT3 file with the appropriate valid local tool paths for Virtuoso.

   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
   ```

3. Save and close the file.

4. Source the updated class setup file.

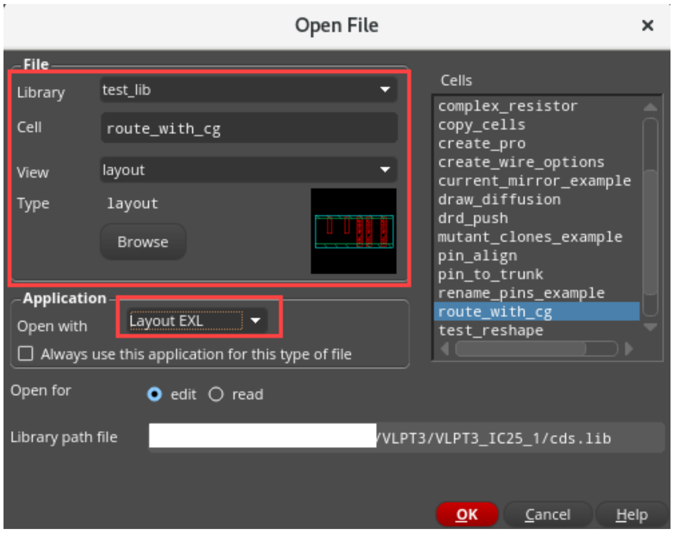
   ```
   Source .cshrc_VLPT3
   ```

Starting the session

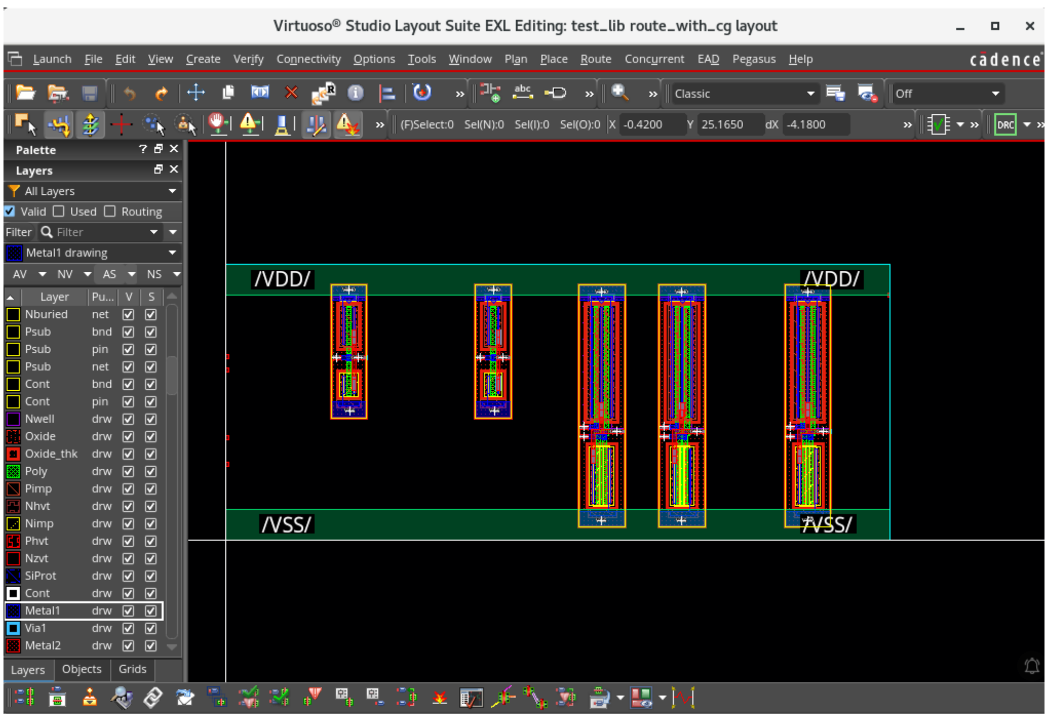
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

   ```
   cd ./VLPT3_IC_25_1/VLPT3_M04_VCG
   ```


2. In the CIW, choose **File** – **Open** and enter the following values in the form, as shown here:



The *test_lib/route_with_cg/schematic* and *test_lib/route_with_cg/layout* cellviews open. Minimize the schematic cellview and maximize the layout cellview.



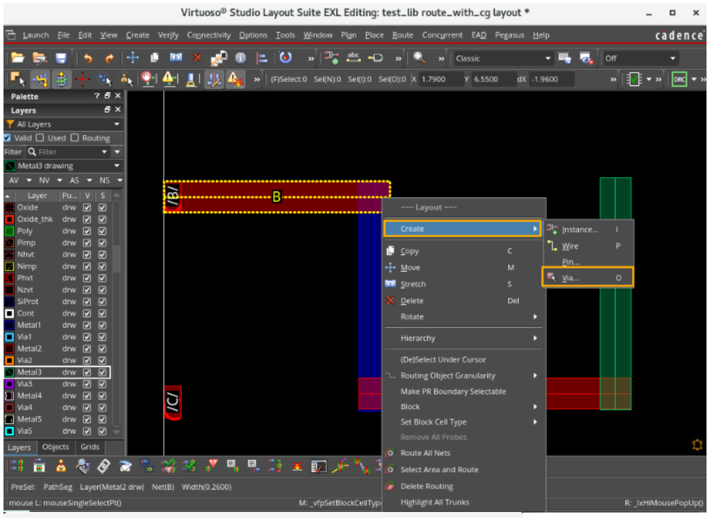
3. Invoke the Create Wire form by choosing either the **Create** – **Wiring** – **Wire** pull-down menu or pressing the bindkey **P** or by clicking the **Create Wire** icon on the Create toolbar. If the Create Wire form does not pop up, press **F3** to bring up the form.

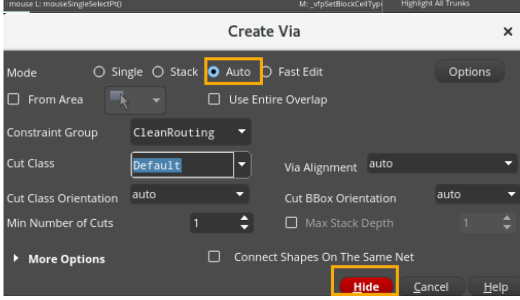
   The Create Wire form appears.

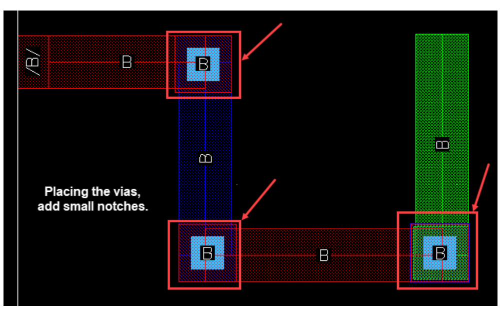
4. Start interactive routing with the **Metal2** layer from **Pin B** and complete the connection, as shown here.

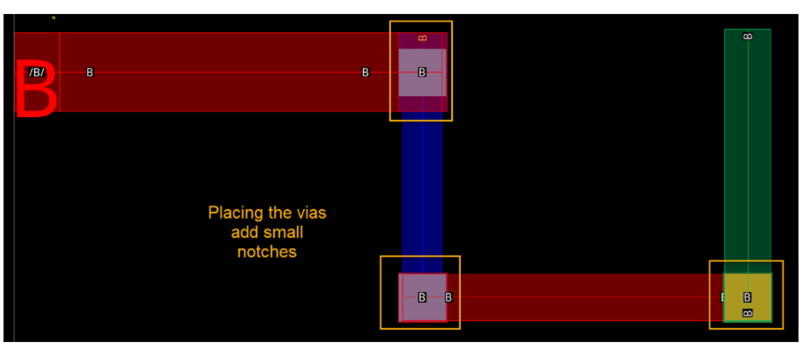
   You can drop via using the **Right Mouse Button** (**RMB**), and go to **Create** – **Via**, which pops up the Create Single Wire form. You can use the bindkeys listed below as per your need:

   You can drop via using ‘o’ Bindkey, place the via wherever required (as shown)









This wire has been created using “minimum” rules. As you can see, using these rules introduces small notches every time a via is placed while creating wire from Pin **B**.

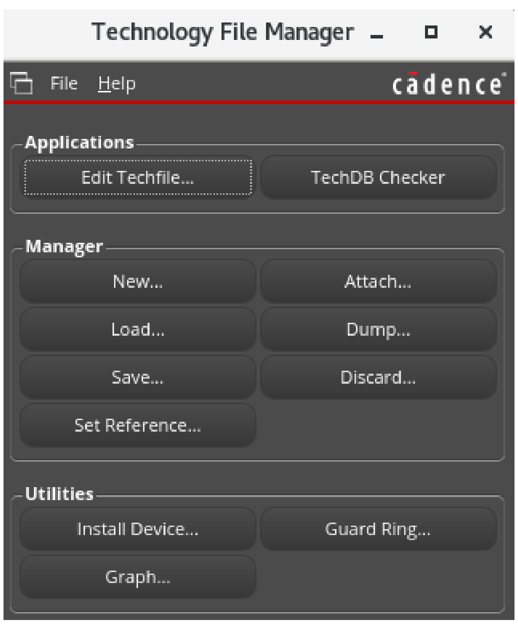
To get rid of these notches, you will introduce a new **Constraint Group** (**CG**) where the widths are slightly larger, thus allowing placing vias without any notches.

5. You can have a look at the new constraint group you will add by opening the file **new_constraint_groups.tf** with a text editor (for example, “nedit”).

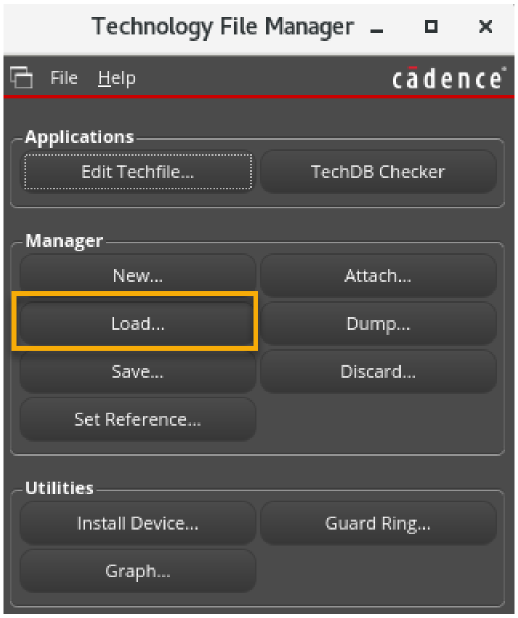
```
;------------------------------------------------------------------------------
; Technology File gpdk090
;------------------------------------------------------------------------------
; This file can be used to create additional constraint groups, which introduce
; a relaxed design style (part of Virtuoso Layout Pro T3 training).
;------------------------------------------------------------------------------

;********************************
; CONSTRAINT GROUPS
;********************************
constraintGroups(

  ( "CleanRouting"      nil
    spacings(
     ( minWidth  "Poly"   0.120 ) ; avoid cont notches
     ( minWidth  "Metal1" 0.260 ) ; same width as Via1
     ( minWidth  "Metal2" 0.260 ) ; avoid via2 notches
     ( minWidth  "Metal3" 0.260 ) ; same width as Via2
    ) ; spacings
  ) ; CleanRouting

  ( "DoubleVias"        nil
    spacingTables(
      ( minNumCut "Cont"  (( "width" nil nil)) (0.0 2))
      ( minNumCut "Via1"  (( "width" nil nil)) (0.0 2))
      ( minNumCut "Via2"  (( "width" nil nil)) (0.0 2))
      ( minNumCut "Via3"  (( "width" nil nil)) (0.0 2))
    ) ;spacingTables
  ) ; DoubleVias

) ; constraintGroups
```

6. In the CIW, choose **Tools** – **Technology File Manager**.

   The Technology File Manager form opens.

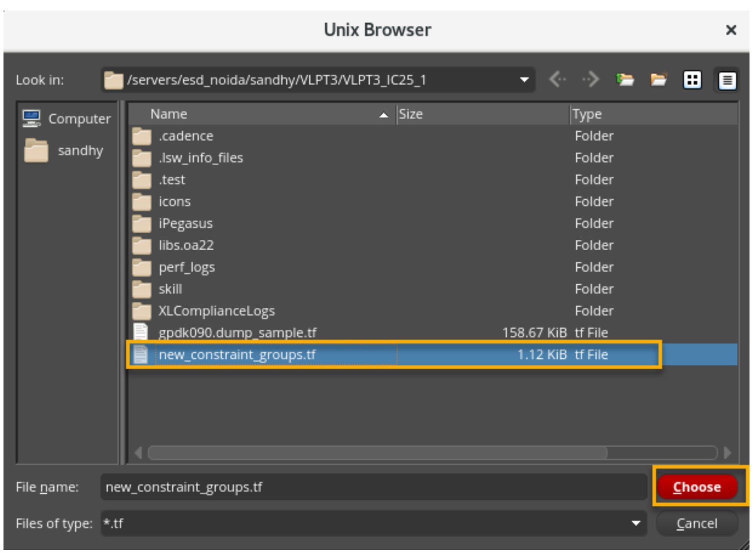


7. In the Technology File Manager form, click the **Load** button.



8. The Load Technology File form pops up; perform the following actions, as shown here.

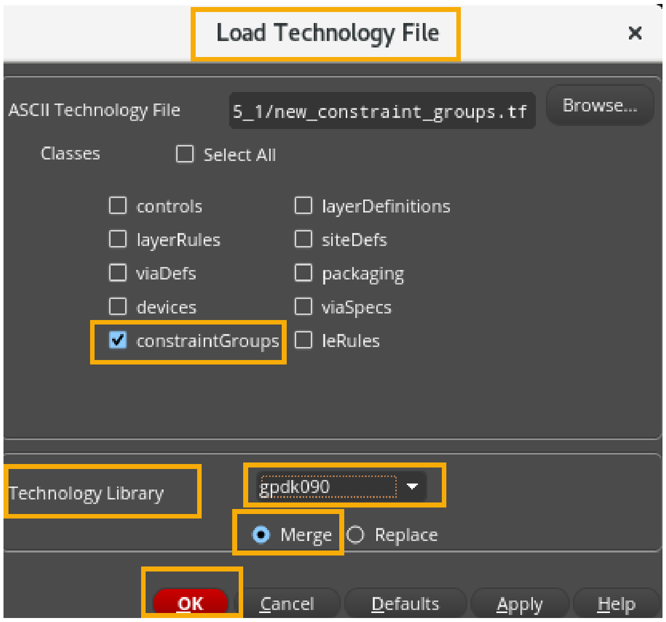
   a. Browse to the **new_constraint_groups.tf** file in the ASCII Technology File section.



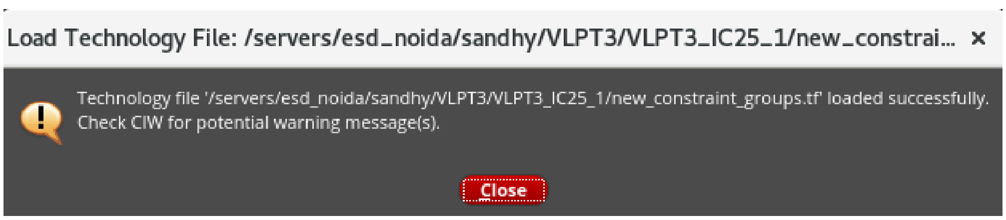
   b. Ensure that only the **constraintGroups** checkbox is selected.

   c. Select the Technology **Library – gpdk090**.

d. Choose the **Merge** option.



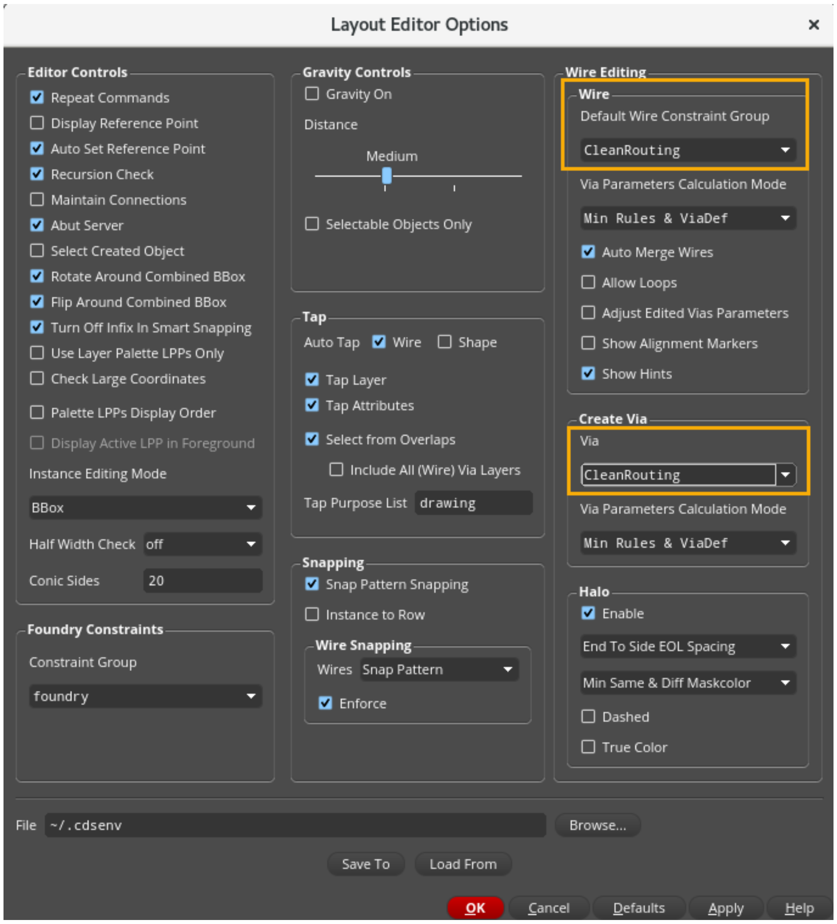
9. Click **OK**.



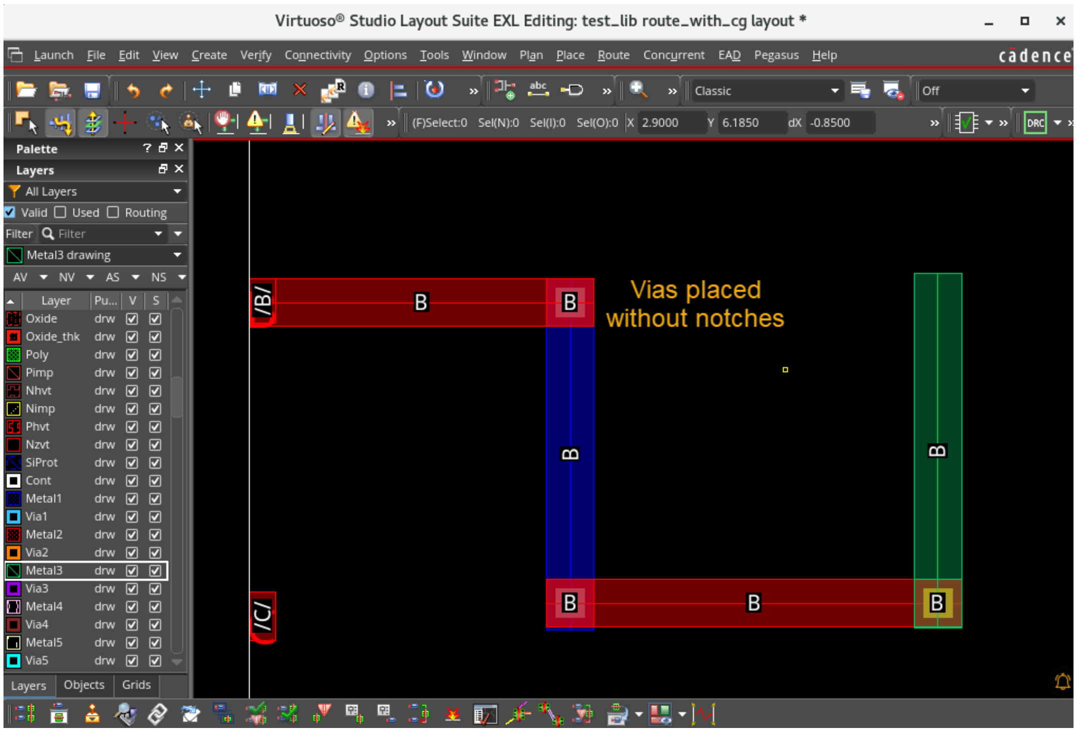
10. You will now use the newly introduced **Constraint Group (CG)** while routing.

11. Discard all edits done in the layout window using **File** – **Discard Edits** and clicking **Yes** in the Discard Edits pop-up form.

12. Now, activate the new **CG** using **Options** – **Editor** or pressing the bindkey **Shift+E** and then select **CleanRouting** as the **CG** to use for both **Wire** and **Via creation** in the Layout Editor Options form, as shown here:

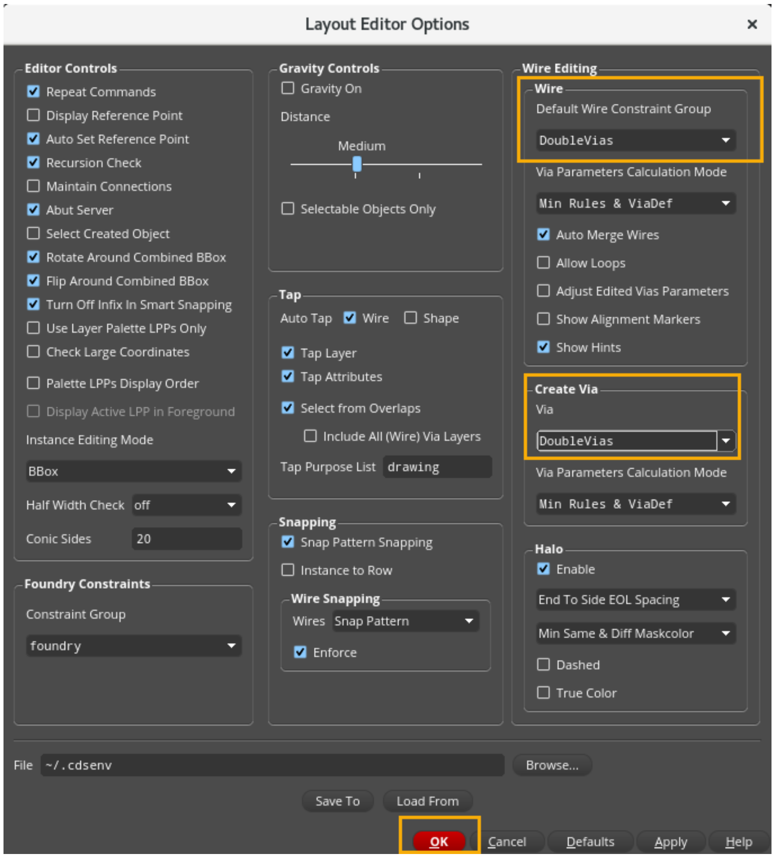


13. Repeat the **Create Wire** steps reported before from **Pin B**. Refer to **Steps 2** and **3**.

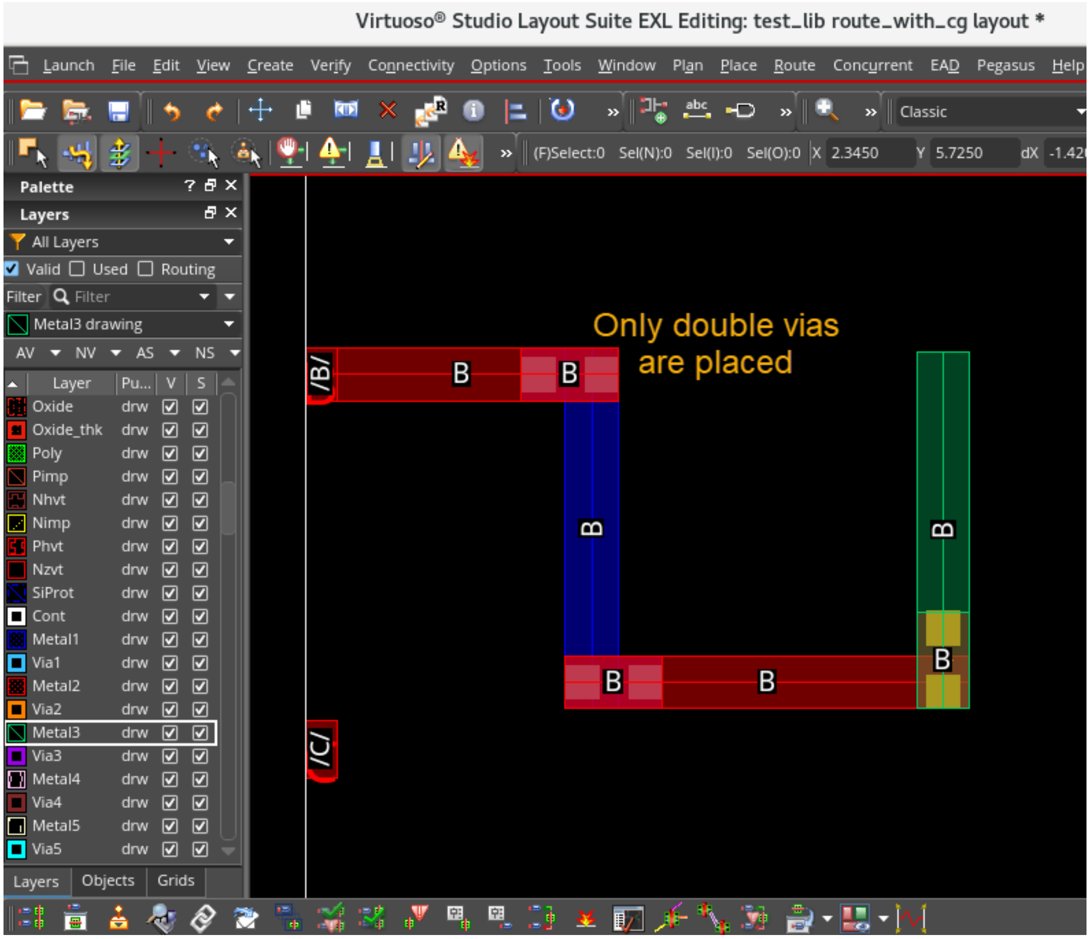


You can see that, with **CleanRouting** as the **CG**, you are able to place vias without any notches.

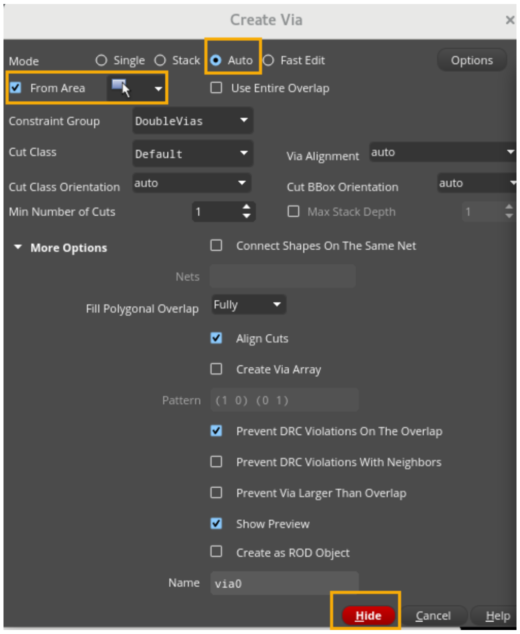
14. In addition to the **CleanRouting CG**, a **DoubleVias CG** has been created. To activate, go to **Shift+E**, and then choose this **CG** for both **Wire** and **Via creation**. You can ensure that only double vias are placed during the interactive routing, as shown here:

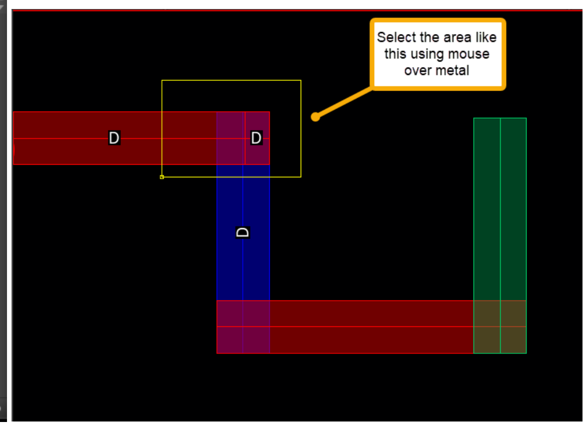


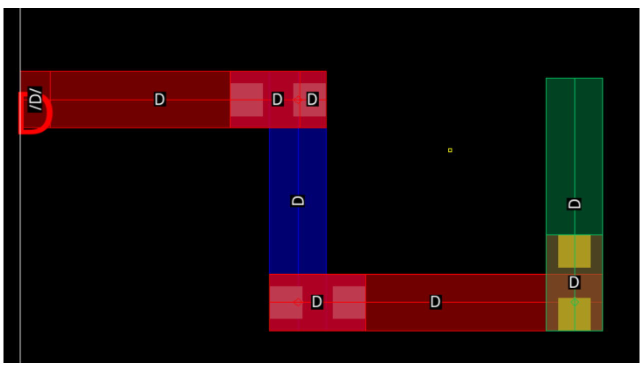
15. Start interactive routing with the **Metal2** layer from **Pin D** and complete the connection, as shown here:



Note: Select the "From Area" while placing the via, and select the area where you want the via. It will auto-place the via without a notch. Drag and select the area over the wire.







You can observe that only double vias are placed during the interactive routing.

16. Close all the cellviews without saving the changes.

17. Close the Virtuoso platform.

Summary

In this lab, you learned:

- How Constraint Groups are defined.
- How to select the proper Constraint Group for interactive routing.

End of Lab

